# ACADEMIC REGULATIONS
# &
# SYLLABUS 2025 -26

## Faculty of Computer Science & Applications

## Smt. Chandaben Mohanbhai Patel Institute of Computer Applications

### Bachelor of Science in Information Technology

### (BSc-IT) Programme

### (as per NEP-2020)

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
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# ACADEMIC RULES

To ensure uniform system of education, duration of under graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

## 1. System of Education

The Semester system of education should be followed across the Charotar University of Science and Technology (CHARUSAT) at Bachelor's levels. Each semester will be at least 90 working days' duration. The Curriculum and Credit Framework for Undergraduate Programmes (CCFUP) is based on the University Grants Commission's (UGC) guidelines, with a student-centric focus, flexible choice-based credit system, and a multidisciplinary approach. This framework also provides multiple entry and exit options, facilitating students to align their education with their career aspirations by choosing subjects/fields of their interest. Every enrolled student will be required to take a specified load of course work in the chosen course of specialization and also complete a project/dissertation if any.

## 2. Certification and Duration of Study with Multiple Entry Multiple Exit (MEME)

The undergraduate degree will have a flexible duration of either three or four years, with multiple exit options during this period. The table below outlines the various certifications a student can earn at different stages of their undergraduate study:

## Table-1 Certification and Duration on MEME

| Duration of Study | Semesters Completed | Certification Earned |
|---|---|---|
| 4 Years | Eight Semesters (4 Academic Years) | Bachelor's Degree (Honours) |
| | | Bachelor's Degree (Honours with Research) |
| 3 Years | Six Semesters (3 Academic Years) | Bachelor's Degree |
| 2 Years | Four Semesters (2 Academic Years) | Undergraduate Diploma |
| 1 Year | Two Semesters (1 Academic Year) | Undergraduate Certificate |

In consonance with the National Education Policy (NEP) 2020 and the guidelines of the University Grants Commission (UGC), Charotar University of Science and Technology (CHARUSAT) implements the Multiple Entry and Multiple Exit (MEME) scheme in their Bachelor of Science in Information Technology(BSc-IT) programme. This structure provides students with the flexibility to enter, exit, and re-enter the programme at various stages, each with its corresponding certification mentioned above in Table-1.

## Table 2: Multiple Entry Multiple Exit options

| NCrF Credit Levels | Qualification Title | Credit Requirement | Programme Exit | Additional Requirement | Re-entry to Degree Programme |
|---|---|---|---|---|---|
| 4.5 | UG Certificate | 48 | After Year 1 | Completion of one 4- credit of summer internship in core specific NSQF defined course during summer vacation of Year 1 | Within three years of exit |

| | | | | | |
|---|---|---|---|---|---|
| 5 | UG Diploma | 92 | After Year 2 | Completion of one 4- credit of summer internship in core specific NSQF defined course during summer vacation of Year 2 | Within three years of exit |
| 5.5 | Three Year Bachelor Degree | 132 | After Year 3 | Compliance with minimum credit requirements specified in regulations | Within three years of exit |
| 6 | Bachelor‟s Degree with Honors | 176 | After Year 4 | Compliance with minimum credit requirements specified in regulations | NA |
| 6 | Bachelor‟s Degree with Honors with Research | 176 | After Year 4 | Compliance with credit requirements as specified in regulations | NA |

Students who are opting for exit at any level, shall re-enter the institution to complete the UG Degree, where they had left off. They can re-enter within three years of exit and complete the degree programme within the stipulated maximum period of seven years from the date of admission to UG programme.

## 3. CREDIT FRAMEWORK FOR BSc-IT(Bachelor of Science in Information Technology)

Arrangement of Credit Distribution Framework for three/four years Honors/Honors with Research Degree Programme with Multiple Entry and Exit Options.

(As per GR No: KCG/admin/2023-24/0607/kh.1, Sachivalaya, Gandhinagar, Date-11/07/2023).

| Sr.No. | Categories of Courses | Credit Requirement for each Category | | | | |
|---|---|---|---|---|---|---|
| | | Certificate (1 Year) | Diploma (2 Years) | 3-Year UG | 4-Year UG (Honors) | 4-Year UG (Honors+Research) |
| 1 | Major - Core Courses | 16 | 40 | 64 | 88 | 88 |
| 2 | Minor-Discipline Specific Electives | 8 | 12 | 24 | 32 | 32 |
| 3 | Multidisciplinary Courses Open Electives | 8 | 10 | 12 | 12 | 12 |
| 4 | Ability Enhancement Courses(AEC) | 4 | 8 | 10 | 10 | 10 |
| 5 | Skill Enhancement Courses(SEC) | 4 | 10 | 14 | 14 | 14 |
| 6 | Value Added Courses (VAC) | 4 | 8 | 8 | 8 | 8 |
| 7 | Summer Internship/ Research Project /Dissertation | - | - | - | 12 | 12 |
| 8 | Exit Courses | 4 | 4 | - | - | |
| | **Total** | **48** | **92** | **132** | **176** | **176** |

## 4. Structure of BSc-IT (Bachelor of Science in Information Technology)

| NCrF Credit Level | Semester | Major Core | Minor | Multi/Inter Disciplinary | Ability Enhancements Courses(AEC) | Skill Enhancement Courses (SEC) | Value Added Courses(VAC)/IKS | Research Project/Dissertation | Total | Qualification/Certificate |
|---|---|---|---|---|---|---|---|---|---|---|
| **4.5 First Year** | **I** | **8** | **4** | **4** | **2** | **2** | **2** | **-** | **22** | **UG Certificate** |
| | **II** | **8** | **4** | **4** | **2** | **2** | **2** | **-** | **22** | |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **1st Year Credit Total** | | **16** | **8** | **8** | **4** | **4** | **4** | **-** | **44** | |
| **Exit 1**: Award of UG certificate in Major course with 44 credits with additional 4 credits of Summer Internship in core specific NSQF defined course OR continue with Major and Minor course for t h e next NCrF credit level | | | | | | | | | | |
| **NCrF Credit Level** | **Semester** | **Major Core** | **Minor** | **Multi/Inter Disciplinary** | **Ability Enhancements Courses(AEC)** | **Skill Enhancement Courses (SEC)** | **Value Added Courses(VAC)/IKS** | **Research Project/Dissertation** | **Total** | **Qualification/Certificate** |
| **5.0 Second Year** | **III** | **12** | **-** | **2** | **2** | **4** | **2** | **-** | **22** | **UG Diploma** |
| | **IV** | **12** | **4** | **2** | **-** | **2** | **2** | **-** | **22** | |
| **2nd Year Credit Total** | | **24** | **4** | **6** | **2** | **6** | **4** | **-** | **44** | |
| **Grand Total 1st and 2nd Year Credit** | | **40** | **12** | **12** | **6** | **10** | **8** | **-** | **88** | |
| **Exit 2**: Award of UG Diploma in Major course with 88 credits with additional 4 credits of Summer Internship in core specific NSQF defined course OR continue with Major and Minor course for t h e next NCrF credit level | | | | | | | | | | |
| **NCrF Credit Level** | **Semester** | **Major Core** | **Minor** | **Multi/Inter Disciplinary** | **Ability Enhancements Courses(AEC)** | **Skill Enhancement Courses (SEC)** | **Value Added Courses(VAC)/IKS** | **Research Project/Dissertation** | **Total** | **Qualification/Certificate** |
| **5.5 Third Year** | **V** | **12** | **8** | **-** | **-** | **2** | **-** | **-** | **22** | **UG Degree** |
| | **VI** | **12** | **4** | **-** | **2** | **4** | **-** | **-** | **22** | |
| **3rd Year Credit Total** | | **24** | **12** | **-** | **2** | **6** | **-** | **-** | **44** | |
| **Grand Total 1st to 3rd Year Credit** | | **64** | **24** | **12** | **08** | **16** | **8** | **-** | **132** | |
| **Exit 3**: Award of UG Degree in Major course with 132 credits. | | | | | | | | | | |
| **NCrF Credit Level** | **Semester** | **Major Core** | **Minor** | **Multi/Inter Disciplinary** | **Ability Enhancements Courses(AEC)** | **Skill Enhancement Courses (SEC)** | **Value Added Courses(VAC)/IKS** | **Research Project/Dissertation** | **Total** | **Qualification/Certificate** |
| **6 Fourth Year** | **VII** | **12** | **4** | **-** | **-** | **-** | **-** | **6( OJT)** | **22** | **UG Honors Degree** |
| | **VIII** | **12** | **4** | **-** | **-** | **-** | **-** | **6(OJT)** | **22** | |
| **4th Year Credit Total** | | **24** | **8** | **-** | **-** | **-** | **-** | **12** | **44** | |
| **Award of UG Honors Degree in Major course with 176 credits.** | | | | | | | | | | |
| **6** | **VII** | **12** | **4** | **-** | **-** | **-** | **-** | **6( RP)** | **22** | **UG** |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Fourth Year** | **VIII** | **12** | **4** | **-** | **-** | **-** | **-** | **6(RP)** | **22** | **Honors With Research Degree** |
| **4$^{th}$ Year Credit Total** | | **24** | **8** | **-** | **-** | **-** | **-** | **12** | **44** | |
| **Grand Total 1$^{st}$ to 4$^{th}$ Year Credit** | | **88** | **32** | **12** | **08** | **16** | **8** | **12** | **176** | |
| **Award of UG Honors with Research Degree in Major course with 176 credits.** | | | | | | | | | | |
| ***OJT – On the Job Training** | | | | | | *** RP – Research Project With Major Core Courses Only** | | | | |

2 credit course on Community Service/NSS/NCC/Sports as a compulsory course to be undertaken by students during the initial two years of the study. This course will not affect the overall SGPA/CGPA of the students.

5. Eligibility & Mode of Admissions

   Eligibility of a candidate and mode of admission to the programme will be according to the regulations for admission committee decided by Government of Gujarat from time to time.

6. Programme structure and Credits

   A student admitted to a program should study the course and earn credits specified in the course structure.

7. Attendance

   7.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

   7.2 Student attendance in a course should be 80%.

8. Course Evaluation

8.1 The performance of every student in each course will be evaluated as follows:

**8.1.1** Continuous and Comprehensive Evaluation using various components such as Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc. by the course faculty member(s). The weightage of these components is 50% of the marks for the course; and

**8.1.2** Semester-End-Evaluation by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 50% of the marks for the course.

8.2 University Examination

**8.2.1** The final examination by the University for 50% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.

**8.2.2** In order to earn the credit in a course a student has to obtain grade other than FF.

**8.2.3** Performance of Continuous and Comprehensive Evaluation(CCE) and Semester-End-Examination(SEE) Examination will be done on the grading system.

9. Grading

The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are defined as follows:

| **Grading Scheme Range of Marks (%)** | 96.0-100 | 86.0-95.9 | 76.0-85.9 | 66.0-75.9 | 56.0-65.9 | 46.0 – 55.9 | 36.0 – 45.9 | Below 36.0 | Absent |
|---|---|---|---|---|---|---|---|---|---|
| **Letter Grade** | **O** (Outstanding) | **A+** (Excellent) | **A** (Very Good) | **B+** (Good) | **B** (Above Average | **C** (Average) | **P** (Pass) | **F** (Fail) | **Ab** (Absent) |
| **Grade Point** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **0** | **0** |

SGPA = ΣCiGi/ΣCi where Ci is the number of credits of course i Gi is the Grade Point for the course i and i = 1 to n, n = number of courses in the semester.

CGPA =Σ CiGi/ΣCi where Ci is the number of credits of course i Gi is the Grade Point for the course and i = 1 to n, n = number of courses of all semesters up to which CGPA is computed

The minimum passing marks for each pattern of evaluation are 36% .

10. Detention Rule

A student will be promoted to next year only if he/she has cleared all the courses of the year he/she is studying in. Awards of Degree.

10.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

**10.1.1** He should have earned at least minimum required credits as prescribed in course structure; and

**10.1.2** He should have cleared all evaluation components in every course;

11. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

| | |
|---|---|
| **Distinction:** | **CGPA ≥7.0 & ≤10.0** |
| **First class:** | **CGPA ≥ 6.0 & <7.0** |
| **Second Class:** | **CGPA ≥ 5.0 & <6.0** |
| **Pass Class:** | **CGPA >= 3.60 & <5.0** |

12. Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

## Choice of Courses

Undergraduate students will make course selections from the various lists and that will fall into categories as outlined below:

Major Course:

This is the primary subject area a student wishes to delve into during their undergraduate course. It is the area of study that the student is most passionate about and may wish to pursue as a career.

Minor Course:

This is a secondary area of study that complements the student's chosen Major. Students who complete a sufficient number of courses in a discipline or an interdisciplinary area of study other than their chosen Major will qualify for a Minor in that discipline or interdisciplinary area

Multidisciplinary Course:

All undergraduate students are required to complete three introductory-level courses relating to any of the broad disciplines. These courses aim to broaden the student's intellectual experience and form part of liberal arts and science education.

Ability Enhancement Course (AEC):

These are Modern Indian Language (MIL) & English language courses focused on enhancing language and communication skills. The courses aim to enable students to acquire and demonstrate core linguistic skills, including critical reading and expository and academic writing skills. The courses also emphasize the development and enhancement of skills such as communication, and the ability to participate in/conduct discussion and debate.

Skill Enhancement Courses (SEC):

These courses aim to impart practical skills, hands-on training, soft skills, etc., to enhance the employability of students. The institution may design courses as per the students' needs and available institutional resources.

Value Added Course (VAC):

These courses are designed to provide extra skills or knowledge beyond the standard curriculum, often tailored towards enhancing employability, promoting entrepreneurship, or developing personal and professional skills.

The choice based credit system provides flexibility in designing curriculum and assigning credits based on the course content and hour of teaching. The choice based credit system provides an opportunity for the students to choose courses from the prescribed courses comprising core, elective and open elective courses. The CBCS provides a cafeteria type approach in which the students can take courses of their choice and adopt an interdisciplinary approach to learning. The courses shall be evaluated on the grading system, which is considered to be better than the conventional marks system.

## Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

**A. University Core Courses(UC):**

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme.

**B. Programme Core Courses(PC):**

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

## Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialized or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

**A. University Elective Courses(UE):**

The pool of elective courses offered across all faculties / programmes.

**B. Institute Elective Course (IE)**

Institute elective courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialization.

**C. Programme Elective Courses(PE):**

The programme specific pool of elective courses offered by respective programme.

## Credit Hour Allocation for Different Course Types

### Definition and Standardization

The workload associated with a course will be measured in terms of credit hours. One credit hour is equivalent to one hour of instructional time per week over the duration of a semester (minimum 15 weeks).

### Types of Course Components

Courses can be comprised of one or more of the following components:

1. Lecture

2. Practicum

3. Seminar/Internship/ Studio Activities/Field Practice/ Projects / Engagement and Service.

### Credit Hours per Course Component

The weekly and semester-wise instructional hours associated with each credit for different components are as follows:

**Credit Hours per Course *Component***

| Course Component | Weekly Hours per Credit | Total Hours in a 15-week Semester |
|---|---|---|
| Lecture | 1 Hour | 15 Hours |
| Practicum | 2 Hours | 30 Hours |
| Internship, Field Practice/Projects, Community Engagement and Service/NCC/NSS/Sports | 2 Hours | 30 Hours |

## Course Categories

Courses in the study program are categorised based on the nature of the learning activities. These include:

***Lecture courses:*** These involve expert-led sessions related to a specific field.

***Practicum or Laboratory work:*** These apply previously learned principles/theories in practical projects or lab activities.

***Internship:*** These involve professional activities or work experiences, typically supervised by experts from the relevant external entities.

***Field practice/projects:*** These involve field-based learning or projects supervised by external experts.

***Community engagement and service/NCC/NSS/Sports:*** These involve activities exposing students to societal socio-economic issues, facilitating the application of theoretical and practical knowledge to real-life problems. It also involves skills and physical fitness among students through indoor & outdoor sports, field & track events.

# Vision, Mission, PEOs, POs and PSOs

## Vision

To become a leading institution in the field of computer applications and contribute in national efforts of computerizing public systems

## Mission

To produce competent computer professionals with the ability to face future challenges.

## PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

The graduates will:

**PEO1**: Be able to understand the requirement of computing problem and implement an effective solution.

**PEO2**: Be able to successfully take up various available career options.

**PEO3**: Be able to continuously learn in their preferred domains.

**PEO4**: Be able to acquit themselves in ethical and professional manner while demonstrating effective skills.

## PROGRAM OUTCOMES (POs)

The Graduate of Computer Science and Applications will be able to:

**PO1**: **Computational Knowledge**: Apply knowledge of computing fundamentals, computing specialization, mathematics, and domain knowledge to provide effective solution in the area of computing.

**PO2**: **Problem Analysis**: Identify, formulate, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.

**PO3**: **Design /Development of Solutions:** Design and evaluate solutions for computing problems, and design and evaluate systems, components, or processes that meet specified needs by considering several societal aspects.

**PO4**: **Modern Tool Usage:** Create, select, adapt and apply appropriate techniques, resources, and modern computing tools for any computing activities, with an understanding of the limitations.

**PO5**: **Professional Ethics:** Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.

**PO6**: **Life-long Learning**: Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.

**PO7**: **Project management:** Demonstrate knowledge and understanding of the computing and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

**PO8**: **Communication Efficacy:** Communicate effectively with the computing community, and with society at large, about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.

**PO9**: **Societal and Environmental Concern:** Understand and assess societal and environmental issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.

**P10**: **Individual and Team Work:** Function effectively as an individual and as a member or a leader in diverse teams and in multidisciplinary environments**.**

**P11: Innovation and Entrepreneurship:** Identify a timely opportunity and using innovation to pursue that opportunity to create value and wealth for the betterment of the individual and society at large.

## PROGRAM SPECIFIC OUTCOMES

At the end of the programme, the student should be able to

**PSO1:** Analyze, evaluate and formulate workable computing solutions to real world problems using computing methods.

**PSO2:** Apply proficiencies of communication, teamwork and management skills with spirit of entrepreneurship.

## SALIENT FEATURES

- Four Years unique programme as per National Education Policy-2020.
- Have a flexible duration of either three or four years, with multiple exit options during this period.
- Provides students with the flexibility to enter, exit, and re-enter the programme at various stages, each with its corresponding certification.
- Award of Degree:
    - **UG Certificate** (Candidate wish to Exit after the First Year with additional requirements of 4 credits summer internship)
    - **UG Diploma (**Candidate wish to Exit after the Second Year additional requirements of 4 credits summer internship)
    - **UG Degree (**Candidate wish to Exit after the Third Year)
    - **UG Honors Degree or UG Honors with Research Degree (**Candidate finishes all Four Years)
- Programme is designed with various categories of courses such as Major, Minor, Multidisciplinary, Ability Enhancement, Skill Enhancement, Value Added, Community engagement and service/NCC/NSS/Sports based on Choice Based Credit System(CBCS).
- Focus on student-centered learning.
- Problem-based learning and industry-relevant curriculum.
- Industry visits, seminars, workshops, and guest lectures by eminent researchers and industry practitioners.
- Active student clubs and bodies.
- The multi-cultural and vibrant teaching environment.

**Total Credits of Programme (8 Semesters): 176**

# TEACHING SCHEME AND EXAMINATION SCHEME

# FOR

# B.Sc. (IT) PROGRAMME

# EFFECTIVE FROM ACADEMIC YEAR 2025-26

## Credit Framework for 3 Years/4 Years UG Programme

| NCrF Credit Levels | Qualification Title | Credit Requirement | No. of Semesters | Year |
|---|---|---|---|---|
| 4.5 | UG Certificate | 48 | 2 | 1 |
| 5 | UG Diploma | 92 | 4 | 2 |
| 5.5 | Three Year Bachelor Degree | 132 | 6 | 3 |
| 6 | Bachelor's Degree with Honors **OR** Bachelor's Degree with Honors with Research | 176 | 8 | 4 |
| • 1 credit = 1 Hour of Theory<br>• 1 credit = 2 Hour of Practical/Project | | | | |

## Degree programs offered by Faculty

Bachelor of Science in Information Technology (Honors) / Bachelor of Science in Information Technology (Honors with Research) (4-Year Programme) and maximum duration of the programme is 7 Years.

## CREDIT FRAMEWORK For B.Sc. (IT)

## (Bachelor of Computer Science in Information Technology)

Arrangement of Credit Distribution Framework for three/four years Honors/Honors with Research Degree Programme with Multiple Entry and Exit Options

(As per GR No: KCG/admin/2023-24/0607/kh.1, Sachivalaya, Gandhinagar, Date-11/07/2023)

| **Sr. No.** | **Categories of Courses** | **Credit Requirement for each Category** | | | | |
|---|---|---|---|---|---|---|
| | | 1 Year (Certificate) | 2 Years (Diploma) | 3-Year (UG) | 4-Year UG (Honors) | 4-Year UG (Honors + Research) |
| 1 | Major - Core Courses | 16 | 40 | 64 | 88 | 88 |
| 2 | Minor-Discipline Specific Electives | 8 | 12 | 24 | 32 | 32 |
| 3 | Multidisciplinary Courses Open Electives | 8 | 12 | 12 | 12 | 12 |
| 4 | Ability Enhancement Courses (AEC) | 4 | 8 | 10 | 10 | 10 |
| 5 | Skill Enhancement Courses (SEC) | 4 | 8 | 14 | 14 | 14 |
| 6 | Value Added Courses (VAC) | 4 | 8 | 8 | 8 | 8 |
| 7 | Summer Internship/ Research Project /Dissertation | - | - | - | 12 | 12 |
| 8 | Exit Courses | 4 | 4 | - | - | |
| | **Total** | **48** | **92** | **132** | **176** | **176** |

## Structure of B.Sc. (IT) (Bachelor of Computer Science in Information Technology)

| NCrF Credit Level | Semester | Major Core | Minor | Multi / Inter Disciplinary | Ability Enhancements Courses (AEC) | Skill Enhancement Courses (SEC) | Value Added Courses (VAC) / IKS | Research Project / Dissertation | Total | Qualification / Certificate |
|---|---|---|---|---|---|---|---|---|---|---|
| 4.5 First Year | I | 8 | 4 | 4 | 2 | 2 | 2 | - | 22 | **UG Certificate** |
| | II | 8 | 4 | 4 | 2 | 2 | 2 | - | 22 | |
| **1st Year Credit Total** | | 16 | 8 | 8 | 4 | 4 | 4 | - | 44 | |
| **Exit 1**: Award of UG certificate in Major course with 44 credits with additional 4 credits of Summer Internship in core specific NSQF defined course OR continue with Major and Minor course for t h e next NCrF credit level | | | | | | | | | | |
| **NCrF Credit Level** | **Semester** | **Major Core** | **Minor** | **Multi / Inter Disciplinary** | **Ability Enhancements Courses (AEC)** | **Skill Enhancement Courses (SEC)** | **Value Added Courses (VAC)/IKS** | **Research Project / Dissertation** | **Total** | **Qualification/ Certificate** |
| 5.0 Second Year | III | 12 | - | 2 | 2 | 4 | 2 | - | 22 | |
| | IV | 12 | 4 | 2 | 2 | - | 2 | - | 22 | |
| **2nd Year Credit Total** | | 24 | 4 | 4 | 4 | 4 | 4 | - | 44 | |
| Grand Total 1st and 2nd Year Credit | | 40 | 12 | 12 | 8 | 8 | 8 | - | 88 | **UG Diploma** |
| **Exit 2**: Award of UG Diploma in Major course with 88 credits with additional 4 credits of Summer Internship in core specific NSQF defined course OR continue with Major and Minor course for t h e next NCrF credit level | | | | | | | | | | |
| **NCrF Credit Level** | **Semester** | **Major Core** | **Minor** | **Multi/Inter Disciplinary** | **Ability Enhancements Courses (AEC)** | **Skill Enhancement Courses (SEC)** | **Value Added Courses (VAC)/IKS** | **Research Project / Dissertation** | **Total** | **Qualification/ Certificate** |
| 5.5 Third Year | V | 12 | 8 | - | - | 2 | - | - | 22 | **UG Degree** |
| | VI | 12 | 4 | - | 2 | 4 | - | - | 22 | |
| **3rd Year Credit Total** | | 24 | 12 | - | 2 | 6 | - | - | 44 | |
| Grand Total 1st to 3rd Year Credit | | 64 | 24 | 12 | 10 | 14 | 8 | - | 132 | |
| **Exit 3**: Award of UG Degree in Major course with 132 credits. | | | | | | | | | | |
| **NCrF Credit Level** | **Semester** | **Major Core** | **Minor** | **Multi/Inter Disciplinary** | **Ability Enhancements Courses (AEC)** | **Skill Enhancement Courses (SEC)** | **Value Added Courses (VAC)/IKS** | **Research Project / Dissertation** | **Total** | **Qualification/ Certificate** |
| 6 Fourth Year | VII | 12 | 4 | - | - | - | - | 6 (OJT) | 22 | **UG Honors Degree** |
| | VIII | 12 | 4 | - | - | - | - | 6(OJT) | 22 | |
| **4th Year Credit Total** | | 24 | 8 | - | - | - | - | 12 | 44 | |
| **Award of UG Honors Degree in Major course with 176 credits.** | | | | | | | | | | |
| 6 Fourth Year | VII | 12 | 4 | - | - | - | - | 6 (RP) | 22 | **UG Honors With Research Degree** |
| | VIII | 12 | 4 | - | - | - | - | 6 (RP) | 22 | |
| **4th Year Credit Total** | | 24 | 8 | - | - | - | - | 12 | 44 | |
| Grand Total 1st to 4th Year Credit | | 88 | 32 | 12 | 10 | 14 | 8 | 12 | 176 | |
| **Award of UG Honors with Research Degree in Major course with 176 credits.** | | | | | | | | | | |
| ***OJT – On the Job Training** | | *** RP – Research Project With Major Core Courses Only** | | | | | | | | |

# TEACHING SCHEME
# &
# EXAMINATION SCHEME

# FOR

# B.Sc. (IT) PROGRAMME

# EFFECTIVE FROM
# ACADEMIC YEAR 2025-26

## B.Sc. (IT) Semester-I

| Course Code | Course Name | Theory/ Practical /Others | Marks (Theory) | | Marks (Practical) | | Credit | NEP -2020 Classification |
|---|---|---|---|---|---|---|---|---|
| | | | CCE | SEE | CCE | SEE | | |
| **CTUC101** | Introduction to Programming | Theory and Practical | 50 | 50 | 25 | 25 | 06 (04-Theory) (02-Practical) | Major Core |
| **CTUC102** | Relational Database Administration | Practical | - | - | 25 | 25 | 02 | Major Core |
| <br>**CTUE101**<br>**CTUE102**<br>**CTME107**<br>**CTME108** | (Select any one)<br>1. Fundamentals of Business Organization<br>2. Information Technology in Business<br>3. Privacy and Security in Online Social Media<br>4. Computer Architecture and Organization | Theory | 50 | 50 | - | - | 04 | Minor |
| **CTUD101** | Fundamentals of Digital Electronics | Theory | 50 | 50 | - | - | 04 | MDC |
| **CTUS101** | Mathematics for Computer Science | Theory | 25 | 25 | - | - | 02 | SEC |
| **CLUV101** | Environmental Sciences | Practical | - | - | 25 | 25 | 02 | VAC |
| **HSUA101** | Communicative English | Practical | - | - | 25 | 25 | 02 | AEC |
| <br>**CUUV101**<br>**CUUV102** | (Select any one)<br>Community Engagement and Sustainable Development<br>Physical Education and Sports | Practical | - | - | 25 | 25 | 02 | VAC |

*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)
*SEE- Semester –End- Evaluation

## B.Sc. (IT) Semester-II

| Course Code | Course Name | Theory / Practical / Others | Marks (Theory) | | Marks (Practical) | | Credit | NEP -2020 Classification |
|---|---|---|---|---|---|---|---|---|
| | | | CCE | SEE | CCE | SEE | | |
| **CTUC103** | Introduction to Object Oriented Programming | Theory and Practical | 50 | 50 | 25 | 25 | 06 (04-Theory) (02-Practical) | Major Core |
| **CTUC104** | Fundamentals of Web Designing | Practical | - | - | 25 | 25 | 02 | Major Core |
| **CTUE103**<br>**CTUE104**<br>**CTME103**<br>**CTME104** | (Select any one)<br>1. Business Models for e-Commerce<br>2. Managerial Economics<br>3. E-Business<br>4. Business Development: From Start to Scale | Theory | 50 | 50 | - | - | 04 | Minor |
| **CTUD102** | Data Communication and Networks | Theory | 50 | 50 | - | - | 04 | MDC |
| **CTUV101** | Fundamentals of Accounting | Theory | 25 | 25 | - | - | 02 | VAC |
| **HSUS101-117** | (Select any one)<br>A course on Liberal Arts<br>HSUS101 – Painting<br>HSUS102– Photography<br>HSUS103– Sculpting<br>HSUS104– Pottery and Ceramic Art<br>HSUS105-Media and Graphic Design<br>HSUS106-Art and Craft<br>HSUS109- Dramatics<br>HSUS110-Contemporary Dance<br>HSUS111-Music (Vocal)<br>HSUS112-Music (Instrumental)-Tabla<br>HSUS113-Music (Instrumental)-Guitar<br>HSUS114-Music (Instrumental)-Harmonium<br>HSUS115-Music (Instrumental)-Flute<br>HSUS116- Indian Classical Dance-Kathak<br>HSUS117- Indian Classical Dance-Bharatnatyam | Practical | - | - | 25 | 25 | 02 | SEC |
| **CTUA101** | Presentation Skills | Practical | - | - | 25 | 25 | 02 | AEC |
| **CTUI101** | Summer Internship and Viva - I | VIVA | - | - | 50 | 50 | 04 | Exit Course |

*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)
*SEE- Semester –End- Evaluation

## B.Sc. (IT) Semester-III

| Course Code | Course Name | Theory / Practical/ Others | Marks (Theory) | | Marks (Practical | | Credit | NEP -2020 Classification |
|---|---|---|---|---|---|---|---|---|
| | | | CCE | SEE | CCE | SEE | | |
| **CTUC201** | Introduction to Data Structures | Theory and Practical | 50 | 50 | 25 | 25 | 06 (04-Theory) (02-Practical) | Major Core |
| **CTUC202** | System Analysis and Design | Theory | 50 | 50 | - | - | 04 | Major Core |
| **CTUC203** | Programming the Internet | Practical | - | - | 25 | 25 | 02 | Major Core |
| **CTUS201**<br><br>**CTUS202** | (Select any one)<br>1. Computer Oriented Numerical Methods<br>2. Computer Oriented Management System | Theory | 50 | 50 | - | - | 04 | SEC |
| **CTUA201** | Life Management | Theory | 25 | 25 | - | - | 02 | AEC |
| | University Elective-I | Practical | - | - | 25 | 25 | 02 | MDC |
| **HSUV201** | Creativity, Problem Solving and Innovation | Practical | - | - | 25 | 25 | 02 | VAC |

*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)
*SEE- Semester –End- Evaluation

**University Elective - I**

| Sr. No. | Course Code | Course Name | Department/Faculty |
|---|---|---|---|
| 1 | MEUE201 | Engineering Graphics and Design | Engineering |
| 2 | EEUE201 | Fundamentals Of Electrical Engineering | Engineering |
| 3 | CLUE201 | Environment and Development | Engineering |
| 4 | ECUE201 | Scientific Computing Using Matlab | Engineering |
| 5 | CEUE201 | The Joy of Computing using Python | Engineering |
| 6 | ITUE201 | Introduction to Quauntam Computing | Engineering |
| 7 | CSUE201 | Python For Data Science | Engineering |
| 8 | AIUE201 | An Introduction to Artificial Intelligence | Engineering |
| 9 | BMUD201 | Money And Banking | Management |
| 10 | CAUD203 | Human Computer Interactions | Computer Science |
| 11 | PTUD191 | Basics of Health Promotion and Education Intervention | Physiotherapy |
| 12 | NRMD251 | First Aid Masterclass- A complete guide to first aid | Nursing |

## B.Sc. (IT) Semester-IV

| Course Code | Course Name | Theory / Practical / Others | Marks (Theory) | | Marks (Practical) | | Credit | NEP - 2020 Classification |
|---|---|---|---|---|---|---|---|---|
| | | | CCE | SEE | CCE | SEE | | |
| **CTUC204** | Fundamentals of Operating Systems | Theory and Practical | 50 | 50 | 25 | 25 | 06 (04-Theory) (02-Practical) | Major Core |
| **CTUC205** | Fundamentals of Visual Programming | Theory and Practical | 50 | 50 | 25 | 25 | 06 (04-Theory) (02-Practical) | Major Core |
| **CTUE201**<br>**CTUE202**<br>**OCBSIT1007** | (Select any one)<br>1. Introduction to Artificial Intelligence<br>2. Computer Network Security<br>3. The Joy of Computing using Python | Theory | 50 | 50 | - | - | 04 | Minor |
| **CTUA202** | Social Media and Blog Writing | Practical | - | - | 25 | 25 | 02 | AEC |
| | University Elective-II | Practical | - | - | 25 | 25 | 02 | MDC |
| **HSUV203** | Indian Knowledge System | Practical | - | - | 25 | 25 | 02 | VAC |
| **CTUI201** | Summer Internship and Viva - II | VIVA | - | - | 50 | 50 | 04 | Exit Course |

*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)
*SEE- Semester –End- Evaluation

**University Elective - II**

| Sr. No. | Course Code | Course Name | Department/Faculty |
|---|---|---|---|
| 1 | MEUE202 | Nature and Properties of Materials | Engineering |
| 2 | EEUE202 | Solar Energy Engineering and Technology | Engineering |
| 3 | CLUE202 | Ecology and Environment | Engineering |
| 4 | ECUE202 | Introduction to Internet of Things | Engineering |
| 5 | CEUE202 | Software Conceptual Design | Engineering |
| 6 | ITUE202 | Ethical Hacking | Engineering |
| 7 | CSUE202 | Google Cloud Computing Foundations | Engineering |
| 8 | AIUE202 | Social Network Analysis | Engineering |
| 9 | BMUD251 | Economics Of Health And Health Care | Management |
| 10 | CAUD204 | Modern Application Development | Computer Science |
| 11 | PTUD192 | Ergonomics Workplace Analysis | Physiotherapy |
| 12 | NRMD261 | Mindfulness and Well-being: Living with Balance and Ease | Nursing |

## B.Sc. (IT) Semester-V

| Course Code | Course Name | Theory / Practical/ Others | Marks (Theory) | | Marks (Practical) | | Credit | NEP - 2020 Classification |
|---|---|---|---|---|---|---|---|---|
| | | | CCE | SEE | CCE | SEE | | |
| **CTUC301** | Object Oriented Programming Using JAVA | Theory and Practical | 50 | 50 | 25 | 25 | 06 (04-Theory) (02-Practical) | Major Core |
| **CTUC302** | Open Source Technology | Theory and Practical | 50 | 50 | 25 | 25 | 06 (04-Theory) (02-Practical | Major Core |
| **CTUE301** **CTUE302** | (Select any one) 1. Data Science Essentials 2. Basics of Digital Image Processing | Theory | 25 | 25 | - | - | 02 | Minor |
| **CTUE303** | Object Oriented Analysis and Design | Theory | 50 | 50 | - | - | 04 | Minor |
| **CTUE304** **CTUE305** | 1. Advanced Internet of Things 2. Introduction to Game Development | Practical | - | - | 25 | 25 | 02 | Minor |
| **HSUS301** | French | Practical | - | - | 25 | 25 | 02 | SEC |

*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)
*SEE- Semester –End- Evaluation

## B.Sc. (IT) Semester-VI

| Course Code | Course Name | Theory / Practical / Others | Marks (Theory) | | Marks (Practical) | | Credit | NEP - 2020 Classification |
|---|---|---|---|---|---|---|---|---|
| | | | CCE | SEE | CCE | SEE | | |
| **CTUC303** | Project Work | VIVA | 150 | 150 | - | - | 12 | Major Core |
| **CTUE306** **CTUE307** | (Select any one) 1. Multi-paradigm Programming Language 2. Frameworks and Applications | Theory and Practical | 25 | 25 | 25 | 25 | 04 (02-Theory) (02-Practical | Minor |
| **CTUS301** | Mobile Application Development | Theory and Practical | 25 | 25 | 25 | 25 | 04 (02-Theory) (02-Practical | SEC |
| **HSUA302** | Professional Communication, Soft Skills and Personality Development | Practical | - | - | 25 | 25 | 02 | AEC |

*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)
*SEE- Semester –End- Evaluation

## B.Sc. (IT) Semester-VII

| Course Code | Course Name | Theory / Practical/ Others | Marks (Theory) | | Marks (Practical) | | Credit | NEP -2020 Classification |
|---|---|---|---|---|---|---|---|---|
| | | | CCE | SEE | CCE | SEE | | |
| **CTUC401** | Enterprise Computing using Java EE | Theory and Practical | 50 | 50 | 25 | 25 | 06 (04-Theory) (02-Practical | Major Core |
| **CTUC402** | Database Technologies | Theory and Practical | 50 | 50 | 25 | 25 | 06 (04-Theory) (02-Practical | Major Core |
| **CTUE401**<br>**CTUE402**<br>**CTUE403** | (Select any one)<br>1. Cloud Computing<br>2. Cryptography and Network Security<br>3. Blockchain Essentials | Theory | 50 | 50 | - | - | 04 | Minor |
| **CTUP401**<br>**CTUR401** | (Select anyone)<br>1. On-Job Training-I<br>2. Research Project-I | VIVA | - | - | 75 | 75 | 06 | Research Project / Dissertation |

*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)
*SEE- Semester –End- Evaluation

## B.Sc. (IT) Semester-VIII

| Course Code | Course Name | Theory / Practical / Others | Marks (Theory) | | Marks (Practical) | | Credit | NEP -2020 Classification |
|---|---|---|---|---|---|---|---|---|
| | | | CCE | SEE | CCE | SEE | | |
| **CTUC403** | Full Stack Web Development | Theory and Practical | 50 | 50 | 25 | 25 | 06 (04-Theory) (02-Practical | Major Core |
| **CTUC404** | Advanced Mobile Programming | Theory and Practical | 50 | 50 | 25 | 25 | 06 (04-Theory) (02-Practical | Major Core |
| **CTUE404** | Web Development using Open Source Technologies | Practical | - | - | 25 | 25 | 2 | Minor |
| **CTUE405**<br>**CTUE406**<br>**CTUE407** | (Select any one)<br>1. Python Web framework<br>2. HTTP Web Service for Enterprise Applications<br>3. Game Development using Unity | Practical | - | - | 25 | 25 | 2 | Minor |
| **CTUP402**<br>**CTUR402** | (Select any one)<br>On Job Training-II<br>Research Project-II | VIVA | - | - | 75 | 75 | 6 | Research Project / Dissertation |

*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)
*SEE- Semester –End- Evaluation

# TEACHING SCHEME & DETAILED SYLLABUS

# FOR

# B.Sc. (IT) PROGRAMME
# (1st SEMESTER)
# AS PER NEP 2020

# EFFECTIVE FROM ACADEMIC YEAR 2025-26

## B.Sc. (IT) Semester-I

| Course Code | Course Name | Theory/ Practical/ Others | Marks (Theory) | | Marks (Practical) | | Credit | NEP -2020 Classification |
|---|---|---|---|---|---|---|---|---|
| | | | CCE | SEE | CCE | SEE | | |
| **CTUC101** | Introduction to Programming | Theory and Practical | 50 | 50 | 25 | 25 | 06 (04-Theory) (02-Practical) | Major Core |
| **CTUC102** | Relational Database Administration | Practical | - | - | 25 | 25 | 02 | Major Core |
| **CTUE101**<br>**CTUE102**<br>**CTME107**<br>**CTME108** | (Select any one)<br>1. Fundamentals of Business Organization<br>2. Information Technology in Business<br>3. Privacy and Security in Online Social Media<br>4. Computer Architecture and Organization | Theory | 50 | 50 | - | - | 04 | Minor |
| **CTUD101** | Fundamentals of Digital Electronics | Theory | 50 | 50 | - | - | 04 | MDC |
| **CTUS101** | Mathematics for Computer Science | Theory | 25 | 25 | - | - | 02 | SEC |
| **CLUV101** | Environmental Sciences | Practical | - | - | 25 | 25 | 02 | VAC |
| **HSUA101** | Communicative English | Practical | - | - | 25 | 25 | 02 | AEC |
| **CUUV101**<br>**CUUV102** | (Select any one)<br>Community Engagement and Sustainable Development<br>Physical Education and Sports | Practical | - | - | 25 | 25 | 02 | VAC |

*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)
*SEE- Semester –End- Evaluation

# CTUC101 - Introduction to Programming

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components (Theory) :** Unit Tests, Assignments, Case Study etc.<br>**Experiential Learning Components (Practical) :** Lab Assignments, Practical Test, VIVA | | | | | |

**Pre-requisite: None**

**Methodology, Pedagogy & Andragogy:** During theory lectures, facilitate problem-solving skills by employing algorithms and flowcharts, enabling the understanding of the C programming language's structural aspects, and fostering the capability to create basic programs incorporating data types, variables, and constants. Expand comprehension of diverse conditional and iterative statements and expressions, while also delving into the manipulation of arrays encompassing both numerical and character elements. Develop proficiency in comprehending user-defined functions and explore advanced programming concepts such as utilizing arrays with pointers and structures promoting independent learning and application in real-world scenarios.

**Outline of the course (Theory):**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Programming Basics | 08 |
| 2 | Introduction to C | 12 |
| 3 | Expressions and Control Statements | 10 |
| 4 | Arrays and Strings | 10 |
| 5 | Introduction to Functions | 10 |
| 6 | Pointers & Basic of the structures | 10 |

**Total Hours (Theory): 60**
**Total hours (Practical) : 60**
**Total Contact Hours : 120**

**Detail Syllabus:**

**Unit I: Programming Basics** **Hours 08**

Types of Programming languages and its History, features and application. Introduction to algorithm, Key features of algorithm, Introduction to flow charts, Significance of flow chart, Advantages and limitation of flow chart, Introduction to programming languages, Introduction to editor, compiler, and translator

**Unit II: Introduction to C** **Hours 12**

Introduction, characteristics of C, Structure of C Program, writing first C program, Files used in C program, Compiling and Executing C program. Basic data types in C, User define data types, Variables in C, Declaring and initializing variables in C, Constants, Input / Output statements in C, Operators - arithmetic, relational, logical, assignment, increment - decrement, conditional, Bitwise, comma operator, size-of operator, operator precedence chart.

**Unit III: Expressions and Control Statements** **Hours 10**

Arithmetic expressions, evaluation of expressions, type conversions in expressions, operator precedence and associativity, mathematical functions. Type conversion and casting; Introduction to decision control statements, Conditional and branching statements, loop, nested loop, Break, continue and go to statement.

**Unit IV: Arrays and Strings** **Hours 10**

Arrays: One-dimensional, two-dimensional, Handling of Character Strings: Declaring and initializing string variables, reading string from terminal, writing string to screen, string handling functions, table of strings.

**Unit V: Introduction to Functions** **Hours 10**

Need for user-defined functions, the form of c function, return values and their types, calling a function, category of functions, handling of non-integer functions, nesting of Functions, recursion, functions with arrays.

**Unit VI Pointers & Basic of the structures Hours 10**

Understanding the Computer's Memory, Introduction to Pointers, Declaring Pointer Variables, Pointer Expressions and Pointer Arithmetic, Null Pointers, Generic Pointers, passing arguments to Function using Pointers, Pointers and Arrays, Passing an Array to a Function, Difference between Array name and Pointer, Pointers and Strings, Array of Pointers. Defining a structure, declaring structure variables, accessing structure members, structure initialization.

**Outline of the Course (Practical) :**

| Week No | Practical | Description |
|---|---|---|
| 1 | Introduction to Programming and IDE | Overview of programming languages, algorithms, flowchart, Setting up the development environment,<br><br>Introduction of development environment. |
| 2-3 | Basic of C programming | Structure of C Program, writing first C program, Files used in C program, Compiling and Executing C program; Basic data types in C. Variables in C, Declaring and initializing variables in C, Constants, Input / Output statements in C |
| 4-5 | Operators, Expressions evaluation & type conversions. | Operators - arithmetic, relational, logical, assignment, increment - decrement, conditional, Bitwise and special, comma operator, size-of operator, operator precedence chart. Arithmetic expressions, evaluation of expressions, type conversions. |
| 6-7 | Decision-making statements and looping statements | Decision-making statements (if, else if, else) and nested decision-making statements, Switch statement, Looping statements (for, while, do-while), nested looping statements Break and continue statements. |
| 8-9 | Arrays | Understanding arrays, one-dimensional array, two-dimensional arrays, practical exercises of one-dimensional and two-dimensional array. |
| 10-11 | String and String functions | Declaring and initializing string variables, reading string from terminal, writing string to screen, string-handling functions. |

| | | |
|---|---|---|
| **12-13** | User defined functions | Introduction to functions, Function declaration, definition, and calling, Parameters and return values. Scope and lifetime of variables, Recursion. |
| **14-15** | Pointers & Structures | Introduction to pointers, Pointers and arrays, Pointer arithmetic, Defining and declaring structures, Accessing structure members |

**Total hours: 60**

**Core Books:**

1. ReemaThareja: Computer Fundamentals and Programming in C, 2nd Edition, Oxford University press, 2017
2. PradipDey and Manas Ghosh. "Programming in C", Second Edition, Oxford University Press.
3. E. Balagurusamy: Programming In C, 7th Edition.
4. Asok N Kamthane, Pearson,Programming in C
5. Anita Goel, Pearson, Computer Fundamentals
6. Schaum Series, Gottfried B.S.,Tata McGraw Hill,Programming with C

**Reference Books:**

1. Brian W. Kerighan and Dennis M. Ritchie : The C Programming Language,Prentice Hall.
2. PradipDey, Manas Ghosh: Programming in C, 4th Edition, Oxford University press.
3. Herbert Schildt: The complete reference C, Fourth edition, Tata McGraw Hill
4. YashwantKanetkar: Let us C, 13th Edition, BPB publication.
5. Rajaraman V, PHI, Computer Basics and Programming in C

**Web References:**

1. https://cprogrampracticals.blogspot.com/p/basic-concepts.html [For basic c programing
2. practical]
3. https://www.programtopia.net/c-programming/docs/operators-expressions [For operators and expressions in C]
4. https://www.javatpoint.com/functions-in-c [For functions]
5. https://www.programiz.com/c-programming [For basic c programing practical]
6. https://www.javatpoint.com/c-pointers [ pointers in c]
7. https://www.programiz.com/c-programming/c-structures [ structures in c]

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Be able to solve problems using algorithms and flowcharts. |
| CO2 : | Learn about the structure of C program and able to develop simple programs using the C Programming language with data types, variables and constants. |
| CO3 : | Gain the knowledge of different conditional and iterative statements and expressions. |
| CO4 : | Explore the utilization and manipulation of numbers and characters arrays both. |
| CO5 : | Be able to understand user define function. Also, learn Programing-using Array with Pointer and Structure. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Programming Basics | √ | | | | |
| 2 | Introduction to C | | √ | | | |
| 3 | Expressions and Control Statements | | | √ | | |
| 4 | Arrays and Strings | | √ | √ | √ | |
| 5 | Introduction to Functions | | √ | √ | | √ |
| 6 | Pointers & Basic of the structures | | √ | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 2 | 3 | 2 | 2 | 1 | 2 | 2 | 2 | 1 | 1 | 3 | 3 |
| CO2 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| CO3 | 2 | 2 | 3 | 3 | 2 | 2 | 2 | 2 | 2 | 1 | 1 | 3 | 2 |
| CO4 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 2 | 3 | 3 | 2 |
| CO5 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 2 | 3 | 3 | 2 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

# CTUC102 - Relational Database Administration

| Credit | Component | Instruction / Contact Hours (Per Week) | Instruction / Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components:** Lab Assignments, Practical Test, VIVA , certification courses, development of projects etc. | | | | | |

**Prerequisites:** Basic knowledge of working with computer

**Methodology, Pedagogy & Andragogy:** Learn about computer data. Understand what data is, learn its meaning, examine the types of data sources, and see the difference between data and information. The database applications used in the real world will be discussed with necessary examples. During the laboratory hours' students' will implement the concepts. Apply SQL commands to mine data from databases and data visualization techniques to address organization information needs.

### A. Outline of the Course:

| Week No | Practical | Description |
|---|---|---|
| 1 | Introduction to Database System | Overview of data, information, file, database, database systems, DBMS, Purpose of DBMS over file system. |
| 2-3 | Data modelling and architecture | Various data models: ER model and Relational model, Three level architecture, structure of DBMS and database actors and workers, roles of database administrator |
| 4-5 | Database Design Methodology | CODD Rules, Functional Dependency and Normalization for Database Informal Design Guidelines for Relational Schemas, Functional Dependencies, Normal Forms (1NF, 2NF, 3NF, BCNF, 4NF, 5NF) |
| 6-7 | Schema Definition, Constraints, Queries – I | Basic Data types, Create Table Command, Modifying the structure of tables, renaming |

| | | |
|---|---|---|
| | | table, truncating table, destroying table, Insert, Delete and Update Statements in SQL |
| **8** | Schema Definition, Constraints, Queries – II | Data Constraints and Functions: - Pseudo columns, Null values, TAB table, DUAL table Operators, Data constraints, Type of data constraints, modifying constraints, working with data dictionary and use of USER_CONSTRAINTS Functions introduction |
| **9** | Inbuilt functions | Merits and demerits of functions,<br>Types of functions: Numeric functions, Character functions, Date functions, Conversion functions, Aggregate functions |
| **10-11** | Querying the data | Different operators: LOGICAL, range searching and pattern matching, Viewing Data in the tables, sorting |
| **12-13** | Advanced Query Processing-I | Co-related Nested Queries, Group By clause, having clause, Data Control Language Commands: GRANT and REVOKE |
| **14-15** | Advanced Query Processing-II | Joins (Inner Join, Outer Join, Self-Join, Equi Join, Cross Join), Creation and manipulation of database objects indexes, views, sequences and synonym. |

**Total hours: 60**

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**Core Books:**

1. Ivan Bayross : SQL, PL/SQL The programming Language Oracle ($4^{th}$ Revised edition).
2. Ramkrishnan, Gehrke : Database Management Systems, 3rd Edition, McGrawHill Publication.
3. RamezElmasri, Shamkant B. Navathe : Fundamentals of Database Systems , 5th Edition, , Pearson Publication.

**Reference Books:**

1. Silberschatz, Korth, Sudarshan : Database System Concepts, 5th Edition, McGraw Hill.
2. C.J.Date, a Kannan, S Swaminathan : An Introduction to Database Systems, 8th Edition, Pearson Education,(Equivalent Reading).

3. Scoot Urban : Oracle 9i, PL/SQL Programming, Oracle Press.
4. S. K. Singh : Database Systems: Concepts, Design and Applications, Pearson Education
5. Peter Rob, Carlos Coronel: Database Systems: Design, Implementation and Management, 7th Edition, Cengage Learning, 2007.
6. Anjali Jivani and Amisha Shingala, "Practice book on SQL and PL/SQL with solutions", Roopal and Nirav publication.
7. Leon and Leon : Database management Systems, Vikas Publication.


**Web References:**


1. https://www.javatpoint.com/er-model-vs-relational-model [For difference of ER and Relationmodel]
2. http://www.microsoftvirtualacademy.com/trainingcourses/databasefundamentals#?fbid=tbZ92pOp _Tt [For overall subject]
3. http://www.ntu.edu.sg/home/ehchua/programming/sql/Relational_Database_Design.html [for relational database design]
4. http://docs.oracle.com/cd/A97335_02/apps.102/a81358/05_dev1.htm [For ER Diagram]
5. http://plsql-tutorial.com/[For PL/SQL]


**Course Outcome:** Upon successful completion of the course, student will:

| | |
|---|---|
| CO1: | Understand basic concepts regarding data, database systems and various data models. |
| CO2: | Understand various data modeling techniques and entity relationship models. |
| CO3: | Learn about relational data model and related concepts. |
| CO4: | Gain insights into database design, SQL and normalization concepts. |
| CO5: | Advanced query processing |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 3 | 1 | 1 | - | 3 | - | - | - | - | - | 2 | - |
| CO2 | 1 | 3 | 3 | 3 | - | 3 | 1 | - | - | - | - | 1 | 3 |
| CO3 | 3 | 3 | 3 | 3 | - | 3 | 2 | - | - | 2 | - | 2 | |
| CO4 | - | 3 | 3 | 3 | - | 3 | 3 | - | - | 3 | 3 | - | 3 |
| CO5 | 2 | 3 | 3 | 3 | 2 | 3 | 3 | - | - | - | 3 | 2 | 3 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

# CTUE101 - Fundamentals of Business Organization

| Credit | Component | Instruction / Contact Hours (Per Week) | Instruction / Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| **Experiential Learning Components:** Unit Tests, Assignments, Case Study etc. | | | | | |

**Pre-requisite:** None

**Methodology, Pedagogy & Andragogy:** During theory lectures the emphasis will be given on the Fundamentals of Business Organization. Students will be introduced to Business Fundamentals, Business Enterprises, Business Services, Emerging Trends of Business, Managing Financial Resources, and Business in Global Environment. Students will give practical exposure in form of case study and by visit to E-commerce / Digital Marketing company and/or supply chain Centre.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Business Fundamentals | 09 |
| 2 | Business Enterprises | 11 |
| 3 | Business Services | 09 |
| 4 | Emerging Trends of Business | 11 |
| 5 | Managing Financial Resources | 09 |
| 6 | Business in Global Environment | 11 |

**Total Hours (Theory): 60**
**Total Contact Hours: 60**

**Detail Syllabus:**

**Unit I : Foundation of Information Technology Hours 09**

Introduction to Information Technology, Acquisition of various types of data-Introduction to Number and Textual Data, Image Data, Audio and Video Data Numbers and Textual Data, Data Storage and Organization

**Unit II : Business Fundamentals Hours 11**

Business – Concept, nature and scope, business as a system, business objectives, business and environment interface, distinction between business, commerce and trade.

**Unit III : Business Enterprise Hours 09**

Forms of business organization: Sole proprietorship, Joint Hindu Family Firm, partnership firm, joint stock Company, types of cooperative society: Limited Liability Partnership, Choice of form of organization, multinational corporations.

**Unit IV : Emerging Trends of Business Hours 11**

E – business – Meaning, Scope and benefits, Resource required for successful E – Business implementation, on – line transactions, payment mechanism, Security and safety of business transactions, Outsourcing – Concept, need and scope

**Unit V : Managing Financial Resources Hours 09**

Functions of money, financial institutions, role of financial manager, understanding security markets, career in finance.

**Unit VI : Business in Global Environment Hours 11**

Globalization of business, threats and opportunities in global business, global business environment, trade controls, reducing International trade barriers, career in international business

**Core Books:**

1. Koontz:Principles Of Management (Ascent Series) Paperback: Tata McGraw Hill Education, 2004.
2. Y.K. Bhushan: Fundamentals of Business Organization, Sultan Chand Publisher, 2018

**Reference Books:**

1. Dr. Shveta Klara, Dr. Neha Singhal: Business Organization and Management, Scholar Tech Press, 2020.

2. Karen M. Collins, Jacqueline Shemko: Exploring Business, Pearson Education, 2008.
3. Stephen J. Skripak : Fundamentals of Business : 3rd Edition, Virginia Tech Libraries, 2018.

**Web References:**

1. https://vtechworks.lib.vt.edu/bitstream/handle/10919/70961/Chapter%204%20 Business %20in%20a%20Global%20Environment.pdf [Business in Global Environment ]
2. https://www.managementstudyguide.com/financial-management.htm [Managing Financial Resources]
3. https://leverageedu.com/blog/emerging-modes-of-business/ [ Emerging Trends of Business]
4. https://open.umn.edu/opentextbooks/textbooks/319 [ebook for reference]
5. http://thuvienso.bvu.edu.vn/bitstream/TVDHBRVT/15850/1/An-Introduction-toBusiness-and-Business-Planning.pdf [Introduction to Business and Business Planning]

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Able to understand business fundamentals |
| CO2 : | Able to understand different types of business proprietorship. |
| CO3 : | Able to get the knowledge of functionality of various online business services. |
| CO4 : | Able to understand the basics to manage finance for business |
| CO5 : | Learned the concepts and the scope of international business enterprise. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Business Fundamentals | √ | | | | |
| 2 | Business Enterprises | | √ | | | |
| 3 | Business Services | | | √ | | |
| 4 | Emerging Trends of Business | | | √ | | |
| 5 | Managing Financial Resources | | | | √ | |
| 6 | Business in Global Environment | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | - | - | - | - | - | 3 | - | - | - | - | - | - | 1 |
| CO2 | - | - | - | - | - | 3 | - | - | - | - | 2 | - | 2 |
| CO3 | - | 2 | 2 | 2 | - | 3 | 2 | - | - | 2 | 2 | - | 2 |
| CO4 | - | - | - | - | - | 3 | - | - | - | - | - | - | 2 |
| CO5 | - | - | - | - | - | 3 | - | - | 1 | 1 | 2 | - | 2 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUE102 - Information Technology in Business

| Credit | Component | Instruction / Contact Hours (Per Week) | Instruction / Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| **Experiential Learning Components:** Unit Tests, Assignments, Case Study etc. | | | | | |

**Pre-requisite:** None

**Methodology, Pedagogy & Andragogy:** During theory lectures the emphasis will be given on the Information Technology in Business. Students will be introduced to Foundation of Information Technology, Internet Applications, Internet Service Trends in Business, E-Business, Data Analytics, Modern Business Management, Emerging Trends, and Societal Impact of IT. Students will give practical exposure in form of case study and by visit to E-commerce / Digital Marketing company and/or supply chain Centre.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Foundation of Information Technology | 09 |
| 2 | Internet Applications | 11 |
| 3 | Internet Service Trends in Business | 09 |
| 4 | E-Business and Data Analytics | 11 |
| 5 | Modern Business Management | 09 |
| 6 | Emerging Trends and Societal Impact of IT | 11 |

**Total Hours (Theory): 60**
**Total Contact Hours:60**

**Detail Syllabus:**

**Unit I : Foundation of Information Technology** **Hours 09**

Introduction to Information Technology, Acquisition of various types of data-Introduction to Number and Textual Data, Image Data, Audio and Video Data Numbers and Textual Data, Data Storage and Organization

**Unit II : Internet Applications** **Hours 11**

Internet: Introduction, Internet basics, Internet protocols, Internet addressing, Browser WWW, E-mail, telnet, FTP, application, benefits and limitation of internet, electronic conferencing, and teleconferencing.

**Unit III : Internet Service Trends in Business** **Hours 09**

Fundamentals of Cloud Computing, Types of Cloud based Services, Varieties of Cloud Computing, Types of Cloud Computing, Advantages and Disadvantage of Cloud Computing, Cloud Computing Providers, Application of Cloud Computing in Business Area. E-commerce, E-Marketing, E-Branding, E-Advertising.

**Unit IV : E-Business and Data Analytics** **Hours 11**

E-Business: Introduction, Components, Difference between E-Commerce and E-Business ; Types of Data – Qualitative and Quantitative, Data Analytics basics, Types of Analytics, Common technologies used in data analytics, Tools for Data Analytics

**Unit V : Modern Business Management** **Hours 09**

E-SCM: Introduction, Supply Chain Management, E- Supply Chain Management, Component of Modern E-SCM Major Trends in E-SCM, Example of E-SCM, Architecture of E-Supply Chain Models

E-CRM: Customer Relationship Management Concept, E-CRM Solution Advantages of E-CRM Solutions, Advantages of E-CRM, E-CRM Capabilities, Example of E-CRM, E-CRM Framework

**Unit VI : Emerging Trends and Societal Impact of IT** **Hours 11**

Internet of Things, Virtual and Augmented Reality, Industry 4.0, Image Processing; Introduction to Societal Impact, Social Use of World Wide Web, Privacy, Security and Integration of Information, Disaster Recovery, Intellectual Property Right, Career in IT

**Core Books:**


1. V Rajaraman : Introduction to Information Technology, 2ndEdition, PHI Learning Private Limited, 2013.
2. P.T. Joseph, S.J.: E-Commerce An Indian Perspective, 5th Edition, PHI Learning Private Limited, 2015.
3. R. Kalakota & M. Robinson, e-Business 2.0, 2nd Edition, Pearson Publisher, 2012.
4. Thomas Erl, Z. Mahmood and R. Puttini: Cloud Computing Concepts, Technology & Architecture, Pearson Publication, 2014


**Reference Books:**


1. Syed Muhammad Fahad Akhtar, Big Data Architect's Handbook, Packt Publishing ltd, 2018.
2. Kamlesh K. Bajaj, Debjani Nag: E-Commerce, The Cutting Edge of Business, 2nd Edition, McGraw-Hill Education, 2014.


**Web References:**


1. https://blog.k2datascience.com/the-basics-of-data-analytics-77e5cc7ea741 [For Data Analytics]
2. https://www.investopedia.com/terms/c/cloud-computing.asp [For Cloud Computing]
3. https://www.javatpoint.com/cloud-computing-applications [For Cloud Computing]
4. https://www.geeksforgeeks.org/difference-between-e-commerce-and-e-business/ [ For E-business vs E-Commerce]
5. https://www.temok.com/blog/what-is-e-business/ [For E- business]
6. https://files.eric.ed.gov/fulltext/ED536788.pdf [For Data Analytics]
7. https://vdocument.in/introduction-to-e-scm.html [For E-SCM]
8. http://nek.istanbul.edu.tr:4444/ekos/TEZ/42370.pdf [For E-CRM]
9. https://wps.prenhall.com/wps/media/objects/10704/10961611/Online_Appendix_B.pdf [For E-CRM]


**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Understand the basics of Information Technology. |
| CO2 : | Aware about Internet based Applications for any business context. |
| CO3 : | Understand trends of Internet services in Business. |
| CO4 : | Understands basics of Data Analytics and E-business concepts. |
| CO5 : | Understand modern business management concepts, emerging trends and societal Impact of Information Technology. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Foundation of Information Technology | √ | | | | |
| 2 | Internet Applications | | √ | √ | | |
| 3 | Internet Service Trends in Business | | | √ | √ | |
| 4 | E-Business and Data Analytics | | | | √ | |
| 5 | Modern Business Management | | | | | √ |
| 6 | Emerging Trends and Societal Impact of IT | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 1 | 1 | - | - | 2 | - | - | - | - | - | 2 | 2 |
| CO2 | 3 | 1 | 2 | 1 | - | 3 | - | - | - | - | 2 | 2 | 2 |
| CO3 | 3 | 1 | 2 | 1 | - | 3 | - | - | - | - | 2 | 2 | 2 |
| CO4 | 3 | 1 | 2 | 1 | - | 3 | - | - | - | - | 2 | 2 | 2 |
| CO5 | 3 | 1 | 2 | 1 | 2 | 2 | - | - | 2 | - | - | 2 | 2 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

## CTME107 - Privacy and Security in Online Social Media

**Description:**

**Credit and Week:**

| Teaching Scheme | Week | Marks | Credit |
|---|---|---|---|
| | 12 | 100 | 3 |

**About the course:**

With increase in the usage of the Internet, there has been an exponential increase in the use of online social media and networks on the Internet. Websites like Facebook, YouTube, LinkedIn, Twitter, Flickr, Instagram, Google+, FourSquare, Pinterest, Tinder, and the likes have changed the way the Internet is being used. However, widely used, there is a lack of understanding of privacy and security issues on online social media. Privacy and security of online social media need to be investigated, studied and characterized from various perspectives (computational, cultural, psychological, etc.). Student completing the course will be able to appreciate various privacy and security concerns (spam, phishing, fraud nodes, identity theft) on Online Social Media and Student will be able to clearly articulate one or two concerns comprehensively on one Online Social Media, this will be achieved by homework.

**Pre-requisites**:

Basic / Intermediate programming course. Understanding of Python will be necessary for the course. Should be able to quickly learn APIs, and to collect data from social networks.

**Industry support:**

Any company which is interested in social media / networks data will be interested in recruiting the students finishing the course.

**Course layout**

**Week 1:** What is Online Social Networks, data collection from social networks, challenges, opportunities, and pitfalls in online social networks, APIs .

**Week 2:** Collecting data from Online Social Media.

**Week 3:** Trust, credibility, and reputations in social systems.

**Week 4:** Trust, credibility, and reputations in social systems.

**Week 5:** Online social Media and Policing.

**Week 6:** Information privacy disclosure, revelation and its effects in OSM and online social networks.

**Week 7:** Phishing in OSM & Identifying fraudulent entities in online social networks.

**Week 8:** Refresher for all topics.

**Week 9 to 12:** Research paper discussion.

**Books and references**


1. Michael E. Whitman, Herbert J. Mattord, (2018). Principles of Information Security, 6th edition, Cenage Learning, N. Delhi.
2. Darktrace, "Technology" https://www.darktrace.com/en/technology/#machine- learning, accessed November 2018.
3. Van Kessel, P. Is cyber security about more than protection? EY Global Information Security Survey 2018-2019.
4. Johnston, A.C. and Warkentin, M. Fear appeals and information security behaviors: An empirical study. MIS Quarterly, 2010.
5. Arce I. et al. Avoiding the top 10 software security design flaws. IEEE Computer Society Center for Secure Design (CSD), 2014.
6. Smith, H. J., Dinev, T., & Xu, H. Information privacy research: an interdisciplinary review. MIS Quarterly, 2011.
7. Subramanian R. Security, privacy and politics in India: a historical review. Journal of Information Systems Security (JISSec), 2010.
8. Acquisti, A., John, L. K., & Loewenstein, G. **What** is privacy worth? The Journal of Legal Studies, 2013
9. Xu H., Luo X.R., Carroll J.M., Rosson M.B. The personalization privacy paradox: An exploratory study of decision making process for location-aware marketing. Decision Support Systems, 2011.


**Criteria to get a certificate:**

**If You wish to get certified on this course you must register and write the proctored exam after payment of exam fee.**

30 Marks will be allocated for Internal Assessment and 70 Marks will be allocated for end term proctored examination.

Securing 40% in both separately is mandatory to pass the course and get Credit Certificate.

## CTME108 - Computer Architecture and Organization

**Description:**

**Credit and Week:**

| Teaching Scheme | Week | Marks | Credit |
|---|---|---|---|
| | 12 | 100 | 3 |

**About the course:**

This course will discuss the basic concepts of computer architecture and organization that can help the participants to have a clear view as to how a computer system works. Examples and illustrations will be mostly based on a popular Reduced Instruction Set Computer (RISC) platform. Illustrative examples and illustrations will be provided to convey the concepts and challenges to the participants. Starting from the basics, the participants will be introduced to the state-of-the-art in this field.

**Pre-requisites**:

Basic concepts in digital circuit design, Familiarity with a programming language like C or C++.

**Industry support:**

TCS ; Wipro ; CTS ; Google ; Microsoft ; HP ; Intel

**Course layout:**

**Week 1:** Evolution of Computer Systems.

**Week 2:** Instruction Set Architecture.

**Week 3:** Quantitative Principles of Computer Design.

**Week 4:** Control Unit Design.

**Week 5:** Memory System Design.

**Week 6:** Design of Cache Memory Systems.

**Week 7:** Design of Arithmetic Unit.

**Week 8:** Design of Arithmetic Unit (contd.)

**Week 9:** Input-Output System Design.

**Week 10:** Input-Output System Design (contd.)

**Week 11:** Instruction Set Pipelining.

**Week 12:** Parallel Processing Architectures.

**Books and References:**

1. D.A. Patterson and J.L. Hennessy, "Computer Architecture: A Quantitative Approach, 5/E", Morgan Koffman, 2011.
2. D.A. Patterson and J.L. Hennessy, "Computer Organization and Design: The Hardware/Software Interface, 5/E", Elsevier India, 2016.
3. W. Stallings, "Computer Organization and Architecture: Designing for Performance", Pearson, 2015.
4. C. Hamacher, Z. Vranesic and S. Zaky, "Computer Organization, 5/E", McGraw Hill, 2011.
5. J.P. Hayes, "Computer Architecture and Organization, 3/E", McGraw Hill, 1998.

**Criteria to get a certificate:**

**If You wish to get certified on this course you must register and write the proctored exam after payment of exam fee.**

30 Marks will be allocated for Internal Assessment and 70 Marks will be allocated for end term proctored examination.

Securing 40% in both separately is mandatory to pass the course and get Credit Certificate.

# CTUD101 - Fundamentals of Digital Electronics

| Credit | Component | Instruction / Contact Hours (Per Week) | Instruction / Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| **Experiential Learning Components:** Unit Tests, Assignments, Case Study etc. | | | | | |

**Pre-requisite:** None

**Methodology, Pedagogy & Andragogy:** In order to achieve the course objectives, students will be introduced to digital technologies. Various digital modules used to create digital computer devices like gates, flip flops, decoder, encoder etc. are to be studied. Mathematical base is created to understand the organization of digital computer.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Digital Systems and Number Systems | 12 |
| 2 | Boolean Algebra and Logic Gates | 12 |
| 3 | Combinational Circuit Designing | 10 |
| 4 | Sequential Circuit Designing | 10 |
| 5 | Memory Interfaces and Data Representation | 08 |
| 6 | The Arithmetic Logic Unit and Control Unit | 08 |

**Total Hours (Theory): 60**
**Total Contact Hours:60**

**Detail Syllabus:**

**Unit I: Digital Systems and Number Systems Hours 12**

Introduction and Evolution of Digital computers, Major components of digital computer, Interfaces and Buses, compiler, interpreter, Assembler, Introduction to various number systems (Decimal, Binary, Octal, Hexadecimal, BCD), Conversion of number system, Binary operations, signed binary numbers, 1's complement and 2's complement.

**Unit II: Boolean Algebra and Logic Gates Hours 12**

Fundamental concepts and rules of Boolean algebra, , Boolean Algebra Laws, Boolean expressions and its simplification with and without Truth Tables, Canonical form and standard form, D-Morgan's Theorem, Basic duality laws, Derivation of Boolean expressions, Introduction to logic gates (AND, OR, NOT, NAND, NOR, XOR, XNOR), Implementation of Boolean expressions using logic gates, Sum of product and Product of Sum forms, Universal gates, Implementation of other gates using universal gates, Karnaugh Map method for two, three, four and five variables, K-map with don't care conditions, Tabulation method, determination and selection of prime applicants.

**Unit III: Combinational Circuit Designing Hours 10**

Introduction to combinational circuits, Design procedures, Adders, Subtractors, code conversion circuits, Analysis procedures, Multilevel NAND circuits, Multilevel NOR circuits, EX-OR functions, Binary adder and Subtractors, Decimal adder and Subtractors, Magnitude comparator, Decoder and encoders, Multiplexer and de-multiplexer.

**Unit IV: Sequential Circuit Designing Hours 10**

Introduction to Sequential circuits, Concepts of Flip-flops, triggering flip-flops, Clock signals, Analysis of clock sequential circuits, State Reduction and assignments, Flip flop excitation table, Design procedure of asynchronous and synchronous counters (Ripple counter, Binary counter, BCD counter), Design of registers, Shift registers, Timing sequences.

**Unit V: Memory Interfaces and Data Representation Hours 08**

Memory, Types of ROM, Design of ROM, Types of RAM, Magnetic disc memory, Magnetic tape, Digital recording techniques, Linear-select memory organization, Types of digital codes (Gray code, 8421 code, Alphanumeric codes), Parity checking codes, Floating point representation, Fixed point representation.

**Unit VI: The Arithmetic Logic Unit and Control Unit Hours 08**

Basic operations of Arithmetic Logic Unit, Construction of the Arithmetic Logic Unit, Representation of Instruction word, Control registers, Instruction and execution cycles of Control registers, Sequence operation of Control Registers.

**Core Books:**

1. M. Morris Mano: Digital Logic and Computer Design, Third Edition, Pearson Education, 2016.
2. Thomas C. Bartee: Digital Computer Fundamentals, Sixth Edition, Tata McGraw Hill Publishing, 2012.
3. M. Morris Mano: Computer System Architecture, Third Edition, Pearson Education, 2011.
4. Balagurusamy : Fundamentals of Computers

**Reference Books:**

1. Andrew S. Tanenbaum: Structured Computer Organization, Fourth Edition, Pearson Education, 2005.
2. Albert Paul Malvino and Jerald A. Brown: Digital Computer Electronics, Third Edition, Tata McGraw Hill Publishing, 2008.
3. V. Rajaraman: Fundamentals of Computers

**Web References:**

1. https://www.tutorialspoint.com/digital_circuits/index.htm [Digital Circuits Tutorial]
2. https://www.geeksforgeeks.org/digital-electronics-logic-design-tutorials/ [Digital Electronics and Logic Design Tutorials]
3. https://studymaterialz.in/digital-logic-and-computer-design-by-morris-mano/ [Digital Logic and Computer Design book]
4. https://cupola.gettysburg.edu/oer/1/ [Digital Circuit Projects]

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Able to get the idea about digital system and numbering system. |
| CO2 : | Able to study logic gates for digital circuit designing. |
| CO3 : | Able to learn to design various combinational circuits. |
| CO4 : | Able to learn to design various sequential circuits. |
| CO5 : | Able to learn memory design concepts and to study computer Arithmetic Logic Unit and Control Unit. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Digital Systems and Number Systems | √ | | | | |
| 2 | Boolean Algebra and Logic Gates | | √ | | | |
| 3 | Combinational Circuit Designing | | | √ | | |
| 4 | Sequential Circuit Designing | | | | √ | |
| 5 | Memory Interfaces and Data Representation | | | | | √ |
| 6 | The Arithmetic Logic Unit and Control Unit | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | 2 | 2 | - | 3 | 2 | - | - | 2 | 1 | 3 | 2 |
| CO2 | 3 | 2 | 2 | 2 | - | 3 | 2 | - | - | 2 | 1 | 3 | 2 |
| CO3 | 3 | 3 | 3 | 2 | - | 2 | 2 | - | - | 2 | 2 | 3 | 2 |
| CO4 | 3 | 3 | 3 | 2 | - | 2 | 2 | - | - | 2 | 2 | 3 | 2 |
| CO5 | 3 | 3 | 3 | 2 | - | 3 | 2 | - | - | 2 | 2 | 3 | 2 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUS101 - Mathematics for Computer Science

| Credit | Component | Instruction / Contact Hours (Per Week) | Instruction / Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 02 | Theory | 02 | 30 | 25 | 55 |
| **Experiential Learning Components:** Unit Tests, Assignments, Case Study etc. | | | | | |

**Pre-requisite:** Basic algebra and geometry, as well as a good understanding of logic and proofs, mathematical statistics.

**Methodology, Pedagogy & Andragogy:** During theory lectures the emphasis will be given on the fundamentals of mathematics and statistics. Problem solving: Students will demonstrate the ability to solve problems, including applications outside of mathematics. Mathematical communication: Students will demonstrate the ability to communicate mathematical ideas clearly. They will use correct mathematical terminology and proper mathematical notation.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Sets and functions | 04 |
| 2 | Vectors and Matrices | 04 |
| 3 | Fundamentals of Calculus | 06 |
| 4 | Permutation & Combination | 05 |
| 5 | Probability | 06 |
| 6 | Graph Theory | 05 |

**Total Hours (Theory): 30**
**Total Contact Hours:30**

**Detail Syllabus:**

**Unit I : Sets and Functions** **Hours 04**

Introduction to set theory, methods of representation of a set, operations on set, algebra of sets, De 'Morgan's law, function definition, domain, range, one-to-one function, onto function.

**Unit II : Vectors and Matrices** **Hours 04**

Definition of vector, addition and subtraction of vector, magnitude of a vector, unit vectors, dot product and cross product, definition of matrix, equal matrices, diagonal elements of matrix, row matrix, column matrix, Symmetric matrix, skew symmetric matrix, orthogonal matrix, diagonal matrix, identity matrix, operations on matrix.

**Unit III : Fundamentals of Calculus** **Hours 06**

Limits, Continuity and Differentiability, Successive Differentiation, Partial Differentiation, Tangents and Normals, Maxima and Minima.

**Unit IV : Permutation and Combination** **Hours 05**

Meaning of permutation, Formula of permutation, Permutation of n different things, Permutation of similar things, Permutation of repeated things, Circular Permutation, Combination: Meaning of Combination, Formula of Combination.

**Unit V : Probability** **Hours 06**

Probability: Random Experiment, Sample Space, Event, Mutually exclusive event, Exhaustive event, Equally likely event, Probability Classical definition. (Simple examples of Probability).

**Unit VI : Graph Theory** **Hours 05**

Introduction to Graph, Graph Definition, Vertices, Edges, Loops, Parallel Edges, Simple Graph, Finite Graph, Adjacent vertices, Incidence between vertex and edge, Degree of a vertex, Isolated Vertex, Pendent Vertex, Null Graph. Isomorphism, Labeled Graph

**Core Books:**

1. D. C. Sancheti, V. K. Kapoor: Business Mathematics, Sultan Chand & sons.
2. Lipschutz & Marc Lipson: DISCRETE MATHEMATICS, Tata McGraw Hill
3. Narsingh Deo: Graph Theory with application to engineering and computer science, Prentice Hall of India Pvt. Ltd

**Reference Books:**

1. Gupta and Gupta: Business Statistics, Sultan Chand and Sons.

**Web References:**

1. https://www.statlect.com/matrix-algebra/vectors-and-matrices
2. https://www.geeksforgeeks.org/permutations-and-combinations/
3. https://www.khanacademy.org/math/statistics-probability/probability-library/basic-theoretical-probability/a/probability-the-basics

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Students will learn basics of sets and functions |
| CO2 : | Students will be familiar with various linear algebra. |
| CO3 : | Students will be able to compute limits, derivatives, and integrals. Analyze functions using limits, derivatives, and integrals |
| CO4 : | Students will be able to define a permutation and explain how to calculate probability |
| CO5 : | Students will be able to understand and apply the fundamental concepts in graph theory |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Sets and Functions | √ | | | | |
| 2 | Vectors and Matrices | | √ | | | |
| 3 | Fundamentals of Calculus | | | √ | | |
| 4 | Permutation and Combination | | | | √ | |
| 5 | Probability | | | | √ | |
| 6 | Graph Theory | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 3 | 3 | 2 | - | 2 | 1 | - | - | - | - | 2 | - |
| CO2 | 3 | 3 | 3 | 2 | - | 2 | 1 | - | - | - | - | 2 | - |
| CO3 | 3 | 3 | 3 | 2 | - | 2 | 1 | - | - | - | - | 2 | - |
| CO4 | 3 | 3 | 3 | 2 | - | 2 | 1 | - | - | - | - | 2 | - |
| CO5 | 3 | 3 | 3 | 2 | - | 2 | 1 | - | - | - | - | 2 | - |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# TEACHING SCHEME & DETAILED SYLLABUS

# FOR

# B.Sc. (IT) PROGRAMME
# (2nd SEMESTER)
# AS PER NEP 2020

# EFFECTIVE FROM ACADEMIC YEAR 2025-26

## B.Sc. (IT) Semester-II

| Course Code | Course Name | Theory / Practical / Others | Marks (Theory) | | Marks (Practical) | | Credit | NEP -2020 Classification |
|---|---|---|---|---|---|---|---|---|
| | | | CCE | SEE | CCE | SEE | | |
| **CTUC103** | Introduction to Object Oriented Programming | Theory and Practical | 50 | 50 | 25 | 25 | 06 (04-Theory) (02-Practical) | Major Core |
| **CTUC104** | Fundamentals of Web Designing | Practical | - | - | 25 | 25 | 02 | Major Core |
| **CTUE103**<br><br>**CTUE104** | (Select any one)<br>5. Business Models for e-Commerce<br>6. Managerial Economics | Theory | 50 | 50 | - | - | 04 | Minor |
| **CTUD102** | Data Communication and Networks | Theory | 50 | 50 | - | - | 04 | MDC |
| **CTUV101** | Fundamentals of Accounting | Theory | 25 | 25 | - | - | 02 | VAC |
| **HSUS101-117** | (Select any one)<br>A course on Liberal Arts<br>HSUS101 – Painting<br>HSUS102– Photography<br>HSUS103– Sculpting<br>HSUS104– Pottery and Ceramic Art<br>HSUS105-Media and Graphic Design<br>HSUS106-Art and Craft<br>HSUS109- Dramatics<br>HSUS110-Contemporary Dance<br>HSUS111-Music (Vocal)<br>HSUS112-Music (Instrumental)-Tabla<br>HSUS113-Music (Instrumental)-Guitar<br>HSUS114-Music (Instrumental)-Harmonium<br>HSUS115-Music (Instrumental)-Flute<br>HSUS116- Indian Classical Dance-Kathak<br>HSUS117- Indian Classical Dance-Bharatnatyam | Practical | - | - | 25 | 25 | 02 | SEC |
| **CTUA101** | Presentation Skills | Practical | - | - | 25 | 25 | 02 | AEC |
| **CTUI101** | Summer Internship and Viva - I | VIVA | - | - | 50 | 50 | 04 | Exit Course |

*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)
*SEE- Semester –End- Evaluation

# CTUC103 - Introduction to Object Oriented Programming

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components (Theory) :** Unit Tests, Assignments, Case Study etc.<br>**Experiential Learning Components (Practical) :** Lab Assignments, Practical Test, VIVA | | | | | |

**Pre-requisite:** Introduction to Programming

**Methodology, Pedagogy & Andragogy:** The theory sessions will be enriched with real-world examples to demonstrate how object-oriented programming (OOP) enhances problem-solving. These examples will showcase the practical application of OOP concepts such as classes, objects, inheritance, polymorphism, and exception handling. By applying these concepts to real-world scenarios, students will gain a deeper understanding of how OOP can be used to solve complex problems efficiently and effectively. During the practical sessions of the Practical Programming Subject course, students will apply the concepts through a variety of assignments and case studies to assess their comprehension and progress.

**Outline of the course(Theory):**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Introduction to Object Oriented Programming | 08 |
| 2 | Classes and Objects | 11 |
| 3 | Inheritance | 13 |
| 4 | Polymorphism | 10 |
| 5 | Exception Handling and Type Cast Operators | 10 |
| 6 | Managing I/O formats and operations | 08 |

**Total Hours (Theory): 60**
**Total hours (Practical): 60**
**Total Contact Hours: 120**

**Detailed Syllabus:**

**Unit - I: Introduction to Object Oriented Programming Hours: 08**

Structured programming vs. object oriented programming, Introduction to C++: structure of a C++ program Tokens, inline function, pass by reference, default arguments, OOP characteristics.

**Unit - II: Classes and Objects Hours: 11**

Define class, objects, visibility modes, static members, friend function, Constructors and Destructors, Default Constructor, Copy constructors, Parameterized Constructor, this keyword.

**Unit - III: Inheritance Hours: 13**

Objective of Inheritance, types of inheritance, impact of access modes during inheritance, Virtual base classes, Dynamic Binding, Virtual functions, Pure virtual functions, Abstract class, Early vs. Late binding.

**Unit - IV: Polymorphism Hours: 10**

Introduction of Static Binding, Function Overloading and ambiguity raised in function overloading, Operator overloading through member function and friend functions.

**Unit - V: Exception Handling and Type Cast Operators Hours: 10**

Introduction to Exception handling, catching class types, using multiple catch statements, exception handling options, creating custom exceptions, introduction to cast operators, const cast, static cast, reinterpret cast, dynamic cast.

**Unit - VI: Managing I/O Formats and Operations Hours: 08**

Understand and apply concepts from various header files like iostream, iomanip, fstream, etc. opening and closing file, reading and writing text files and binary files, performing random access on files, handle command line arguments.

**Outline of the course(Practical):**

| Week No | Practical | Description |
|---|---|---|
| 1 | Introduction to Object Oriented Programming and Setting up IDE | Introduction to C++, Setting up a development environment and its introduction, structure of a C++ program Tokens, inline function, pass by reference, default arguments |
| 2 | Classes and Objects | Define class, objects, visibility modes, static members |
| 3 | Friend Functions | Introduction and hands on friend functions |
| 4 | Constructors and Destructors | Constructors and Destructors, Default Constructor, Types of constructs |
| 5 | Inheritance | Exposure on Inheritance programs and their types |
| 6 | Virtual Function | Programs on Virtual Function, Pure Virtual functions, Abstract Class, Early Binding and Late Binding |
| 7 | Function Overloading | Introduction of Static Binding, Function Overloading and ambiguity raised in function overloading |
| 8 | Operator Overloading | Operator overloading through member function and friend functions |
| 9 | Exception Handling | Introduction to Exception handling, catching class types, |
| 10 | Multiple Catch Statements | Using multiple catch statements, exception handling options, creating custom exceptions |
| 11 | Type Cast Operators | Introduction to cast operators, const cast, static cast, reinterpret cast, dynamic cast |
| 12 | Managing I/O Formats | Understand and apply concepts from various header files like iostream. Iomanip ,fstream, etc |
| 13 | File Handling | Opening and closing file, reading and writing text files and binary files |
| 14 | File Handling Operations | opening and closing file, reading and writing text files and binary files, performing random access on files |
| 15 | Command Line Arguments | handle command line arguments. |

**Total hours: 60**

**Core Books:**

1. Herbert Schildt: C++: A Beginner's Guide, 2nd Edition Paperback, McGraw-Hill,2003.
2. E Balagurusamy: Object Oriented Programming with C++ Paperback, McGraw Hill India,2017.

**Reference Books:**

1. Yashavant Kanetkar: Let Us C++, Paperback, BPB Publications,2020.
2. Robert Lafore: Object Oriented Programming in C++, 4th Edition, Sams Publications,2002.
3. Bruce Eckel:Thinking in C++, Volume 1, 2nd Edition, Pearson Education,2006.

**Web References:**

1. https://www.geeksforgeeks.org/c-plus-plus/ [For Entire Syllabus]
2. https://google.github.io/styleguide/cppguide.html [ For coding standards and concepts]
3. http://www.tutorialspoint.com/cplusplus/cpp_basic_input_output.htm [For Practical's]

**Course Outcomes:** Upon successful completion of the course, the students will:

| | |
|---|---|
| CO1 : | Understand the basic structure of C++ program and understand the concept of object oriented programming characteristics. |
| CO2 : | Understand the concept of Constructor and Destructor and able to create class and objects. |
| CO3 : | Understand the need of Polymorphism and implement static polymorphism. |
| CO4 : | Understand the purpose of Inheritance, identify relations between classes, implement inheritance and perform dynamic binding. |
| CO5 : | Able to handle the runtime errors or exceptions in C++ and use type cast operators. Able to read input from device, file, command line arguments and apply formats to input and output. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | **Co1** | **Co2** | **Co3** | **Co4** | **Co5** |
| 1 | Introduction to Object Oriented Programming | √ | | | | |
| 2 | Classes and Objects | | √ | | | |
| 3 | Inheritance | | | √ | | |
| 4 | Polymorphism | | | | √ | |
| 5 | Exception Handling and Type cast operators | | | | | √ |
| 6 | Managing I/O formats and operations | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 3 | 2 |
| CO2 | 3 | 3 | 3 | 2 | 2 | 2 | 3 | 2 | 2 | 2 | 2 | 3 | 2 |
| CO3 | 3 | 2 | 3 | 2 | 2 | 2 | 3 | 2 | 2 | 2 | 2 | 3 | 2 |
| CO4 | 3 | 3 | 3 | 2 | 2 | 2 | 3 | 2 | 2 | 2 | 2 | 3 | 2 |
| CO5 | 3 | 2 | 3 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 3 | 2 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

# CTUC104 - Fundamentals of Web Designing

| Credit | Component | Instruction / Contact Hours (Per Week) | Instruction / Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components :** Lab Assignments, Practical Test, VIVA | | | | | |

**Prerequisites:** Basic concepts of web.

**Methodology, Pedagogy & Andragogy:** This course focuses on providing hands-on experience to students for design and develop entire web sites using several web designing tools and HTML scripting language.

**Outline of the Course:**

| Week No | Practical | Description |
|---|---|---|
| 1 | Basic Construction of an HTML & HTML5 Page,<br>HTML Text Editors, HTML Building blocks. | Learn basic concepts of web page design. |
| 2 | Basic HTML tags – Text formatting tags,<br>Heading tags, paragraph tags, division tag, Span tags,<br>Break line tag, Binding space in HTML<br>. | Get familiar with different tags and attributes which is used to add the text in a web page. |
| 3 | Controlling Font size and color, Adding Table – Table tag. | Learn user interface designing using Table Layout, change font style, color etc. |
| 4 | Grouping and merging table rows and columns. Adding List | Able to display the data in various formats and design different types of list in the table. |

| | | |
|---|---|---|
| | – ordered, unordered and definition list. | |
| **5** | Adding Image, Adding links - Changing link colors, Email link. | Learn to add images and link in the web page. |
| **6** | Embedding audio and video, Adding Field sets. | Able to add multimedia in web page using internal and external methods. |
| **7** | Adding Form: Form tag, Action, Method. Basic form controls. | Learn basic form elements and when and where to apply it. |
| **8** | Textbox, Password field, Radio Button, Checkbox, Multiple line text area, file upload. | Use various types basic form controls with types of form input elements. |
| **9** | Various types of buttons – submit, reset, image, simple button. | Learn various types of buttons to make it more interactive. |
| **10** | HTML5 input elements - Color, Number, Date, Month. | Learn new input types in HTML5. |
| **11** | Week, Time, date time-local, Email, url, range, search. | Learn to add defining inputs and minimum and maximum range to make the form attractive. |
| **12** | Data list, output, progress, meter. | Learn to add list and provide progress horizontal view and range with output. |
| **13** | Regular expression, HTML5 new attributes – Placeholder, Required. | Learn HTML 5 form element to provide user side web form validation. |
| **14** | Pattern, Autocomplete, Autofocus, novalidate, formnovalidate. | Learn HTML 5 new form properties. |
| **15** | formaction, formmethod, spellcheck, contenteditable. | Learn HTML 5 to user submit the form and redirect the form data to selective method. |

**Total hours: 60**

**Text Books:**

1. Powel Thomas: HTML - The Complete Reference, 3rd Edition, McGraw-Hill Education – Europe,2018.
2. DT Editorial Services: HTML 5 Black Book,2nd Edition, Dreamtech Press, 2016
3. Matt West: HTML 5 Foundations, Wiley publication, 2013
4. Farrar: HTML Example book, BPB,2007.

**Reference Books:**

1. David DuRocher : HTML and CSS QuickStart Guide, Paperback,2021
2. Laura Lemay, Rafe Colburn, Jennifer Kyrnin : Mastering Html, CSS & JavaScript Web Publishing, Paperback, 2016.
3. Whyte: Basic HTML, 2nd Edition, Payne-Gallway, Oxford, 2003.
4. Shelly Woods: HTML introductory concepts and techniques, 5th Edition, Course Technology,2009.
5. Jon Dukett: Beginning Web Programming with HTML, XHTML and CSS, Wrox Publication.

**Web References:**

1. www.w3schools.com/html [ For HTML tutorials]
2. html.net/tutorials/html/ [ For HTML tutorials]
3. www.htmlgoodies.com/ [ For HTML resources]
4. https://developer.mozilla.org/en-US/learn/html [ For HTML best practices]

**Course Outcome:** Upon successful completion of the course, student will:

| | |
|---|---|
| CO1 : | Able to understand basic construction of HTML & HTML5 document and started using basic tags. |
| CO2 : | Able to Add content using various tags and format the content in table & list. |
| CO3 : | Able to design links and multimedia elements such as image, audio, video. |
| CO4 : | Able to design a web page form using HTML & HTML5. |
| CO5 : | Able to apply validation using HTML5 attributes. |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 1 | 2 | 1 | 3 | 2 | 2 | 1 | 1 | - | 1 | 1 | 1 | 1 |
| CO2 | - | 1 | 2 | 3 | 1 | 3 | 1 | - | - | 2 | 2 | 2 | 2 |
| CO3 | 1 | 1 | 2 | 3 | 1 | 3 | 1 | - | - | 2 | 2 | 2 | 2 |
| CO4 | 1 | 1 | 2 | 3 | 1 | 1 | 1 | 1 | - | 1 | - | 1 | 1 |
| CO5 | 2 | 2 | 3 | 3 | 2 | 3 | 3 | 2 | 1 | 2 | 2 | 3 | 2 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

# CTUE103 - Business Models for e-Commerce

| Credit | Component | Instruction / Contact Hours (Per Week) | Instruction / Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| **Experiential Learning Components:** Unit Tests, Assignments, Case Study etc. | | | | | |

**Pre-requisite:** None

**Methodology, Pedagogy & Andragogy:** During theory lectures the emphasis will be given on the fundamentals of E-commerce. Students will be introduced to E-Commerce, E-Marketing, E-Security and E-Payment, Mobile Commerce, CRM and ERP, and E-Commerce Applications. Students will give practical exposure in form of case study and by visit to E-commerce / Digital Marketing company and/or supply chain Centre.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Introduction to E-Commerce | 09 |
| 2 | E-Marketing | 11 |
| 3 | E-Security and E-Payment | 09 |
| 4 | Mobile Commerce | 11 |
| 5 | E-CRM and ERP | 09 |
| 6 | E-Commerce Applications | 11 |

**Total Hours (Theory): 60**
**Total Contact Hours:60**

**Detail Syllabus:**

**Unit I : Introduction to E-Commerce** **Hours 09**

Emergence and use of Internet, origin of web and world wide web, advantages and disadvantages of E-commerce, Features of E-Commerce, Electronic Commerce over the Internet, Types of Ecommerce.

**Unit II : E-Marketing** **Hours 11**

Traditional Marketing, Online Marketing, E- advertising, Internet Marketing Trends and strategies, E-branding.

**Unit III : E-Security and E-Payment** **Hours 09**

Security on the Internet, E-Business risk management issues, Digital token based E-Payment System, Properties of Electronic Cash, Digital Signature.

**Unit IV : Mobile Commerce** **Hours 11**

Definition of M-Commerce, Features of M-Commerce, Advantages and Disadvantages of M-Commerce, Areas of M-Commerce Applications, Payment Systems and Models in M-Commerce.

**Unit V : E-CRM and ERP** **Hours 09**

Customer Relationship Management, CRM capabilities and Customer life cycle, Introduction to ERP, Reasons for the growth of the ERP Market, Advantages and Disadvantages of ERP.

**Unit VI : E-Commerce Applications** **Hours 11**

Retail and Wholesale, Finance, Manufacturing, Online Booking, Online Publishing, Digital Advertising, Digital Shopping, Digital Media, Auctions, A case study need to be discussed for these application. Hands-on experience in setting up and managing an e-commerce store. Developing a business plan for an ecommerce startup.

**Core Books:**

1. P.T.Joseph, S.J.: E-Commerce- An Indian Perspective, 3rd Edition, PHI learning Private Limited.
2. Kamlesh K. Bajaj, Debjani Nag: E-Commerce, The Cutting Edge of Business, 2nd Edition, McGraw-Hill Education.
3. V. Rajaraman: E-Commerce.

**Reference Books:**

1. Anita Rosen: The e-commerce Question and Answer Book: A Survival Guide for Business Managers, 2nd Edition, Amacom.
2. Janice Reynolds: The Complete E-Commerce Book: Design, Build & Maintain a Successful Web-based Business, 2nd Edition, CRC Press.
3. By Philippe Humeau&Matthieu Jung : The White Book of Ecommerce Solutions, NBS System.
4. IshitaLahiri and Sujit Kumar Ghose : Principals of Marketing and E-Commerce
5. Kenneth C. Laudon & Carol Traver : E-Commerce: Business, Technology, Society

**Web References:**

1. http://www.ecommercetutorial.net/ [For Tutorial]

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | To introduce the concept of Electronic Commerce |
| CO2 : | To understand the concept of E–marketing |
| CO3 : | To understand the role of E–Security & E–Payment |
| CO4 : | To understand M–Commerce Payment System |
| CO5 : | To learn CRM, ERP and E–commerce Applications. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Introduction to E-Commerce | √ | | | | |
| 2 | E-Marketing | √ | √ | | | |
| 3 | E-Security and E-Payment | | √ | √ | | |
| 4 | Mobile Commerce | | √ | √ | | |
| 5 | E-CRM and ERP | | | | √ | |
| 6 | E-Commerce Applications | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 2 | 3 | 2 | 2 | 1 | 2 | 2 | 2 | 1 | 1 | 3 | 3 |
| CO2 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| CO3 | 2 | 2 | 3 | 3 | 2 | 2 | 2 | 2 | 2 | 1 | 1 | 3 | 2 |
| CO4 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 2 | 3 | 3 | 2 |
| CO5 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 2 | 3 | 3 | 2 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUE104 - Managerial Economics

| Credit | Component | Instruction / Contact Hours (Per Week) | Instruction / Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| **Experiential Learning Components :** Unit Tests, Assignments, Case Study etc. | | | | | |

**Pre-requisite:** None

**Methodology, Pedagogy & Andragogy:** During theoretical lessons, real-world examples will be used to discuss economic topics such as demand, supply, production, cost function, market structure, factor pricing, and macroeconomics. Students will be offered appropriate case studies in order to provide actual exposure to theoretical economic ideas.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Basic Economic Concepts | 08 |
| 2 | Demand and Supply | 11 |
| 3 | Production, Cost and Revenue | 11 |
| 4 | Inflation | 11 |
| 5 | Market Structures & Pricing | 10 |
| 6 | Market Failures and Price Regulations | 09 |

**Total Hours (Theory): 60**
**Total Contact Hours:60**

**Detail Syllabus:**

**Unit I: Basic Economic Concepts** **Hours 08**

Basic of Economics, Nature of Economics, Scope of Economics, Business Economics, Difference Between Micro and Macro Economics, Laws of Economics, Business Cycle, Inflation, Needs, wants.

**Unit II: Demand and Supply** **Hours 11**

Demand and supply, Types of Demand, Determinants of Demand and supply, Law of Demand and supply Demand and supply Schedule, Demand Function. Own-price elasticity of demand, cross-price elasticity of demand, and income elasticity of demand.

**Unit III: Production, Cost and Revenue** **Hours 11**

Production function, law of variable proportion and laws of returns to scale, different types of costs – variable cost, fixed cost, total cost, average cost, average fixed cost, average variable cost and marginal cost, Total revenue, average revenue and marginal revenue, profit function.

**Unit IV: Inflation** **Hours 11**

Meaning, Depreciation of money, Monetary inflation concepts, types of inflation, Causes and effect of inflation different sectors of the economy, Causes of Inflation, Advantages, Formula and Disadvantages of inflation.

**Unit V: Market Structures & Pricing** **Hours 10**

Concept of market and equilibrium- characteristics of perfect competition, Features of Market Structure, monopoly, monopolistic competition and oligopoly–price determinations, Measure of market structure with example.

**Unit VI: Market Failures and Price Regulations** **Hours 09**

Market failures and need for regulation, Regulations and market structure, Firm behavior, Price regulation. More on Externalities, and role of government, importance of competition in market failure and price regulations.

**Core Books:**

1. P.L Mehta: Managerial Economics, Sultan Chand and Sons, (2014)
2. Keat Paul , K Young Philip , Erfle Steve , College Dickinson, Banerjee Sreejatha , Managerial Economics, Pearson Education; 6th ed..

**Reference Books:**


1. Yogesh Maheswari, Managerial Economics, Phi Learning, New Delhi, 2005 Gupta G.S.
2. Moyer &Harris. Managerial Economics, Tata Mcgraw-Hill, New Delhi
3. Geetika, Ghosh & Choudhury. Managerial Economics, Cengage Learning, New Delhi, 2005
4. Dominick Salvatore, Managerial Economics, adapted by Ravikesh Srivastava, Oxford University press
5. Mote V.L., Samuel Paul and G.S. Gupta, Managerial Economics Concepts and Cases, Tata McGraw Hill Publishing Company Ltd., New Delhi, 2001


**Web References:**


1. https://www.geektonight.com/managerial-economics-notes/
2. https://www.learningclassesonline.com/2020/10/pedagogy-of-economics.html
3. https://mrcet.com/downloads/MBA/Managerial%20Economics.pdf


**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Able to understand basic economic concepts. |
| CO2 : | Able to understand market. |
| CO3 : | Able to learn basics of inflation. |
| CO4 : | Able to deal with situation between cost and revenue. |
| CO5 : | Able to understand basics of Demand and Supply. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Basic Economic Concepts | √ | | | | |
| 2 | Demand and Supply | | | | | √ |
| 3 | Production, Cost and Revenue | | | | √ | |
| 4 | Inflation | | | √ | | |
| 5 | Market Structures & Pricing | | √ | | | |
| 6 | Market Failures and Price Regulations | | √ | | | |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | - | - | - | - | 1 | 2 | 1 | - | - | - | 1 | 1 | 2 |
| CO2 | - | 1 | 1 | - | 1 | 2 | 2 | - | - | 1 | 3 | 1 | 2 |
| CO3 | - | 1 | 1 | - | 1 | 2 | 2 | - | - | - | 2 | 1 | 2 |
| CO4 | 1 | 1 | 1 | - | 1 | 2 | 2 | - | - | 1 | 2 | 1 | 2 |
| CO5 | 1 | 1 | 1 | - | 1 | 2 | 2 | - | - | 1 | 2 | 1 | 2 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

# CTME103 - E-Business

**Description:**

Credit and Week:

| Teaching Scheme | Week | Marks | Credit |
|---|---|---|---|
| | 12 | 100 | 4 |

**About the course:**

The Internet has changed the way companies carry out their businesses. The primary objective of this course is to introduce concepts, tools and approaches to electronic business to the post- graduate and undergraduate students. Further, the subject will help the students to develop skills to manage businesses in the digital world. The course will cover following aspects of E-Business Systems.

- Part 1: Foundations of E-Business systems
- Part 2: Infrastructure
- Part 3: Functional Areas
- Part 4: Decision Support for E-Business Systems

The course provides a balance approach including concepts from technology and management.

**Pre-requisites:** None

**Industry support**: All the companies

**Course layout**

**Week 1:** Introduction to E-Business

**Week 2:** Making Functional Areas E-Business Enabled : Value chain and supply chain, inter and intra organizational business processes, ERP

**Week 3:** Making Functional Areas E-Business Enabled : E-Procurement

**Week 4:** Making Functional Areas E-Business Enabled : E-marketing, E-Selling, E-Supply Chain Management

**Week 5:** Technologies for E-Business: Internet and Web based system

**Week 6:** Technologies for E-Business: Security and payment systems

**Week 7:** Technologies for E-Business: Supply chain integration technologies (EDI, RFID, Sensors, IoT, GPS, GIS)

**Week 8:** Technologies for E-Business: Supply chain integration technologies (Web services and cloud)

**Week 9:** Decision Support in E-Business: Web analytics

**Week 10:** Decision Support in E-Business: Customer behavior modeling

**Week 11:** Decision Support in E-Business: Auctions

**Week 12:** Decision Support in E-Business: Recommender systems

## Books and references

1. Management Information Systems: Managing the Digital Firm, Laudon and Laudon, Pearson
2. Scaling for E-Business, Menasce & Almeida, PHI
3. eBusiness & eCommerce – Managing the Digital Value Chain, Meier & Stormer, Springer
4. eBook is available in springerlink.com
Some reference books, Internet Resources, and Research Papers

**Criteria to get a certificate:**

**If You wish to get certified on this course you must register and write the proctored exam after payment of exam fee.**

30 Marks will be allocated for Internal Assessment and 70 Marks will be allocated for end term proctored examination.

Securing 40% in both separately is mandatory to pass the course and get Credit Certificate.

# CTME104 - Business Development: From Start to Scale

**Description:**

Credit and Week:

| Teaching Scheme | Week | Marks | Credit |
|---|---|---|---|
| | 12 | 100 | 4 |

**About the course:**

This 12-week, 60 lecture course titled ""Business Development: From Start to Scale"" equips the learners with various concepts and frameworks for establishing and growing businesses. Focusing on customers and markets, the course covers the foundational as well as advanced constructs of business development. Multiple practical examples and case studies are provided. This versatile course will be useful for students and working professionals, and relevant for startups and entrepreneurial firms as well as established small, medium, and large companies for crafting and executing their growth journey. This course, in addition, will be an ideal next-step course for those who would complete the Entrepreneurship course.

**Pre-requisites:** Graduation of any discipline; No requirement to do any other course as a pre-requisite to taking this course

**Industry support**: Likely to be recognized by all types of industries and all types of firms as starting up and scaling up of business is a universal and perpetual need.

**Course layout**

Week 1: Business Fundamentals

- Understanding Business Development
- Marketing and Business Development
- Markets and Marketing
- Strategy Formulation
- Business Development Cases

Week 2: Business Development Strategies

- Successful Businesses
- Industry and Market
- Vision, Mission and Strategy

- Goals
- Case Studies of Business Development Excellence

### Week 3: Industry Structure and Company Analysis

- Industry and Business
- Porter's Five Forces Theory
- Industrial Transformations
- Competitive Strategies
- Company Analysis

### Week 4: Market and Competitor Analysis

- Industry, Market and Business
- Industry and Market Analysis
- Market Structures
- Demand Forecasting
- Competitor Analysis

### Week 5: Connecting with Customers

- Customer Characteristics
- Customer Typologies
- Market Research and Design Thinking
- Customer bonding
- Customer Relationship Management

### Week 6: Business and Market Segments

- Market and Market Descriptors
- Market and Product Segmentation
- Product-Market Segmentation
- Segmentation Deep Dive
- Market Attractiveness and Competitive Positioning

### Week 7: Branding and Pricing

- Branding
- Brand Organisation
- Advertising and Communication
- Servitization
- Pricing

### Week 8: Corporate Development

- A New IT Start-up
- An FMCG Start-up
- A Logistics Start-up
- A Nutraceuticals Start-up
- A Telecom Fightback

### Week 9: Business Development Structures

- Collaborations
- Strategic Alliances
- Joint Ventures
- Subsidiaries
- Mergers and Acquisitions

### Week 10: Business Development Competencies

- Value Chain Competencies
- Functional Competencies
- Negotiation Skills
- Cultural Skills
- Leadership Attributes

### Week 11: Strategies for Markets and Industries

- Growth Strategies
- Growth Examples
- Fragmented Industries and Emerging Industries
- Mature Industries and Declining Industries
- Global Industries and New Businesses

### Week 12: Business Development Case Studies

- Business Transformation
- Strategic Alliances for Growth
- Business Turbulence
- Creating Value
- From Start to Scale
- In Closing

### Books and references

1. Michael E. Porter, Competitive Strategy, New York: Free Press, 1980

2. Orville C. Walker, Jr., and John W. Mullins, Marketing Strategy: A Decision- Focused Approach, McGraw Hill, 2011
3. Philip Kotler and Gary Armstrong, Principles of Marketing, Englewood Cliffs, NJ, Prentice Hall, 1989
4. Philip Kotler, Leven Lane Keller, Marketing Management, Pearson, 2017W. Chan Kim and Renee Mauborgne, Blue Ocean Strategy, Harvard Business Review Press, 2017
5. Charles W L Hill, Gareth R Jones and Melissa A Schilling, Strategic Management, Cengage Learning, 2015
6. Thomas L. Wheelen, J. David Hunger, Alan N. Hoffman, and Charles E. Bamford, Strategic Management and Business Policy: Globalization, Innovation, and Sustainability, Pearson, 2015
7. Rao, C B, Strategic Management, Practice and Philosophy for India Inc, Notion Press, 2021

**Criteria to get a certificate:**

**If You wish to get certified on this course you must register and write the proctored exam after payment of exam fee.**

30 Marks will be allocated for Internal Assessment and 70 Marks will be allocated for end term proctored examination.

Securing 40% in both separately is mandatory to pass the course and get Credit Certificate.

# CTUD102 - Data Communications and Networks

| Credit | Component | Instruction / Contact Hours (Per Week) | Instruction / Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| **Experiential Learning Components:** Unit Tests, Assignments, Case Study, Certification courses, etc. | | | | | |

**Pre-requisite:** Basic knowledge of computers and their working.

**Methodology, Pedagogy & Andragogy:** During the lecture sessions, the students will learn about role of computer network in modern communication systems, underlying principles of network communication, need for standardization through network reference model. The teacher will also discuss communication issues at different layers and concepts related to network security. The students will adopt the latest certification courses from reputed and recognized platforms. Furthermore, the students will be engaged in problem-solving, emphasizing real-world applications and relevance.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Introduction to Computer Networks | 10 |
| 2 | Physical Layer Communication | 08 |
| 3 | Network reference model | 11 |
| 4 | Data Link Layer Functions and Protocol | 12 |
| 5 | Network Layer Functions and Protocol | 10 |
| 6 | Network security | 09 |

**Total Hours (Theory):60**
**Total Contact Hours:60**

**Detail Syllabus:**

**Unit I: Introduction to computer network** **Hours 10**

Definition of computer network, aim and objective of computer network: applications of computer network, merits & demerits of computer network, examples of computer network, components of computer network, types of networks, network topologies, network devices: hub, switch, repeater, router, gateway, bridge.

**Unit II: Physical Layer Communication** **Hours 08**

Analog and digital signal, Definition of bandwidth, Maximum data rate of a channel, Line encoding schemes, Transmission modes, Modulation techniques, Multiplexing techniques- FDM and TDM, Transmission media-Guided and Unguided, Switching techniques- Circuit switching, Packet switching, Connectionless datagram switching, Connection-oriented virtual circuit

**Unit III: Network reference model** **Hours 11**

Need for network reference model, history of TCP/IP and OCI model, layers of OCI and TCP/IP model, comparison between TCP/IP and OSI model, Protocol Hierarchy, Network Protocols: Simple Mail Transfer Protocol (SMTP), File Transfer Protocol (FTP), Multiple Internet Mail Extension (MIME), Post Office Protocol (POP), Telnet, Domain Name Service (DNS), Dynamic Host Configuration Protocol (DHCP), Address Resolution Protocol (ARP), Reverse Address Resolution Protocol (RARP).

**Unit IV: Data Link Layer Functions and Protocol** **Hours 12**

Definition of Framing, Framing methods, Error detection techniques, Error correction techniques, Flow control mechanisms- Simplex protocol, Stop and Wait ARQ, Go-Back-N ARQ, Point to Point protocol.

**Unit V: Network Layer Functions and Protocol** **Hours 10**

Connection oriented vs Connectionless services, Definition of Routing, Routing algorithms, IP protocol, IP addresses, ARP, RARP

**Unit VI: Network security** **Hours 09**

Introduction to cryptography, symmetric key and public key encryption systems, digital signature and key management concepts, communication security.

**Core Books:**

1. Andrew S. Tanenbaum: Computer Networks, 6th edition, Pearson Publication: 2021.

2. Behrouz Forouzan: Data communications and networking, 4th Edition, McGraw Hill, 2018.
3. William Stallings: Data and computer communication: 5th Edition, Pearson Publication, 2019.


**Reference Books:**


1. James F. Kurose, Keith W. Ross: Computer networking – A top-down approach: 3rd Edition: Pearson Publication: 2017
2. Narsimha Karumanchi: Elements of computer networking: An Integrated Approach: 1st Edition: Careermonk Publication: 2014
3. Ed Tittel: Computer networking: 1st Edition, Schum's Publication: 2020


**Web References:**


1. https://computer.howstuffworks.com/computer-networking-channel.htm [How Stuff Works]
2. https://learn.microsoft.com/en-us/previous-versions/windows/it-pro/windows-server-2012-R2-and-2012 /dn313100(v=ws.11)?redirectedfrom=MSDN [Microsoft Networking]
3. https://help.sap.com/docs/SAP_COMMERCE/d0224eca81e249cb821f2cdf45a82ace/8c74877b866910148cc9ba5f39a2fd28.html?version=6.7.0.0 [For multicast and unicast]


**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Student will be able to understand role of computer network in digital communication |
| CO2 : | Student will grasp underlying concepts related with network communication |
| CO3 : | Student will understand role of reference models in standardizing communication |
| CO4 : | Students will learn about intra and inter network communication and technology behind it. |
| CO5 : | Students will learn about different aspects of network security. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Introduction to Computer Networks | √ | | | | |
| 2 | Physical Layer Communication | | √ | | | |
| 3 | Network reference model | | | √ | | |
| 4 | Data Link Layer Functions and Protocol | | | | √ | |
| 5 | Network Layer Functions and Protocol | | | | √ | |
| 6 | Network security | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 3 | 2 | 1 | - | 1 | 1 | 1 | - | - | 1 | 2 | - |
| CO2 | 2 | 3 | 3 | 2 | - | 1 | 1 | 1 | - | - | 1 | 3 | 1 |
| CO3 | 3 | 3 | 3 | 2 | - | 1 | 1 | 1 | 1 | - | 1 | 3 | 3 |
| CO4 | 1 | 3 | 3 | 2 | - | 1 | 1 | 1 | 1 | - | 1 | 3 | 2 |
| CO5 | 3 | 3 | 3 | 2 | - | 1 | 1 | 1 | 1 | - | 1 | 3 | 2 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

# CTUV101 - Fundamentals of Accounting

| Credit | Component | Instruction / Contact Hours (Per Week) | Instruction / Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 02 | Theory | 02 | 30 | 20 | 50 |
| **Experiential Learning Components:** Unit Tests, Assignments, Case Study etc. | | | | | |

**Pre-requisite: None**

**Methodology, Pedagogy & Andragogy:** During the classroom sessions, the emphasis will be given on the fundamentals of accounting. Students will be introduced to book keeping systems in modern accounting, financial accounting, and cost accounting. Students will prepare financial statements such as journal, ledger, trail balance, trading and profit and loss account, and balance sheet. Students will also get hands on exposure in make or buy decisions in cost accounting. Same concepts will be made concrete through assignments, case studies, and unit tests.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Introduction to Accounting | 05 |
| 2 | Book Keeping System | 05 |
| 3 | Financial Accounting - I | 05 |
| 4 | Financial Accounting - II | 07 |
| 5 | Cost Accounting - I | 04 |
| 6 | Cost Accounting - II | 04 |

**Total Hours (Theory): 30**
**Total Contact Hours:30**

**Detail Syllabus:**

**Unit I : Introduction to Accounting** **Hours 05**

Introduction and users of Accounting, Branches of Accounting, Objectives, Accounting Concepts and Conventions, Accounting Terms

**Unit II : Book Keeping System** **Hours 05**

Single and Double Entry Book Keeping Systems, Advantages of Double Entry Book Keeping System, Limitations of Double Entry Book Keeping System, Types of Accounts: Personal, Real, and Nominal, Terms of Book Keeping Systems

**Unit III : Financial Accounting – I** **Hours 05**

Stages of Accounting Cycle, Types of Transactions, Forms of Economic Transactions, Cash and Credit Transactions, Introduction to Journal, Journal Entries

**Unit IV : Financial Accounting – II** **Hours 07**

Introduction to Ledger, Relationship Between Journal and Ledger, Rules of Ledger Posting, Balancing of the Accounts in Ledger, Trial Balance, Objectives of Preparing Trial Balance, Trading and Profit and Loss Account, Balance Sheet, Different between Trial Balance and Balance Sheet

**Unit V : Cost Accounting – I** **Hours 04**

Introduction to Cost Accounting: Cost, Costing, and Cost Accounting, Limitations of Financial Accounting, Difference Between Cost Accounting and Financial Accounting, Objectives of Cost Accounting,

**Unit VI : Cost Accounting – II** **Hours 04**

Elements of Cost: Material Cost, Labor Cost, and Expenses, Techniques of Costing, Marginal Costing, Make or Buy Decision, Breakeven Analysis

**Core Books:**

1. Dr. S. N. Maheshwari: Principles of Management Accounting, Sultan Chand Publication. Eighteenth Edition, 2021
2. M. Pande: Financial Management, Vikas Publishing House, 2015
3. M.N. Arora: Cost Accounting, Vikas Publishing House, 2013

**Reference Books:**

1. Dr. S.N. Maheshwari: Advanced Accountancy, Volume 1, Eleventh & Revised Edition, Vikas Publishing House, 2017

**Web References:**

1. https://www.accountingtools.com/articles/basics-of-accounting.html [For Basics of Accounting]
2. https://www.youtube.com/watch?v=vuetn_PQOvM [For Journal Entries – 1]
3. https://www.youtube.com/watch?v=3xCzh3-bm4o&list=RDCMUCNh1egMomGI3hjJoDHExdUg&index=3 [For Journal Entries – 2]
4. https://www.youtube.com/watch?v=z_KO49Pk3DM [For Ledger Posting & Trial Balance]
5. https://www.youtube.com/watch?v=Y4azRCTTWoU [For Trading A/c & Profit & Loss A/c]

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Students will get basic understanding of the field of Accounting |
| CO2 : | Students will be familiar with Accounting Book Keeping Systems |
| CO3 : | Students will learn concepts of financial accounting and different types of accounts with rules of debit and credit |
| CO4 : | Students will be able to generate Journal, Ledger, Trial Balance, and Final Accounts |
| CO5 : | Students will get basic understanding of the field of Cost Accounting |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Introduction to Accounting | √ | | | | |
| 2 | Book Keeping System | | √ | | | |
| 3 | Financial Accounting – I | | | √ | | |
| 4 | Financial Accounting – II | | | | √ | |
| 5 | Cost Accounting – I | | | | | √ |
| 6 | Cost Accounting – II | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 1 | 1 | - | - | 2 | 2 | 3 | 1 | 2 | 1 | 1 | 1 | 1 |
| CO2 | 1 | 2 | - | - | 2 | 2 | 3 | 1 | 2 | 1 | 1 | - | - |
| CO3 | 3 | 3 | 2 | 1 | 1 | 2 | 3 | - | 2 | 2 | 1 | 2 | - |
| CO4 | 3 | 3 | 3 | 1 | 1 | 2 | 3 | - | 2 | 2 | 1 | 2 | - |
| CO5 | 2 | 3 | 2 | 1 | 3 | 2 | 3 | 1 | 2 | 3 | 1 | 2 | 1 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUA101 - Presentation Skills

| Credit | Component | Instruction / Contact Hours (Per Week) | Instruction / Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 02 | Practical | 02 | 30 | 20 | 50 |
| **Experiential Learning Components:** Lab Assignments, Practical Test, VIVA | | | | | |

**Prerequisites:** Basic computer skills and knowledge of English.

**Methodology, Pedagogy & Andragogy:** The aim of this course is to develop effective presentation skills in students. Students will be taken systematically through the key stages of giving presentations, from planning and introducing to concluding and handling questions. The topics for the presentation will range from academic presentations to research presentations.

**B. Outline of the Course:**

| Week No | Practical | Description |
|---|---|---|
| 1 | Introduction | Knowing the Purpose, Knowing the Audience, Internal and External Presentation, Presentation to Heterogenic Group, Greetings, Introducing self and peers, Asking and sharing information |
| 2 | ICT tools for presentation – Brief introduction | Microsoft PowerPoint, Canva, Prezi, AI-powered presentation tools |
| 3 | Presentation using Canva | Introduction to Canva, Creating, presenting, sharing and downloading presentation |
| 4 | Presentation design in Canva | Templates, layouts, Presenter Notes, Duration, Timer |
| 5 | Basic design elements in Canva | Shapes, Graphics, Charts, Tables |
| 6 | Advanced design elements in Canva | Photos, Videos, Audio, Frames, Grids, Stickers |

| | | |
|---|---|---|
| **7** | Intermediate features of PowerPoint | Using Screenshot, Inserting objects, video and audio in presentation, Screen recording |
| **8** | Advanced features of PowerPoint | Using Hyperlink and Action, Slide Show Set-up, Presenter View, Master views |
| **9** | Cloud-based presentation using Google Slides | Creating a new presentation, importing slides, Sharing, Printing, Downloading in different formats, Version History |
| **10** | Basic features of Google Slides | Slideshow, Motion, Theme Builder, Inserting image, audio, video, shape |
| **11** | Advanced features of Google Slides | Spell-check, Linked objects, Q-A History, Notification Settings, Accessibility, Activity Dashboard |
| **12** | Engaging audience using feedback-gathering tools | Use of menitimeter for Word Clouds, Live Polls, Scales, Ranking and Pin It |
| **13** | Ethical practices | Using Citations/References for texts, images and other type of information, Plagiarism check |
| **14** | AI and Presentation | Use of Generative AI tools such as ChatGPT, CoPilot etc. in presentations |
| **15** | Select case-study | Examples of best presentations. |

**Total hours: 30**

**Core Books:**

1. The Presentation Book : How to Create It, Shape It and Deliver It! Improve Your Presentation Skills Now, 2nd Edition, Pearson Education.
2. Beginners Guide for Canva 2024, by Karla J Kane. ISBN: 979-8875801853

**Reference Books:**

1. Microsoft PowerPoint Guide for Success by Kevin Pitch, Top Notch International. ISBN:978-1915331489
2. Getting Started with Google Slides: A Practical Guide to Cloud-Based Presentations by Scott La Counte, SL Editions ISBN:978-1629179513

**Web References:**

1. https://www.canva.com/education/students/
2. https://support.microsoft.com/en-us/powerpoint
3. https://support.google.com/a/users/answer/9282488
4. https://www.mentimeter.com/features

**Course Outcome:** Upon successful completion of the course, student will be able to:

| | |
|---|---|
| CO1 | Understand key elements of an effective presentation and ICT tools for presentation. |
| CO2 | Acquire in-depth knowledge of one of the popular presentation tools. |
| CO3 | Develop intermediate and advanced skills in alternative presentation tool. |
| CO4 | Familiarize with cloud-based presentation tool. |
| CO5 | Create effective presentation with ethical practices and enhance it with AI tools. |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | 2 | 1 | 3 | - | 2 | 1 | - | - | 2 | - | - |
| CO2 | 2 | - | 3 | - | 2 | 1 | - | - | 2 | 2 | 3 | 1 | 1 |
| CO3 | - | 3 | - | 2 | - | 1 | 3 | 2 | - | - | - | 3 | 2 |
| CO4 | 2 | 1 | 2 | 2 | 2 | 1 | 2 | - | 2 | - | 2 | 3 | 2 |
| CO5 | - | 2 | 2 | 3 | 1 | - | 2 | 2 | 2 | 2 | - | 2 | 3 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

# CTUI101 - Summer Internship and Viva – I

## Objective and Scope:

Students who choose to exit the programme after the completion of the first two semesters must undertake internships, community engagement programmes and field based learning/minor projects during the summer term. This internship, worth 04 credits, is a requirement for the award of the Undergraduate Certificate. Students have to undergo internship in reputed Tech Companies, Software Development Firm, IT services companies, Research Institutions and Lab etc. This provision ensures that students gain valuable practical experience and industry exposure, complementing their academic learning and enhancing their career prospects. Following are the intended objectives of internship training:

- **Practical Experience**: Provide hands-on experience with real-world projects to apply theoretical knowledge gained during the course.
- **Skill Development**: Enhance technical skills, such as programming, software development, database management, and other relevant IT skills.
- **Industry Exposure**: Familiarize students with the working environment of the IT industry, including understanding company culture, workflows, and professional practices.
- **Professionalism**: Instil a sense of professionalism and work ethic, including adhering to deadlines, following instructions, and working collaboratively.
- **Transition to Employment**: Facilitate a smoother transition from academic life to professional careers by bridging the gap between theory and practice.

## Guidelines of Internship:

The Summer Internship shall be of 15 (Minimum 60 hours) duration and will be undertaken during the Summer Vacation. Students have to undergo internship in reputed Tech Companies, Software Development Firm, IT services companies, Research Institutions and Lab etc. It is mandatory for the students to seek written approval from the coordinator about the topic and the organization before commencing the Summer Internship. During Summer Internship students are expected to take necessary guidance from the faculty guide allotted by the Institute. To do it effectively they should be in touch with their guide through e-mail or phone. Students must maintain regular reports detailing the work completed, challenges faced, and learning experiences. Students must submit a final report and presentation summarizing the internship experience, project outcomes, and key learnings to guide. The technical specifications for report preparation is as under:

**Technical details:**

1. The report shall be printed on A-4 size white paper.
2. 12 pt. Times New Roman font shall be used with 1.5-line spacing for typing the report.
3. 1" margin shall be left from all the sides.
4. Considering the environmental issues, students are encouraged to print on both sides of the paper.
5. The report shall be spiral bound as per the standard format of the cover page given by the Institute.
6. The report should include a Certificate (on company's letter head) from the company duly signed by the competent authority with the stamp.
7. The report shall be signed by the respective guide(s) & the Principal of the Institute 10 (Ten) days before the viva-voce examinations.
8. Student should prepare two hard bound copies of the Summer Internship Project Report and submit one copy in the institute. The other copy of the report is to be kept by the student for their record and future references.

## Evaluation:

The evaluation of Internship will be done as per the following criteria:

| Sr. No. | Evaluation Criteria | Marks |
|---|---|---|
| 1 | Summer Internship Report | 50 |
| 2 | Presentation and Viva-voce examination | 50 |
| **TOTAL MARKS** | | **100** |

Students must secure 36% passing marks in both components to qualify for the Certificate.

## Internship and Exit:

Any student intending to exit the course must complete the summer internship (Including scoring at least 36% marks in the evaluation) before exit. A student intending to exit shall have to submit an application to Principal through Counsellor before even semester university theory examination. However, any such student desiring to withdraw the exit option may do that before the last paper of theoretical examinations of even semester.

# TEACHING SCHEME & DETAILED SYLLABUS

# FOR

# BSc-IT PROGRAMME
# (3rd SEMESTER)
# AS PER NEP 2020

# EFFECTIVE FROM ACADEMIC YEAR 2025-26

## BSc-IT Semester-III

| Course Code | Course Name | Theory / Practical / Others | Marks (Theory) | | Marks (Practical | | Credit | NEP -2020 Classification |
|---|---|---|---|---|---|---|---|---|
| | | | CCE | SEE | CCE | SEE | | |
| **CTUC201** | Introduction to Data Structures | Theory and Practical | 50 | 50 | 25 | 25 | 06 (04-Theory) (02-Practical) | Major Core |
| **CTUC202** | System Analysis and Design | Theory | 50 | 50 | - | - | 04 | Major Core |
| **CTUC203** | Programming the Internet | Practical | - | - | 25 | 25 | 02 | Major Core |
| **CTUS201**<br>**CTUS202** | (Select any one)<br>3. Computer Oriented Numerical Methods<br>4. Computer Oriented Management System | Theory | 50 | 50 | - | - | 04 | SEC |
| **CTUA201** | Life Management | Theory | 25 | 25 | - | - | 02 | AEC |
| | University Elective-I | Practical | - | - | 50 | 50 | 02 | MDC |
| **HSUV201** | Creativity, Problem Solving and Innovation | Practical | - | - | 25 | 25 | 02 | VAC |

*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)
*SEE- Semester –End- Evaluation

**University Elective - I**

| Sr. No. | Course Code | Course Name | Department/Faculty |
|---|---|---|---|
| 1 | MEUE201 | Engineering Graphics and Design | Engineering |
| 2 | EEUE201 | Fundamentals Of Electrical Engineering | Engineering |
| 3 | CLUE201 | Environment and Development | Engineering |
| 4 | ECUE201 | Scientific Computing Using Matlab | Engineering |
| 5 | CEUE201 | The Joy of Computing using Python | Engineering |
| 6 | ITUE201 | Introduction to Quauntam Computing | Engineering |
| 7 | CSUE201 | Python For Data Science | Engineering |
| 8 | AIUE201 | An Introduction to Artificial Intelligence | Engineering |
| 9 | BMUD201 | Money And Banking | Management |
| 10 | CAUD203 | Human Computer Interactions | Computer Science |
| 11 | PTUD191 | Basics of Health Promotion and Education Intervention | Physiotherapy |
| 12 | NRMD251 | First Aid Masterclass- A complete guide to first aid | Nursing |

# CTUC201 - Introduction to Data Structures

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| **04** | **Theory** | **04** | **60** | **50** | **110** |
| **02** | **Practical** | **04** | **60** | **40** | **100** |
| **Experiential Learning Components (Theory) :** Unit Tests, Assignments, Case Study etc.<br>**Experiential Learning Components (Practical) :** Lab Assignments, Practical Test, VIVA | | | | | |

**Pre-requisite:** Foundation of C Programming, Fundamentals of Object Oriented Programming

**Methodology, Pedagogy & Andragogy:** Theory lectures focus on fundamental concepts of data structures and algorithms, emphasizing the selection of the most efficient structures for various applications. Andragogical principles are integrated to promote self-directed and practical learning for learners. For practical teaching of Data Structures and Algorithms, the approach combines hands-on programming exercises with real-world case studies to reinforce the selection of efficient structures for various applications. This blended method enhances both theoretical understanding and practical skills, ensuring learners can effectively apply concepts in real-world scenarios.

**Outline of the course (Theory):**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| **1.** | Principles of Data Structures and Algorithms | 06 |
| **2.** | Arrays and Sequential Data Structures | 10 |
| **3.** | Advanced Dynamic Data Structures | 12 |
| **4.** | Tree Structures and Their Applications | 11 |
| **5.** | Graph Theory and Utilizations | 11 |
| **6.** | Sorting and Searching | 10 |

**Total Hours (Theory): 60**
**Total hours (Practical) : 60**
**Total Contact Hours : 120**

**Detail Syllabus:**

**Unit I: Principles of Data Structures and Algorithms Hours 06**

Atomic and composite data, Data type, Data object, Data Structure, Abstract Data type, Types of Data Structures, Introduction to Algorithms, Relationship among Data, Data Structures and Algorithms, Analysis of Algorithms, space and time complexity algorithm

**Unit II: Arrays and Sequential Data Structures Hours 10**

Linear Data Structures using sequential organization, Representation of Stacks using sequential organization, Applications of Stack, Application of Recursion, Concepts of Queues, Realization of Queues using sequential organization, circular queue, Multiqueue, Deque, Priority Queue, Applications of Queue.

**Unit III: Advanced Dynamic Data Structures Hours 12**

Linked list, Comparison of sequential and linked organizations, Dynamic Memory Management, types of link list , Applications of linked list, operation on link list.

**Unit IV: Tree Structures and Their Applications Hours 11**

Basic Terminologies, Definition and concepts, Representation of Binary Tree, Operations on Binary Tree and algorithms, Types of Binary Trees , BTree, B+Tree, ,Red and Black Tree, AVL trees,

**Unit V: Graph Theory and Utilizations Hours 11**

Graph Terminology, Representation of Graph, Operations on Graph, Applications of Graph Structure, minimum spanning tree.

**Unit VI: Sorting and Searching Hours 10**

Sorting Notations and Concepts, Sorting Techniques, Sequential Searching, Binary Searching, Search Trees.

**Outline of the Course (Practical):**

| Week No | Practical | Description |
|---|---|---|
| 1 | Basic of C and C++ | foundational concepts in C and C++. |
| 2 | Two dimension array | Understand and utilize arrays effectively. |
| 3 | Stack | Implement stack operations including push, pop, and peek. |
| 4 | Stack and its application | Cover infix, prefix, and postfix notations. |
| 5 | Queue and Circular queue | Implement queue operations such as insert, delete, and display. |
| 6 | Linear and Binary Search | Apply various searching techniques on arrays. |
| 7 | Merge sort, Bubble Sort | Implement and compare different sorting techniques to sort data. |
| 8 | Selection sort , Insertion Sort, | |
| 9 | Quick sort | |
| 10 | Single Link List | Perform insert and delete operations at the beginning, end, and middle of link list data structures. |
| 11 | | |
| 12 | Double Link List | |
| 13 | | |
| 14 | Circular Single Link List | |
| 15 | Circular Double Link List | |

**Total hours: 60**

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## Core Books:

1. Jean-Paul Tremblay, Paul G. Sorenson: An introduction to data structures with applications, 2nd Edition, Tata McGraw Hill Publications, 1991.
2. D. Samanta: Classic Data Structures, 2 Edition, PHI Publications, 2002.

**Reference Books:**


1. Varsha h. Patil : Data Structures Using C++, 1st Edition, Oxford University Press, 2012.
2. Yashvant Kanethkar : Data structures through C++, 2nd Edition, BPB Publications,2003.
3. Mark Allen Weiss: Data Structures and Algorithm Analysis in C++, 3rd Edition, Pearson Education, 2009.
4. Reema Thareja: Data Structures using C, 1st Edition, Oxford University Press, 2012.


**Web References:**

1. https://www.cs.auckland.ac.nz/~jmor159/PLDS210/ds_ToC.html [For materials of Data Structures and Algorithms]
2. http://www.cs.usfca.edu/~galles/visualization/ [For Data Structures Visualization]
3. https://www.cs.auckland.ac.nz/~jmor159/PLDS210/mst.html[For Graph Data Structures ]
4. http://interactivepython.org/runestone/static/pythonds/Trees/trees.html [For Tree Data Structures]
5. https://www.cs.auckland.ac.nz/~jmor159/PLDS210/niemann/s_man.pdf [For Sorting and
    i. Searching Cookbook]
6. http://www.cs.princeton.edu/~rs/AlgsDS07/10Hashing.pdf [ For Hashing Materials]
7. http://nptel.ac.in/video.php?subjectId=106102064 [ For Data Structures videos]

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Comprehend and convey the essential concepts and importance of data structures and algorithms. |
| CO2 : | Analyze and apply static and dynamic linear data structures, such as stack, queue , array and linked lists. |
| CO3 : | Construct and manipulate various tree data structures, including binary trees and AVL trees. |
| CO4 : | Use graph theory concepts to solve complex problems and understand real-world applications. |
| CO5 : | Implement and compare various sorting and searching algorithms for optimal performance. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Principles of Data Structures and Algorithms | √ | | | | |
| 2 | Arrays and Sequential Data Structures | | √ | | | |
| 3 | Advanced Dynamic Data Structures | | √ | | | |
| 4 | Tree Structures and Their Applications | | | √ | | |
| 5 | Graph Theory and Utilizations | | | | √ | |
| 6 | Sorting and Searching | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | 3 | - | 1 | 3 | 1 | - | - | 1 | 1 | 2 | 1 |
| CO2 | 3 | 2 | 3 | 1 | 1 | 3 | 1 | - | - | 1 | 2 | 2 | 1 |
| CO3 | 3 | 2 | 3 | 1 | 1 | 3 | 1 | - | - | 1 | 2 | 2 | 1 |
| CO4 | 3 | 2 | 3 | 1 | 1 | 3 | 1 | - | - | 1 | 2 | 2 | 1 |
| CO5 | 3 | 2 | 3 | 1 | 1 | 3 | 1 | | | 1 | 3 | 2 | 1 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUC202 - System Analysis and Design

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| **Experiential Learning Components :** Unit Tests, Assignments, Case Study etc. | | | | | |

**Pre-requisite:** None.

**Methodology & Pedagogy:** During theory lectures emphasis will be given on understanding of analysing business needs, perform planning and analysis of business system, give examples of how Information Systems are designed and constructed and how major modifications are made to the existing systems.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Basics of Information Systems | 08 |
| 2 | Information Systems Analysis Overview | 11 |
| 3 | Information Gathering and System Requirement specification | 11 |
| 4 | Data flow diagrams and Process Specification | 11 |
| 5 | Overview to Data input and designing output methods | 10 |
| 6 | Implementation and maintenance of information systems | 09 |

**Total Hours (Theory): 60**

**Total Contact Hours:60**

**Detailed Syllabus:**

**Unit - I: Basics of Information Systems** **Hours:08**

Types of Information, Understand the concept of System, Need for a Computer-Based Information
System, Management Structure, Management and Information Requirements, Qualities of Information, Examples of Information Systems, Various Functions in Organizations, Varieties of Information Systems.

**Unit - II: Information Systems Analysis Overview** **Hours: 11**

Overview of Design of an Information System, System Development Life Cycle (SDLC), The Role and Tasks of Systems Analyst, Attributes of a Systems Analyst, Tools Used by Systems Analyst, Approaches to Systems Development.

**Unit III: Information Gathering and System Requirement specification** **Hours: 11**

Strategy to Gather Information, Information Sources, Methods of Searching for Information,
Interviewing Technique, Questionnaires, Other Methods of Information Search, System Requirements Specification, Data Requirements, steps in Systems Analysis, Modularizing Requirements Specifications, deciding on Project Goals, Examining Alternative Solutions, Evaluating Proposed Solution, Cost-benefit Analysis, Payback Period, Feasibility Report, System Proposal.

**Unit IV: Data flow diagrams and Process Specification** **Hours: 11**

Symbols Used in DFDs, describing a System with a DFD, Levelling of DFDs, Levelling Rules,
Logical and Physical DFDs, Case Tools to Draw DFD, Process Specification Methods, Structured English, Decision Table Terminology and Development, Extended Entry Decision Tables, Eliminating Redundant Specifications

**Unit V: Overview to Data input and designing output methods** **Hours: 10**

Data Input, Coding techniques, validating input data, interactive data input, objective of output designing, design of screens and output reports.

**Unit VI: Implementation and maintenance of information systems** **Hours: 09**

Control in Information Systems, The Objectives of Control, Control Techniques, Audit of Information Systems, Security of Information Systems, designing reliable and maintainable systems, design of software, software design and documentation tools, managing quality assurance, user training, training methods and conversion.

**Core Books:**


1. James A Senn: Analysis and Design of Information System, McGraw Hill International, 2003.
2. V. Rajaraman : Analysis and Design of Information Systems, Prentice-Hall of India Private Limited, 2003.
3. Kendall and Kendall: Systems analysis and Design, 5th Edition, Prentice-Hall of India Private Limited, 2003.


**Reference Books:**


1. Jeffrey L. Whitten, Lonnie D. Bentely and Kevin C. Dittman : Systems Analysis and Design Methods, Tata McGraw Hill Publishing Co. Ltd., 2001.
2. Tuthill and Leavy : Knowledge Based Systems : Mangers Perspectives : Tab professional and Reference Books, 1991.


**Web references:**


1. http://lecture-notes-forstudents.blogspot.in/2010/04/system-analysis-anddesignsad.html
2. http://www.slideshare.net/aroravinay/1-introduction-to-ado-infosys
3. http://bcastuff.blogspot.in/p/sad-notes.html#.U6uW-ZzWIeo
4. http://www.eis.mdx.ac.uk/staffpages/geetha/bis2030/DFD.html
5. http://faculty.washington.edu/ytan/is460/notes/LN11.pdf


**Course Outcomes:** Upon successful completion of the course, students will:

| | |
|---|---|
| CO1 | Understand basic concepts of system, types of information, qualities of information and its flow with respect to management structure |
| CO2 | Gain knowledge about how the system will develop and guide by well-experienced person in the organization with its documentation. |
| CO3 | Learn about various tools used to design flows of data, processes and data stores, performed by users of the system in graphical way. |
| CO4 | Understand various designing methods used to solve a problem while taking input and producing output as per given specification. |
| CO5 | Gain knowledge about various concepts of system maintenance and conversions |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | Co1 | Co2 | Co3 | Co4 | Co5 |
| 1 | Basics of Information systems | ✓ | | | | |
| 2 | Information system analysis Overview | | ✓ | | | |
| 3 | Information gathering and System Requirement Specification | | ✓ | | | |
| 4 | Data Flow Diagram and Process specifications | | | ✓ | | |
| 5 | Overview to data input and design output methods | | | | ✓ | |
| 6 | Implementation and maintenance of information systems | | | | | ✓ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 3 | 2 | 2 | 2 | 3 | 3 | 2 | 1 | 3 | 2 | 3 | 2 |
| CO2 | 2 | 2 | 3 | 2 | 3 | 3 | 2 | 2 | 3 | 2 | 3 | 3 | 2 |
| CO3 | 2 | 2 | 3 | 3 | 1 | 2 | 2 | 1 | 2 | 1 | 1 | 3 | 1 |
| CO4 | 2 | 2 | 3 | 1 | 2 | 2 | 1 | 1 | 1 | 1 | 1 | 3 | 2 |
| CO5 | 2 | 2 | 2 | 2 | 1 | 2 | 2 | 1 | 1 | 1 | 1 | 2 | 1 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

# CTUC203 - Programming the Internet

| Credit | Component | Instruction/Contact Hours( Per Week) | Instruction/Contact Hours( Per Course) | Experiential Learning Hours( Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components :** Lab Assignments, Practical Test | | | | | |

**Prerequisites:** Basic concepts of web.

**Methodology, Pedagogy & Andragogy:** This course focuses on providing hands-on experience to students for design and develop dynamic and interactive web sites using HTML5, CSS3 and JavaScript.

### C. Outline of the Course:

| Week No | Practical | Description |
|---|---|---|
| 1 | Basic Construction of an HTML & HTML5 Page, HTML Text Editors, HTML Building blocks. Adding Content: Basic HTML tags – Text formatting tags, heading tags, paragraph tags, division tag, Span tags, Break line tag, Binding space in HTML. | Student can learn basic concepts of web page design. Students get familiar with different tags and attributes which is used to add the text in a web page. |
| 2 | Controlling Font size and color, Adding Table – Table tag, grouping and merging table rows and columns. Adding List – ordered, unordered and definition list. | Student can able to change the font styles and color in more attractive way.<br>Also, they can able to display the information in tabular & List format. |
| 3 | Adding Image, Adding links - Changing link colors, Email link, Embedding audio and video, Adding Field sets. | Student can add multimedia like image, audio, video in a web page. Student can explore various internal and external linking methods. |
| 4 | Adding Form: Form tag, Action, Method. Basic form controls – Textbox, Password field, Radio Button,Checkbox, Multiple line text area, file upload, Various types of | Student can design the various form elements for interaction with the web page users. |

| | | |
|---|---|---|
| | buttons – submit, reset, image, simple button. | |
| **5** | HTML5 input elements - Color, Number, Date, Month,week, Time, datetime-local, Email, url, range, search,<datalist>, <output>, <progress>, <meter> | Student can apply HTML5 input element to make form more attractive. |
| **6** | Regular expression, HTML5 new attributes –Placeholder, Required, Pattern, Autocomplete,Autofocus, novalidate, formnovalidate, formaction, formmethod, spellcheck, contenteditable. | Student can explore the basic HTML5 new attributes and also apply validation on form submission. |
| **7** | CSS Styling the web - Applying CSS Building blocks –CSS Selectors, Cascade & inheritance, CSS Properties, CSS Box model. | Student can apply various types of CSS with its basic properties. |
| **8** | Creating Layout - Header, footer, navigation menu, and sub-navigation. Floats, Flexbox, Grids, Positioning,<br><br>Multiple column layout. | Student can create different layout using semantic elements of HTML5 & CSS classification properties. |
| **9** | Creating Image Gallery and Animation - 2D/3D Transformation, Transitions and Animation, CSS Media query | Student can create layout image gallery & animation. |
| **10** | JavaScript – Adding Internal & External JavaScript,Working with variables, datatype – Number, String,Boolean. | Student can write script for basic programs using condition, Loop. |
| **11** | datatype –Array, Object, Operators, Conditional statements, Looping statements | Student can write script for basic programs using array and object. |
| **12** | Adding Dialog boxes in HTML page – Alert, Confirm and Prompt. Creating user defined function. | Student can able to use in-build and user defined functions of JavaScript |
| **13** | Adding Mouse and Keyboard Event in HTML page, Manipulating String and Date. | Students can make user interaction using various Events |
| **14** | JavaScript Document Object Model hierarchy – Create find and manipulate HTML Element using Objects and methods | Student can explore the DOM Hierarchy. |
| **15** | Form validation, Applying Style using | Student can apply various objects to make web page dynamic with user interactions. |

| | | |
|---|---|---|
| | JavaScript. Creating New window, Accessing & manipulating History of HTML Pages. | |

**Total hours: 60**

**Core Books:**

1. Matthew MacDonald: HTML5 The Missing Manual, O'Reilly Media, August 2011.
2. Peter Gasston: The Book of CSS3, A Developer's Guide to the Future of Web Design, No Starch Press, April 2011.
3. Richard York: Beginning CSS Cascading Style sheets for Web Design, Wrox Press (Wiley Publishing), 2005.

**Reference Books:**

1. Ivan Bayros : Web Enabled Commercial Application Development using HTML, JavaScript, DHTML and PHP, fourth revised edition , BPB Publication.
2. David Mc Farland : CSS The Missing Manual, O'Reilly, 2006.
3. Julie C. Meloni : HTML, CSS and JavaScript All in One, Pearson.

**Web References:**

1. http://www.tutorialspoint.com/html5 [For notes on HTML5 tags]
2. http://www.w3schools.com/html/ [For HTML5, CSS and JavaScript notes and examples]
3. https://in.godaddy.com/help/dreamweaver-cs6-publish-your-website-7811[Publish your website using Dreamweaver]
4. [http://fullbooksfreedownload.blogspot.in/2016/02/html-css-javascript-web-publishing-in.html](http://fullbooksfreedownload.blogspot.in/2016/02/html-css-javascript-web-publishing-in.html) [Book :- HTML, CSS & JavaScript Web Publishing in One Hour a Day, Sams Teach Yourself, 7$^{th}$ Edition PDF]

**Course Outcome:** Upon successful completion of the course, student will:

| | |
|---|---|
| CO1: | Able to understand basic construction of HTML & HTML5 document and started using basic tags. |
| CO2: | Able to understand the implementation of HTML tags and able to create HTML static Web pages. |
| CO3: | Able to use basic properties of CSS and CSS3 in html file and Able to create HTML, CSS layout. |
| CO4: | Able to write client side script – JavaScript to perform basic program. |
| CO5: | Able to use DOM objects for dynamic web pages. |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | 2 | 1 | - | 2 | - | - | - | 2 | - | 1 | 1 |
| CO2 | 3 | 2 | 2 | 1 | - | 2 | - | - | - | 2 | - | 1 | 1 |
| CO3 | 3 | 3 | 3 | 1 | - | 2 | - | - | - | 2 | - | 2 | 1 |
| CO4 | 3 | 3 | 2 | 1 | - | 2 | - | - | - | 2 | - | 2 | 1 |
| CO5 | 3 | 3 | 2 | 1 | - | 2 | - | - | - | 2 | - | 2 | 1 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUS201 - Computer Oriented Numerical Methods

| Credit | Component | Instruction/Contact Hours( Per Week) | Instruction/ Contact Hours( Per Course) | Experiential Learning Hours( Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| **Experiential Learning Components :** Unit Tests, Assignments, Case Study etc. | | | | | |

**Pre-requisite:** None

**Methodology, Pedagogy & Andragogy:** The course covers key topics like finding roots of equations, solving linear equations, interpolation, and solving differential equations. Teaching includes lectures, interactive problem-solving, and project-based assessments with continuous feedback. The focus is on practical applications and real-world problems, using flexible and engaging teaching methods. The aim is to make the content relevant and useful for students' careers and interests.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Introduction to Numerical Methods | 08 |
| 2 | Interpolation | 09 |
| 3 | Numerical Integration and Differentiation | 10 |
| 4 | Solution of Linear Equation | 12 |
| 5 | Solution of Differential Equations | 11 |
| 6 | Solution of Partial Differential Equations | 10 |

**Total Hours (Theory): 60**
**Total Contact Hours: 60**

**Detailed Syllabus:**

**Unit - I: Numerical Methods** **Hours: 08**

Introduction to Numerical Methods, Various methods such as Bisection, False-Position and Newton Raphson methods, Graffes Root Squaring Method, Convergence of Solution

**Unit - II: Interpolation** **Hours:09**

What is an interpolation, Difference tables, Newton forward and backward Interpolation formula, Lagrange’s formula, Newton's Divided Difference Formula

**Unit - III: Numerical Integration and Differentiation** **Hours:10**

Numerical Integration, Trapezoidal rule, Differentiating Continuous function (Two point and Three-point formula), Tabulated function and its maxima and minima, Newton- Cote’s Formula.

**Unit - IV: Solution of Linear Equation** **Hours:12**

Solution of Linear Equation, Gauss-Jordan Elimination methods, Gauss - Seidal and Gauss Jacobi Interactive methods

**Unit - V: D Solution of Ordinary Differential Equations** **Hours:11**

Solution of Ordinary Differential Equations, Taylor Series and Euler Methods, Runge-Kutta Methods.

**Unit - VI: Solution of Partial Differential Equations:** **Hours:10**

Review and examples of partial differential equations, classification of partial differential equations, Difference equation, Laplace’s equation and Poisson’s equation.

**Core Books:**

1. D. N. Datta: Computer Oriented Numerical Methods, Vikas Pub. House, 2012.

2. S.S. Sastry : Introductory methods of Numerical Analysis, Prentice Hall of India,4th edition 2006

**Reference Books:**

1. V. Rajaraman Computer Oriented Numerical Methods, 3rd Edition, Prentice-Hall of India Pvt.Ltd, 3 rd edition, 2016

2. M. Ray, Har Swarup Sharma, Sanjay Chaudhary: Mathematical Statistics, Ram Prasad and sons, 11th edition

3. B S Grewal: Numerical Methods In Engineering & Science (With Programs In C, C++ And MATLAB), Khanna Publishers, 10th edition, 2014

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Able to use Numerical methods in computer technology |
| CO2 : | To learn Interpolation |
| CO3 : | To Learn the concepts of Numerical Integration and Numerical Differentiation |
| CO4 : | To understand linear equations |
| CO5 : | To learn techniques to solve ordinary differential and partial differential equations |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Introduction to Numerical Methods | √ | | | | |
| 2 | Interpolation | | √ | | | |
| 3 | Numerical Integration and Differentiation | | | √ | | |
| 4 | Solution of Linear Equation | | | | √ | |
| 5 | Solution of Differential Equations | | | | | √ |
| 6 | Solution of Partial Differential Equations | | | | | √ |

## Course Articulation Matrix:

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO 2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 3 | 2 | 1 | - | 2 | - | - | - | 2 | - | 2 | 1 |
| CO 2 | 3 | 3 | 2 | 1 | - | 2 | - | - | - | 2 | - | 2 | 1 |
| CO3 | 3 | 3 | 2 | 1 | - | 2 | - | - | - | 2 | - | 2 | 1 |
| CO 4 | 3 | 3 | 2 | 1 | - | 2 | - | - | - | 2 | - | 2 | 1 |
| CO5 | 3 | 3 | 2 | 1 | - | 2 | - | - | - | 2 | - | 2 | 1 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

# CTUS202 - Computer Oriented Management System

| Credit | Component | Instruction/Contact Hours( Per Week) | Instruction/ Contact Hours( Per Course) | Experiential Learning Hours( Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| **Experiential Learning Components :** Unit Tests, Assignments, Case Study etc. | | | | | |

**Pre-requisite:** None

**Methodology, Pedagogy & Andragogy:** The theory sessions will emphasize real-time systems. To enhance comprehension, we will provide illustrations using actual data across various systems. The sessions will meticulously explain the system's functionality through detailed flow descriptions. To bridge the gap between theoretical knowledge and practical application, we have to schedule industrial visits. These visits are designed to offer students a tangible correlation between their studies and real-world practices. Students should be assigned case studies to facilitate the practical application of their acquired knowledge. These case studies aim to deepen understanding and foster the ability to apply theoretical concepts to real-world scenario.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Basics of Information System | 09 |
| 2 | Architecture and Design of Information System | 08 |
| 3 | Transaction Processing System | 09 |
| 4 | Management Information System | 06 |
| 5 | Decision Support System & Executive Support System | 18 |
| 6 | Introduction to ERP and trending technologies | 10 |

**Total Hours (Theory): 60**
**Total Contact Hours:60**

**Detail Syllabus:**

**Unit I : Basics of Information System** **Hours 09**

Role of data and information, Types of Systems, Types of information, Quality of Information, what is an Information System, Functions of Information System, classification of an Information System, Major Types of Information Systems, Interrelationship of Information Systems.

**Unit II : Architecture and Design of Information System** **Hours 08**

Meaning of Architecture, development and maintenance of information system, centralized and decentralized systems, factors of success and failure in information system and risk involved with information system.

**Unit III : Transaction Processing System** **Hours 09**

Introduction of Transaction Processing System (TPS), Transaction Processing System (TPS) Architecture, Transaction Processing System (TPS) Life cycle, Methods of Transaction Processing System (TPS), Information System Availability Control, Illustrations and live demonstration of TPS such as Banking System, Railway reservation system etc.

**Unit IV : Management Information System** **Hours 06**

Introduction to Management: Approaches of Management, Function of Management, System from Functional Perspective, Reports of Management Information System, A business perspective of MIS, Dimensions of MIS, contemporary Approaches to Information System.

**Unit V : Decision Support System and Executive Support System** **Hours 18**

Business value of Improved Decision making, Types of Decision, Decision making Process, The difference between MIS and DSS, Components of DSS, System for Decision Support Group Decision Support System, Business value of GDSS. Important Dimensions of knowledge, Organizational learning and Knowledge Management, The Knowledge Management value chain, Overview of different types of Knowledge Management Systems, Characteristics of ESS, The Role of ESS in the Firm, Business value of ESS, Monitoring Corporate Performance, Working Examples of ESS.

**Unit VI : Introduction to ERP and trending technologies Hours 10**

An overview of ERP, Basic ERP Concepts, Risk and Benefits of ERP, ERP and related technologies, Business Intelligence, Some case studies on ERP or Information system.

**Core Books:**

1. K. C. Laudon and J. P. Laudon: Management Information Systems, 12th Edition, Pearson Education.
2. Alexis Leon: ERP Demystified, Second Edition, Tata McGraw-Hill.

**Reference Books:**

1. W.S. Jawadekar: Management Information Systems, 2nd Edition, Tata McGraw-Hill.

**Web References:**

1. http://www.slideshare.net/NorazilaMat1/laudon-mis12-ppt01-16595885 [Function of Information system, A business perspective of MIS, Dimensions of MIS, Contemporary Approaches to Information System]
2. http://www.uh.edu/~mrana/try.htm [Types and Functions of Information system]
3. http://bisom.uncc.edu/courses/info2130/Topics/istypes.htm [Types of Information system]
4. http://kalyan-city.blogspot.com/2011/05/levels-of-management-top-middle-and.html [classification of information system]
5. http://my.safaribooksonline.com/book/management/9780470916803/operational-planning-and-control-systems/management_levels_comma_functions_comm [classification of information system]
6. http://ambarwati.dosen.narotama.ac.id/files/2011/05/FIS-2011-w2.pdf [Transaction processing system]

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | To understand basic of data & information, types of information system, function of information system and its quality. |
| CO2 : | To understand designing and architecture of information system |
| CO3 : | To understand transaction processing concepts and its working. |
| CO4 : | Learning Management Information system, its types and its implementation. To gain understanding of Decision Support Systems. |
| CO5 : | To learn Knowledge Management, Executive Support Systems, ERP and its technology. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Basics of Information System | √ | | | | |
| 2 | Architecture and Design of Information System | | √ | | | |
| 3 | Transaction Processing System | | | √ | | |
| 4 | Management Information System | | | | √ | |
| 5 | Decision Support System & Executive Support System | | | | √ | |
| 6 | Introduction to ERP and trending technologies | | | | | √ |

## Course Articulation Matrix:

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 1 | 3 | 1 | - | 3 | - | 1 | 2 | 3 | 3 | - | 1 | 1 |
| CO2 | 1 | 3 | 1 | - | 2 | 1 | - | 2 | 3 | 3 | 2 | - | 1 |
| CO3 | 1 | 3 | 1 | 1 | 3 | 3 | - | 2 | 3 | 3 | - | - | 1 |
| CO4 | 1 | 2 | 1 | - | 2 | 3 | 1 | 2 | 2 | 2 | 1 | 1 | 1 |
| CO5 | 1 | 3 | 1 | - | 3 | 3 | - | 1 | 3 | 3 | 1 | - | 1 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUA201 - Life Management

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction/Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 02 | Theory | 02 | 30 | 25 | 55 |
| **Experiential Learning Components :** Unit Tests, Assignments, Case Study etc. | | | | | |

**Pre-requisite:** None

**Methodology, Pedagogy & Andragogy:** During theory lectures the emphasis will be given on the fundamentals of life management. Students will be introduced life styles and its basic, time management, work efficiency, yoga, physical health management, stress and mental health management, and personality skills. Students will give experimental learning exposure in form of assignment and case study.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Life Style and its Basics | 06 |
| 2 | Work Efficiency | 04 |
| 3 | Personality Skills | 06 |
| 4 | Time Management for students | 05 |
| 5 | Physical health management | 04 |
| 6 | Stress and mental health management | 05 |

**Total Hours (Theory):30**
**Total Contact Hours:30**

**Detail Syllabus:**

**Unit I : Life Style and its Basics** **Hours 06**

Purpose of life and its dimensions, Importance of Self-Evaluation; (Daily routine, Food habits, Dressing Sense, Habit formation, Company, Etiquettes), Duties and Commitment of Self, Family and Society, Adjustment with Self and Environment

**Unit II : Work Efficiency** **Hours 04**

Positive way of thinking, Tools & techniques for Positive thinking, Karma and Karma Phal Sidhanta, Behavioural Skill

**Unit III : Personality Skills** **Hours 06**

Self-Assessment Techniques, Adjustment Skills, Art of Positive Thinking, Developing Self Confidence,

**Unit IV : Time Management for students** **Hours 05**

Importance, Steps, Barriers and solutions of Time Management, Reading and Writing Skill, Making Notes and Conceptual Clarity of Subject topic.

**Unit V : Physical health management** **Hours 04**

Importance of physical health and exercise, importance of yoga, physical problems, exercise for health improvements

**Unit VI : Stress and mental health management** **Hours 05**

Importance of mental health, spirituality, improving concentration, importance of Dhyan, Memory problem and Memory Boosting techniques, causes and effects of stress, importance of Prathana, Upasna, Sadhna, and Aradhna

**Core Books:**

1. Time Management, McGraw-Hill, 2003
2. How To Manage Your Life, Vishal Tatwaved, 2012.

**Reference Books:**

1. Living Stress Free - The Secret of How to Manage Stress and Live Life Fully, Sonali Perera, SP MEDIA MARKETING.
2. TIME MANAGEMENT, BRIAN TRACY, American Management Association, 2013.

**Web References:**

1. Life management: a holistic approach to make the most of your life, Dr. Hannah Rose https://nesslabs.com/life-management
2. Life Management, https://www.sciencedirect.com/topics/computer-science/life-management
3. https://edynamiclearning.com/course/health-1-life-management-skills/

**Course Outcomes:** Upon successful completion of the course, students will,

| | |
|---|---|
| CO1 : | Learn basics of life managements |
| CO2 : | Be Familiar with various styles |
| CO3 : | Get the knowledge of time management and stress management |
| CO4 : | Get various aspects of physical and mental health |
| CO5 : | Be able to develop personality and good habits |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Life Style and its Basics | √ | | | | |
| 2 | Work Efficiency | | √ | | | |
| 3 | Personality Skills | | | | | √ |
| 4 | Time Management for students | | | √ | | |
| 5 | Physical health management | | | | √ | |
| 6 | Stress and mental health management | | | | √ | |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 1 | 3 | 1 | - | 3 | - | 1 | 2 | 3 | 3 | - | 1 | 1 |
| CO2 | 1 | 3 | 1 | - | 3 | 1 | - | 1 | 3 | 3 | 1 | - | 1 |
| CO3 | 1 | 3 | 1 | 1 | 3 | 3 | - | 2 | 3 | 3 | - | - | 1 |
| CO4 | 1 | 2 | 1 | - | 2 | 3 | 1 | 2 | 2 | 2 | 1 | 1 | 1 |
| CO5 | 1 | 3 | 1 | - | 3 | 3 | - | 1 | 3 | 3 | 1 | - | 1 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

# TEACHING SCHEME & DETAILED SYLLABUS

# FOR

# BSc-IT PROGRAMME
# ($4^{th}$ SEMESTER)
# AS PER NEP 2020

# EFFECTIVE FROM ACADEMIC YEAR 2025-26

## B.Sc. (IT) Semester-IV

| Course Code | Course Name | Theory / Practical / Others | Marks (Theory) | | Marks (Practical) | | Credit | NEP - 2020 Classification |
|---|---|---|---|---|---|---|---|---|
| | | | CCE | SEE | CCE | SEE | | |
| **CTUC204** | Fundamentals of Operating Systems | Theory and Practical | 50 | 50 | 25 | 25 | 06 (04-Theory) (02-Practical) | Major Core |
| **CTUC205** | Fundamentals of Visual Programming | Theory and Practical | 50 | 50 | 25 | 25 | 06 (04-Theory) (02-Practical) | Major Core |
| **CTUE201**<br>**CTUE202**<br>**OCBSIT1007** | (Select any one)<br>1. Introduction to Artificial Intelligence<br>2. Computer Network Security<br>3. The Joy of Computing using Python | Theory | 50 | 50 | - | - | 04 | Minor |
| **CTUA202** | Social Media and Blog Writing | Practical | - | - | 25 | 25 | 02 | AEC |
| | University Elective-II | Practical | - | - | 50 | 50 | 02 | MDC |
| **HSUV203** | Indian Knowledge System | Practical | - | - | 25 | 25 | 02 | VAC |
| **CAUT201** | Summer Internship and Viva - II | VIVA | - | - | 50 | 50 | 04 | Exit Course |

*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)
*SEE- Semester –End- Evaluation

**University Elective - II**

| Sr. No. | Course Code | Course Name | Department/Faculty |
|---|---|---|---|
| 1 | MEUE202 | Nature and Properties of Materials | Engineering |
| 2 | EEUE202 | Solar Energy Engineering and Technology | Engineering |
| 3 | CLUE202 | Ecology and Environment | Engineering |
| 4 | ECUE202 | Introduction to Internet of Things | Engineering |
| 5 | CEUE202 | Software Conceptual Design | Engineering |
| 6 | ITUE202 | Ethical Hacking | Engineering |
| 7 | CSUE202 | Google Cloud Computing Foundations | Engineering |
| 8 | AIUE202 | Social Network Analysis | Engineering |
| 9 | BMUD251 | Economics Of Health And Health Care | Management |
| 10 | CAUD204 | Modern Application Development | Computer Science |
| 11 | PTUD192 | Ergonomics Workplace Analysis | Physiotherapy |
| 12 | NRMD261 | Mindfulness and Well-being: Living with Balance and Ease | Nursing |

# CTUC204 - Fundamentals of Operating Systems

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components (Theory) :** Unit Tests, Assignments, Case Study etc.<br>**Experiential Learning Components (Practical) :** Lab Assignments, Practical Test, VIVA | | | | | |

**Pre-requisite:** None

**Methodology, Pedagogy & Andragogy:** In order to achieve the course objectives, students will be introduced to the basic operating system concepts and basic functions. Students will be offered appropriate case studies in order to provide actual exposure to theoretical operating system ideas. During the practical session, student will commence the learning from MS-DOS commands in windows environment. The student will also get hands on working with basic Linux commands. Finally using base of Linux commands, shell scripts would be written to demonstrate operating system programming.

**Outline of the course (Theory):**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Operating System Fundamentals | 09 |
| 2 | Memory Management | 11 |
| 3 | Processor Management | 11 |
| 4 | Device Management | 11 |
| 5 | File Management | 10 |
| 6 | Introduction to Shell Script | 09 |

**Total Hours (Theory): 60**
**Total hours (Practical): 60**
**Total Contact Hours:120**

**Detail Syllabus:**

**Unit I: Operating System Fundamentals** **Hours 09**

What is an operating system, Brief history of an operating system, Pyramid structure of an operating system, Operating system hardware and software, Evolution of an operating system, types of operating system, Batch system, Interactive system, Real time system, Hybrid system, Shell, compiler, assembler, interpreter.

**Unit II: Memory Management** **Hours 11**

Memory management of early system, Fixed partition, Dynamic partition, Re-locatable dynamic partition, Best fit and first fit allocation method, De-allocation, Paged memory allocation, Demand paging allocation, segmented memory allocation, Virtual memory, cache memory, Associative memory, Page replacement policies.

**Unit III: Processor Management** **Hours 11**

What is job and Process, Multiprogramming concept, Process scheduler, Job scheduler, various process states, Thread control, Process Control Block (PCB), Processor scheduling policies, FCFS, SJN, SRT, RR, EDF, Multiple level queues.

**Unit IV: Device Management** **Hours 11**

Meaning, Depreciation of money, Monetary inflation concepts, types of inflation, Causes and effect of inflation different sectors of the economy, Causes of Inflation, Advantages, Formula and Disadvantages of inflation.

**Unit V: File Management** **Hours 10**

Fundamentals of File management, File naming conventions, Fixed length and variable length records, Physical file organization, Physical storage allocation, File access methods, File control matrix, File compression.

**Unit VI: Introduction to Shell Script** **Hours 09**

Shell Script, Shell environment, Making Script interactive, Command Line Argument, Evaluating Expression, Operator, Control Statement, Looping statement.

**Outline of the Course (Practical):**

| Week No | Practical | Description |
|---|---|---|
| 1 | Disk operating system -Commands | Introduction to MS DOS environment with basic commands |
| 2 | Introduction to Linux Editor | Introduction to Cygwin as Linux Editor along with its usage |
| 3 | Linux Operating System-Commands | Using basic commands of Linux |
| 4 | Linux Operating System-Advanced Commands | Implementation of advanced Linux commands used in industry |
| 5 | Vi editor-Create Text file | Practice of writing and modifying various files in Vi editor |
| 6 | Introduction of Shell Programming-Create .sh file and Execute file | Creating simple shell script to get introduced with operating system programming |
| 7 | Variable and Rules for variable | Creating and using variables, and constants in shell scripts |
| 8 | Interactive Shell Script, Working with Command Line Arguments | Practicing shell scripts which are responsive to the user input |
| 9 | Working with expression | Arithmetic expressions in shell script |
| 10 | Conditional statement-IF, Working with Operator | Learning conditional statements such as if, elif, case. Using arithmetic, relational, and logical operators. |
| 11 | Looping Statement-While | Writing shell scripts with while loops with real life examples |
| 12 | Looping Statement-Until, For | Writing shell scripts with until and for loops with real life examples |
| | | **Total hours: 60** |

## Core Books:

1. Ann Mclver McHoes, Ida M Flynn: Understanding Operating System, 7 Edition, Cengage Learning,2014.
2. William Stallings: Operating Systems Internals and Design Principles, 5th Edition, PHI, 2005.
3. Sumitabha Das: Unix concepts & application, 4th Edition, Tata McGraw Hill, 2010.
4. Kenneth Rosen, Douglas Host, James Farber and Richard Rosinski: The Complete Reference, Tata McGraw Hill, 1999.
5. Christopher Negus: Linux Bible, 10th Edition, Wiley, 2020.

## Reference Books:

1. Silberschatz: Operating System Concepts, 5th Edition, John Wiley & Sons (ASIA) Ltd., 2008.
2. Mark G. Sobell: A Practical Guide to Linux, Pearson Education, 1997.
3. KJ.George, Operating System Concepts and Principles, Sroff Publishers, 2003.

**Web References:**

1. https://study.com/learn/operating-system.html [Operating System Video Lessons]
2. https://www.studytonight.com/operating-system/ [Operating System Tutorial]
3. http://www.ics.uciedu/-ics] 43/lectures.html [Lecture notes of OS]
4. http://www.tutorialspoint.com/unix/index.htm [ Unix Tutorials]

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | To understand the fundamental concepts of an operating System |
| CO2 : | To understand memory allocation and de-allocation methods |
| CO3 : | To study fundamentals of processor and its scheduling policies |
| CO4 : | To study various supported device and file access technologies |
| CO5 : | To understand the commands of disk operating system, Linux operating system and Learning basic shell programming. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Operating System Fundamentals | √ | | | | |
| 2 | Memory Management | | √ | | | |
| 3 | Processor Management | | | √ | | |
| 4 | Device Management | | | | √ | |
| 5 | File Management | | | | √ | |
| 6 | Introduction to Shell Script | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO 2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 1 | 1 | 2 | 2 | - | 2 | 2 | - | 1 | 1 | - | 3 | 2 |
| CO 2 | 3 | 2 | 2 | 2 | - | 2 | 2 | - | 1 | 1 | - | 3 | 2 |
| CO 3 | 3 | 2 | 2 | 2 | - | 2 | 2 | - | 1 | 1 | - | 3 | 2 |
| CO 4 | 3 | 2 | 2 | 2 | - | 2 | 2 | - | 1 | 1 | - | 3 | 2 |
| CO 5 | 2 | 2 | 3 | 3 | - | 2 | 3 | - | 1 | 1 | - | 3 | 2 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUC205 - Fundamentals of Visual Programming

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components (Theory) :** Unit Tests, Assignments, Case Study etc.<br>**Experiential Learning Components (Practical) :** Lab Assignments, Practical Test, VIVA | | | | | |

**Pre-requisite:** Basic knowledge of C & C++ programming language.
**Methodology, Pedagogy & Andragogy:** During theory lectures illustrations emphasizing the need for advanced features of .Net framework and to develop windows and web-based application using C# language will be given. During practical sessions, students will actively engage in developing both desktop and web applications, applying the concepts discussed in lectures as well as will explore various features of Visual Studio and .NET.

**Outline of the course (Theory):**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1. | Introduction to .NET framework | 09 |
| 2. | Basics of Object Oriented programming using C# | 11 |
| 3. | Building desktop application | 11 |
| 4. | Web programming fundamentals | 10 |
| 5. | Database connectivity through ADO.NET | 11 |
| 6. | Building WPF application | 08 |

**Total Hours (Theory): 60**
**Total hours (Practical): 60**
**Total Contact Hours:120**

**Detail Syllabus:**

**Unit I: Introduction to .NET framework Hours 09**

Introducing .NET Framework, .NET Framework Component, .NET Framework Version Compatibility, Core of .NET Framework: Application Services, Base Class Library and CLR, Types of Applications which can be developedusing MS.NET, MS.NET Base Class Library, MS.NET Namespaces, The Common Language Runtime (CLR), Managed Code, MS.NET Memory Management / Garbage Collection, Common Type System (CTS), Common Language Specification (CLS), Types of JIT Compilers

**Unit II: Basics of Object-Oriented programming using C# Hours 11**

Overview of Object-Oriented Programming, Principles of Object Orientation programming: Encapsulation, inheritance, polymorphism, Object, Defining Properties, Using Auto- Implemented Properties, Defining Methods, Understanding Constructors, Extending .NET Classes via Inheritance, Defining and Implementing Interfaces, Understanding the Role of Interfaces in .NET. C# Datatypes & Variables Declaration, Implicit and ExplicitCasting, Checked and Unchecked Blocks –Overflow Checks, Casting between other datatypes, Boxing and Unboxing, Enum and Constant, Operators, Control Statements, Working with Arrays, C# Application Basics: Command line and VS.NET compilation.

**Unit III: Building desktop application Hours 11**

Building Windows Forms Applications, Setting Form Properties, Understanding the Life-cycle of a Form, Using the Windows Forms Designer, MessageBox Class and .config File, Working with Windows Forms Controls: TextBox, Button, Selection, List, Container, Image, ErrorProvider, TooltipProvider, etc. Handling Events: Understanding the Event-Driven Programming Model, Writing Event Handlers, Sharing Event Handlers.

**Unit IV: Web programming fundamentals Hours 10**

Basic of Http Request and Http Response. o Understand form Tag and Comparison between Get and Post. ASP.NET Page Life Cycle. Structure of an ASP.NET Page: ASPX Page, Code behind File, WebConfig and machine config, Rich Web Control,Validation Controls, Concept of custom control.

**Unit V: Database connectivity through ADO.NET Hours 11**

Evolution of ADO.NET, Understanding the ADO.NET Object Model, Connected vs. Disconnected Access, Use of Connection, Command, DataReader, DataAdapter and Dataset class.

**Unit VI: Building WPF application Hours 08**

Overview of WPF, Features of WPF, Windows Forms vs WPF , Introduction to XAML, WPF Layout: Grid Panel, Stack Panel, Dock Panel, Wrap Panel and Canvas Panel, WPF Controls: TextBlock, PasswordBox, ListBox, Slider, Popup, Menus and Expander

**Outline of the Course (Practical):**

| Week No | Practical | Description |
|---|---|---|
| 1 | Basics of Visual Studio IDE and .NET Introduction | Getting familiar with Visual Studio IDE and basic programming structure of C#. |
| 2 | Simple C# Programs | Writing basic C# programs, understanding syntax, and compiling with Visual Studio. |
| 3 | Console Applications | Developing console-based applications, using input and output statements, and understanding the main method. |
| 4 | Object-Oriented Programming Concepts | Implementing encapsulation, inheritance, and polymorphism in C#. |
| 5 | Control Statements and Arrays | Using operators, control statements, and working with arrays in C#. |
| 6<br>7<br>8 | Building Windows Forms Applications | Working with TextBox, Button, and other common controls in Windows Forms. |
| 9 | Event Handling in Windows Forms | Understanding event-driven programming and writing event handlers. |
| 10<br>11 | Basics of ASP.NET | Understanding HTTP requests and responses, and creating a simple ASP.NET page. |
| 12<br>13 | Database Connectivity with ADO.NET | Connecting to a database, executing commands, and retrieving data using ADO.NET. |
| 14 | A Fully Windows Form Application | Designing and implementing a complete Windows Forms application, integrating various controls, handling events, and database connectivity. |
| 15 | Building a WPF Application | Creating a basic WPF application. |

**Total hours: 60**

**Core Books:**

1. Christian Nagel : Professionals C# and .NET 2021 Edition , Wiley – India, WROX
2. Albahani Josheph: C# 4.0 Pocket Reference, Shroff publication,2010.

**Reference Books:**

1. Jon Skeet, C# in Depth, Managing Publication,2010
2. Drayton David: C# Language Pocket Reference, Shroff publication,2005

**Web References:**

1. .http://csharp.net-informations.com/ [to learn C# and .NET fundamentals]
2. http://www.wpftutorial.net/[tolearn WPF]
3. https://dotnet.microsoft.com/en-us/learn/csharp [tutorial and videos to learn C#]
4. https://dotnettutorials.net/course/csharp-dot-net-tutorials/
5. https://www.javatpoint.com/c-sharp-tutorial

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Understanding .NET Framework core features and Components. |
| CO2 : | To develop object oriented programming skills using C#. |
| CO3 : | To develop windows and web based application using core feature of .NET framework and C# language |
| CO4 : | Access and manipulate data in a Microsoft SQL Server database by using MicrosoftADO.NET. |
| CO5 : | Use modern approach to build windows based application using WPF |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Introduction to .NET framework | √ | | | | |
| 2 | Basics of Object Oriented programming using C# | | √ | | | |
| 3 | Building desktop application | | | √ | | |
| 4 | Web programming fundamentals | | | √ | | |
| 5 | Database connectivity through ADO.NET | | | | √ | |
| 6 | Building WPF application | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | 2 | 3 | - | 2 | 1 | - | - | 1 | 1 | 2 | - |
| CO2 | 2 | 3 | 2 | 2 | - | 3 | 2 | - | - | 2 | - | 1 | - |
| CO3 | 2 | 3 | 3 | 2 | - | 2 | 2 | - | - | 2 | 1 | 2 | 1 |
| CO4 | 2 | 3 | 3 | 2 | - | 3 | 3 | - | - | 2 | 2 | 2 | 1 |
| CO5 | - | 3 | 3 | - | - | - | - | - | - | - | 1 | - | - |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUE201 - Artificial Intelligence (AI) Fundamentals

| Credit | Component | Instruction/Contact Hours( Per Week) | Instruction/ Contact Hours( Per Course) | Experiential Learning Hours( Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| **Experiential Learning Components :** Unit Tests, Assignments, Case Study etc. | | | | | |

**Pre-requisite:** None

**Methodology, Pedagogy & Andragogy:** The theory sessions will be focused on basics of Artificial Intelligence, problem solving paradigms, and search strategies. Areas of application such as Expert Systems, Software Agents, knowledge representation, natural language processing, expert systems, robotics will be explored.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Introduction to Artificial Intelligence | 08 |
| 2 | Problem-solving by Search and constraint satisfaction | 11 |
| 3 | Knowledge representation and reasoning | 11 |
| 4 | Expert Systems | 11 |
| 5 | Agents | 10 |
| 6 | Introduction to AI Techniques and Application Areas | 09 |

**Total Hours (Theory): 60**
**Total Contact Hours:60**

**Detail Syllabus:**

**Unit I: Introduction to Artificial Intelligence** **Hours 08**

Concepts and definitions of Artificial Intelligence (AI) – Brief history of AI – AI and related fields – The AI problems and underlying assumptions, AI approaches: Cognitive Science, Turing Test.

**Unit II: Problem solving by Search and constraint satisfaction** **Hours 11**

Problem spaces; brute-force search; best-first search; two-player games; constraint satisfaction

**Unit III: Knowledge representation and reasoning** **Hours 11**

Review of propositional and predicate logic; resolution and theorem proving; non-monotonic inference; probabilistic reasoning; Bayes theorem, reasoning under uncertainty.

**Unit IV: Expert Systems** **Hours 11**

Introduction – Representing and using domain knowledge – Knowledge acquisition and representation – General structure of Expert Systems – Expert System Shell – Advantages and disadvantages of Expert Systems, Applications of Expert Systems

**Unit V: Agents** **Hours 10**

Definition of agents; successful applications and state-of-the-art agent-based systems; software agents, personal assistants, and information access; multi-agent systems.

**Unit VI: Introduction to AI Techniques and Application Areas** **Hours 09**

Introduction to fuzzy logic – Introduction to various application areas of AI like Natural Language Processing (NLP), Learning, Speech Recognition, Computer Vision, Game Playing, Robotics.

**Core Books:**

1. Norvig, Peter, Paradigms of AI Programming, Morgan Kauffman, 1992.
2. S. Russell and P. Norvig, Modern Approach to Artificial Intelligence, Prentice Hall of India Ltd., 2006.
3. Rich & Knight, Artificial Intelligence, 2nd edition, TMH, 1991.

**Reference Books:**

1. R.Akerker and P. S. Sajja, Knowledge-Based Systems, Jones and Bartlettes, MIT, 2010.
2. George Luger, Artificial Intelligence, 5th Edition, Addison Wesley, 2004.

**Web References:**

1. http://www.tutorialspoint.com/artificial_intelligence/ [Basics of AI]
2. http://intelligence.worldofcomputing.net/ai-branches/expert-systems.html [Expert Systems]
3. http://pages.cs.wisc.edu/~bolo/shipyard/neural/local.html [Neural Network]

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | To understand basic problems of AI and history of AI. |
| CO2 : | Getting deep knowledge of searching algorithms and its applications in AI. |
| CO3 : | Learning Knowledge Representation and reasoning techniques. |
| CO4 : | To gain understanding Expert System and Agent Based Systems. |
| CO5 : | To explore various application areas of AI. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Introduction to Artificial Intelligence | √ | | | | |
| 2 | Problem solving Search and constraint satisfaction | | √ | | | |
| 3 | Knowledge representation and reasoning | | | √ | | |
| 4 | Expert Systems | | | | √ | |
| 5 | Agents | | | | √ | |
| 6 | Introduction to AI Techniques and Application Areas | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | P10 | P11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | - | 2 | 2 | 2 | 1 | - | - | - | - | - | - | 1 | 2 |
| CO2 | 3 | 2 | 2 | 3 | 2 | 2 | - | 3 | - | - | - | 2 | 2 |
| CO3 | - | - | 3 | 3 | 2 | - | - | - | - | - | - | 1 | - |
| CO4 | 3 | 2 | 2 | - | 2 | 2 | - | 3 | - | - | - | 1 | 2 |
| CO5 | 2 | - | - | - | 2 | - | 3 | 3 | - | - | - | 1 | 1 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

# CTUE202 - Computer Network Security

| Credit | Component | Instruction/Contact Hours( Per Week) | Instruction/ Contact Hours( Per Course) | Experiential Learning Hours( Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| **Experiential Learning Components:** Unit Tests, Assignments, Case Study, Certification courses, etc. | | | | | |

**Pre-requisite:** None

**Methodology, Pedagogy & Andragogy:** The methodology for teaching Computer Network and Security involves a hybrid approach that blends theoretical knowledge with practical application, catering to both novice and experienced learners through pedagogy and andragogy. This includes the use of multimedia resources, simulations, and hands-on exercises to facilitate understanding. Andragogically, the course engages adult learners by emphasizing problem-based learning, real-world case studies, and collaborative projects that draw on learners' prior knowledge and professional experiences. This approach ensures learners can critically analyze and apply security measures in complex networking environments, adapt to emerging threats, and design secure network infrastructures.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Introduction to Computer Network Security | 07 |
| 2 | Symmetric Ciphers | 11 |
| 3 | Public Key Cryptography | 12 |
| 4 | Authentication and Authorization | 11 |
| 5 | Network Security | 10 |
| 6 | Web Security | 09 |

**Total Hours(Theory):60**
**Total Contact Hours:60**

## Detail Syllabus:

**Unit I: Introduction to Computer Network Security** **Hours 07**

Need for Security, Security Attacks, Services and Mechanisms, Network Security, Models

**Unit II: Symmetric Ciphers** **Hours 11**

Substitution &Transposition Techniques, Block Cipher, DES, Triple DES, Stream Ciphers, RC4

**Unit III: Public Key Cryptography** **Hours 12**

Need and Principles of Public Key Cryptosystems, RSA Algorithm, Key Distribution and Management
Diffie-Hellman Key Exchange, Digital Signatures

**Unit IV: Authentication and Authorization** **Hours 11**

Authentication Requirements, Message Authentication Codes, Hashes, MD5 & SHA, User Authentication: Password, Certificate based & Biometric Authentication 6. Kerberos

**Unit V: Network Security** **Hours 10**

Firewalls, IP Security , VPN, Intrusion Detection , SSL, TLS

**Unit VI: Web Security** **Hours 09**

Web Security, Security Principles (CIA Triad: Confidentiality, Integrity, Availability), Vulnerability Scanning (e.g., OWASP ZAP, Burp Suite), Content Security Policy (CSP)

## Core Books:

1. Kizza JM. Computer network security. Springer Science & Business Media; 2005 Apr 7.
2. Wang J. Computer network security. Berlin/Heidelberg, Germany: Springer; 2009.

**Reference Books:**


1. Buchmann J. Introduction to cryptography. New York: Springer; 2004 May.
2. Delfs, H., Knebl, H., & Knebl, H. (2002). *Introduction to cryptography* (Vol. 2). Heidelberg: Springer.
3. Garfinkel S, Spafford G. Web security, privacy & commerce. " O'Reilly Media, Inc."; 2002.
4. Garfinkel, Simson, and Gene Spafford. *Web security & commerce*. Cambridge, MA: O'reilly, 1997.


**Web References:**


1. https://www.cisco.com/c/en_in/products/security/what-is-network-security.html
2. https://www.javatpoint.com/computer-network-security
3. https://tndalu.ac.in/econtent/9_Computer_Network_And_Network_Security.pdf


**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Develop Concept of Security needed in Communication of data through computers and networks along with Various Possible Attacks |
| CO2 : | Understand Various Encryption mechanisms for secure transmission of data and management of key required for required for encryption |
| CO3 : | Understand authentication requirements and study various authentication mechanisms |
| CO4 : | Understand network security concepts and study different Web security mechanisms. |
| CO5 : | Understand web  security concepts |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Introduction to Computer Network Security | √ | | | | |
| 2 | Symmetric Ciphers | | √ | | | |
| 3 | Public Key Cryptography | | | √ | | |
| 4 | Authentication and Authorization | | | | √ | |
| 5 | Network Security | | | | | √ |
| 6 | Web Security | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 1 | 1 | - | 2 | 1 | - | - | 1 | - | 2 | 2 | 2 |
| CO2 | 1 | 2 | 2 | 2 | 3 | 1 | - | - | 1 | 1 | 2 | 2 | 2 |
| CO3 | 1 | 2 | 2 | 2 | 2 | 2 | 1 | - | 1 | - | 2 | 2 | 2 |
| CO4 | 2 | 2 | 2 | 2 | 3 | 3 | 2 | - | 2 | 1 | 3 | 3 | 3 |
| CO5 | 1 | 2 | 2 | 1 | 2 | 2 | 1 | - | 1 | 1 | 2 | 2 | 2 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# OCBSIT1007 : The Joy of Computing using Python

**Description:**

**Credit and Week:**

| Teaching Scheme | Week | Marks | Credit |
|---|---|---|---|
| | 12 | 100 | 3 |

**About the course:**

A fun filled whirlwind tour of 30 hrs, covering everything you need to know to fall in love with the most sought after skill of the 21st century. The course brings programming to your desk with anecdotes, analogies and illustrious examples. Turning abstractions to insights and engineering to art, the course focuses primarily to inspire the learner's mind to think logically and arrive at a solution programmatically. As part of the course, you will be learning how to practice and culture the art of programming with Python as a language. At the end of the course, we introduce some of the current advances in computing to motivate the enthusiastic learner to pursue further directions.

**Pre-requisites:** 10th standard/high school

**Industry support**: Every software company is aware of the potential of a first course in computer science. Especially of a first course in computing, done right.

**Course layout:**

- Motivation for Computing
- Welcome to Programming!!
- Variables and Expressions : Design your own calculator
- Loops and Conditionals : Hopscotch once again
- Lists, Tuples and Conditionals : Lets go on a trip
- Abstraction Everywhere : Apps in your phone
- Counting Candies : Crowd to the rescue
- Birthday Paradox : Find your twin
- Google Translate : Speak in any Language
- Currency Converter : Count your foreign trip expenses

- Monte Hall : 3 doors and a twist
- Sorting : Arrange the books
- Searching : Find in seconds
- Substitution Cipher : What's the secret !!
- Sentiment Analysis : Analyse your Facebook data
- 20 questions game : I can read your mind
- Permutations : Jumbled Words
- Spot the similarities : Dobble game
- Count the words : Hundreds, Thousands or Millions.
- Rock, Paper and Scissor : Cheating not allowed !!
- Lie detector : No lies, only TRUTH
- Calculation of the Area : Don't measure.
- Six degrees of separation : Meet your favourites
- Image Processing : Fun with images
- Tic tac toe : Let's play
- Snakes and Ladders : Down the memory lane.
- Recursion : Tower of Hanoi
- Page Rank : How Google Works !!

**Books and references**

1. *Python Crash Course: A Hands-On, Project-Based Introduction to Programming*.
   No Starch Press, 2019.
2. *Think Python: How to Think Like a Computer Scientist*.
   O'Reilly Media, 2015.
3. *Effective Python: 90 Specific Ways to Write Better Python*.
   Addison-Wesley, 2020.
4. *Head First Python: A Brain-Friendly Guide*.
   O'Reilly Media, 2016.
5. https://www.w3schools.com/python/.

**Criteria to get a certificate:**

**If You wish to get certified on this course you must register and write the proctored exam after payment of exam fee.**

30 Marks will be allocated for Internal Assessment and 70 Marks will be allocated for end term proctored examination.

Securing 40% in both separately is mandatory to pass the course and get Credit Certificate.

# CTUA202 - Social Media and BLOG Writing

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction/Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components :** Lab Assignments, Practical Test, VIVA | | | | | |

**Prerequisites:** None.

**Methodology, Pedagogy & Andragogy:** This course focuses on providing hands-on experience to students for understanding social media marketing strategies and BLOG writing skills.

**Outline of the Course:**

| Week No | Practical | Description |
|---|---|---|
| 1 | Introduction to Social Media<br>• Platforms for social media marketing<br>• Roles of social media in digital marketing | Learn basics of Social Media |
| 2 | Optimizing for Social Media Platforms<br>• Facebook<br>• Instagram<br>• X (formerly Twitter) | Learn about various social media platforms and optimizing for them. |
| 3 | Content Calendar for Social Media<br>• Creating content calendar with different focus | |
| 4 | Content creation and Management in Meta Business Suite<br>• Understanding and Managing Meta Business Suite | Understanding Search Engine Result Page, and learn about On-the-Page and Off-the-Page Optimization |
| 5 | Understanding audience and insights in Meta Business | - |
| 6 | Boosting posts from Meta Business | - |
| 7 | Introduction to BLOG<br>• What is BLOG? | - |

| | | |
|---|---|---|
| | • Importance of BLOG | |
| **8** | Popular Blogging Platforms<br>• WordPress<br>• Blogger | - |
| **9** | Using SEO in Blogging<br>• Using AI in Blogging<br>• Understanding SEO in Blogging | - |
| **10** | Using Blogger for Blogging | - |
| **11** | Using WordPress for Blogging – I<br>• Installation of WordPress<br>• Dashboard management | - |
| **12** | Using WordPress for Blogging – II<br>• Creating and Publishing posts<br>• Managing comments and subscription | |
| **13** | Using WordPress for Blogging – III<br>• Exploration of themes for blogging<br>• Exploration of plugins for blogging | |
| **14** | Case Study – I (Social Media) | |
| **15** | Case Study – II (Blogging) | |

**Total hours: 60**

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**Core Books:**

1. Damian Ryan and Calvin Jones: Understanding Digital Marketing, Kogan Page, 2014
2. Stephanie Diamond: Digital Marketing All-in-One for Dummies, Wiley Publication, 2019
3. Amy Lupold Bair, Blogging for Dummies, 7th Edition, Wiley Publication, 2020

**Reference Books:**

1. Philip Kotler, Hermawan Kartajaya, and Iwan Setiawan: Marketing 4.0: Moving from Traditional to Digital, Wiley Publication, 2016
2. Jan Zimmerman and Deborah Ng: Social Media Marketing All-in-One for Dummies, Wiley Publication, 2017

**Web References:**

1. https://www.simplilearn.com/tutorials/digital-marketing-tutorial
2. https://www.tutorialspoint.com/digital_marketing/index.htm
3. https://intellipaat.com/blog/digital-marketing-tutorial/
4. https://www.javatpoint.com/digital-marketing
5. https://www.javatpoint.com/blog

**Course Outcome:** Upon successful completion of the course, student will:

| | |
|---|---|
| **CO1 :** | Able to understand digital marketing and its importance. |
| **CO2 :** | Able to create and manage content using social media platforms |
| **CO3 :** | Able to optimize social media platforms |
| **CO4 :** | Able to blogging and its importance in digital marketing. |
| **CO5 :** | Able to use generative AI in Digital Marketing. |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 1 | 2 | 1 | 3 | 2 | 2 | 1 | 1 | - | 1 | 1 | 1 | 1 |
| CO2 | - | 1 | 2 | 3 | 1 | 3 | 1 | - | - | 2 | 2 | 2 | 2 |
| CO3 | 1 | 1 | 2 | 3 | 1 | 3 | 1 | - | - | 2 | 2 | 2 | 2 |
| CO4 | 1 | 1 | 2 | 3 | 1 | 1 | 1 | 1 | - | 1 | - | 1 | 1 |
| CO5 | 2 | 2 | 3 | 3 | 2 | 3 | 3 | 2 | 1 | 2 | 2 | 3 | 2 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUI201 - Summer Internship and Viva – II

## Objective and Scope:

Students who choose to exit the programme after the completion of the first two semesters must undertake internships, community engagement programmes and field based learning/minor projects during the summer term. This internship, worth 04 credits, is a requirement for the award of the Diploma. Students have to undergo internship in reputed Tech Companies, Software Development Firm, IT services companies, Research Institutions and Lab etc. This provision ensures that students gain valuable practical experience and industry exposure, complementing their academic learning and enhancing their career prospects. Following are the intended objectives of internship training:

- **Practical Experience**: Provide hands-on experience with real-world projects to apply theoretical knowledge gained during the course.
- **Skill Development**: Enhance technical skills, such as programming, software development, database management, and other relevant IT skills.
- **Industry Exposure**: Familiarize students with the working environment of the IT industry, including understanding company culture, workflows, and professional practices.
- **Professionalism**: Instil a sense of professionalism and work ethic, including adhering to deadlines, following instructions, and working collaboratively.
- **Transition to Employment**: Facilitate a smoother transition from academic life to professional careers by bridging the gap between theory and practice.

## Guidelines of Internship:

The Summer Internship shall be of 15 (Minimum 60 hours) duration and will be undertaken during the Summer Vacation. Students have to undergo internship in reputed Tech Companies, Software Development Firm, IT services companies, Research Institutions and Lab etc. It is mandatory for the students to seek written approval from the coordinator about the topic and the organization before commencing the Summer Internship. During Summer Internship students are expected to take necessary guidance from the faculty guide allotted by the Institute. To do it effectively they should be in touch with their guide through e-mail or phone. Students must maintain regular reports detailing the work completed, challenges faced, and learning experiences. Students must submit a final report and presentation summarizing the internship experience, project outcomes, and key learnings to guide. The technical specifications for report preparation is as under:

**Technical details:**

1. The report shall be printed on A-4 size white paper.
2. 12 pt. Times New Roman font shall be used with 1.5-line spacing for typing the report.
3. 1” margin shall be left from all the sides.
4. Considering the environmental issues, students are encouraged to print on both sides of the paper.
5. The report shall be spiral bound as per the standard format of the cover page given by the Institute.
6. The report should include a Certificate (on company’s letter head) from the company duly signed by the competent authority with the stamp.
7. The report shall be signed by the respective guide(s) & the Principal of the Institute 10 (Ten) days before the viva-voce examinations.
8. Student should prepare two hard bound copies of the Summer Internship Project Report and submit one copy in the institute. The other copy of the report is to be kept by the student for their record and future references.

## Evaluation:

The evaluation of Internship will be done as per the following criteria:

| Sr. No. | Evaluation Criteria | Marks |
|---|---|---|
| 1 | Summer Internship Report | 50 |
| 2 | Presentation and Viva-voce examination | 50 |
| **TOTAL MARKS** | | **100** |

Students must secure 36% passing marks in both components to qualify for the Diploma.

## Internship and Exit:

Any student intending to exit the course must complete the summer internship (Including scoring at least 36% marks in the evaluation) before exit. A student intending to exit shall have to submit an application to Principal through Counsellor before even semester university theory examination. However, any such student desiring to withdraw the exit option may do that before the last paper of theoretical examinations of even semester.

# TEACHING SCHEME & DETAILED SYLLABUS

# FOR

# BSc-IT PROGRAMME
# ($5^{th}$ SEMESTER)
# AS PER NEP 2020

# EFFECTIVE FROM ACADEMIC YEAR 2025-26

## B.Sc. (IT) Semester-V

| Course Code | Course Name | Theory / Practical/ Others | Marks (Theory) | | Marks (Practical) | | Credit | NEP - 2020 Classification |
|---|---|---|---|---|---|---|---|---|
| | | | CCE | SEE | CCE | SEE | | |
| **CTUC301** | Object Oriented Programming Using JAVA | Theory and Practical | 50 | 50 | 25 | 25 | 06<br>(04-Theory)<br>(02-Practical) | Major Core |
| **CTUC302** | Open Source Technology | Theory and Practical | 50 | 50 | 25 | 25 | 06<br>(04-Theory)<br>(02-Practical | Major Core |
| **CTUE301**<br>**CTUE302** | (Select any one)<br>1. Data Science Essentials<br>2. Basics of Digital Image Processing | Theory | 25 | 25 | - | - | 02 | Minor |
| **CTUE303** | Object Oriented Analysis and Design | Theory | 50 | 50 | - | - | 04 | Minor |
| **CTUE304**<br>**CTUE305** | 1. Advanced Internet of Things<br>2. Introduction to Game Development | Practical | - | - | 25 | 25 | 02 | Minor |
| **HSUS301** | French | Practical | - | - | 25 | 25 | 02 | SEC |

*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)
*SEE- Semester –End- Evaluation

# CTUC301 - Object Oriented Programming Using JAVA

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components (Theory):** Unit Tests, Assignments, Case Study etc.<br>**Experiential Learning Components (Practical):** Lab Assignments, Practical Test, VIVA | | | | | |

**Pre-requisite:** CTUC103-Basic Knowledge of Introduction to Programming

**Methodology, Pedagogy & Andragogy:** Upon successful completion of the syllabus, students will get basic knowledge of object oriented programming. Concretely, students should be able to create appropriate classes using the Java Programming Language to solve problems using Object Oriented Approach. They should be able to write console based and multithreaded applications using the Java Programming Language. Students should be able to implement the interface and classed of collection framework.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1. | Basic Introduction of Java | 10 |
| 2. | Variable & Data Types | 08 |
| 3. | Class Objects and Methods | 10 |
| 4. | Exception Handling & Packages | 11 |
| 5. | Multithreading | 11 |
| 6. | Basics of Collections Framework | 10 |

**Total Hours (Theory) : 60**
**Total hours (Practical) : 60**
**Total Contact Hours: 120**

**Detail Syllabus:**

**Unit I: Basic Introduction of Java Hours 10**

Basics of Java, History of Java, Advantages of Java, Java Virtual Machine & Byte Code, Java Environment Setup, Java Program Structure, Difference between POP and OOP, Features of Java, Basics of OOP: Class, Objects, Abstraction, Inheritance, Encapsulation, Polymorphism, Create and deploy simple programs using java.

**Unit II: Variable & Data Types Hours 08**

Data types, Variables, Literals, Identifers, Operators, Decision & Control flow Statements, Arrays. Structure of Java program, creating an application in Java, Compiling and executing applications in java, command line arguments.

**Unit III: Class Objects and Methods Hours 10**

Defining classes, variables and methods, creating objects, this keyword, static keyword, method overloading, final keyword, Constructors: Default constructors, Parameterized constructors, passing object as a parameter, constructor overloading, Basics of Inheritance, Types of inheritance, super keyword, concepts of method overriding, dynamic method dispatch, abstract class, Interface.

**Unit IV: Exception Handling & Packages Hours 11**

Exception Handling: types of exceptions, Throwable class, keywords – try, catch, throw, throws and finally, java built in exceptions, user defined exceptions. Creating package, importing package, access rules for packages. Exploring java. lang package, Wrapper classes, Boxing and Unboxing conversions, String and StringBuffer class with its methods.

**Unit IV: Multithreading Hours 11**

Multitasking, Process based and Thread based multitasking, the Java Thread Model, creating a thread, Creating Multiple Threads, Thread class and the Runnable Interface, Thread Life Cycle, Thread Priorities, Synchronization, Inter thread communication.

**Unit – VI: Basics of Collections Framework Hours 10**

The Collections Framework Overview, Collection Interfaces- Collections, List, Set, Sorted Set, The Collection classes- Array List, Linked List, Vector, Hash Set, Tree Set, Priority Queue. Accessing a collection using an Iterator, ListIterator and Enumeration, Map Interfaces and Classes: Map, Map Entry, SortedMap, and NavigableMap, HashMap, LinkedHashMap, TreeMap, Comparable & Comparator Interface.

**Outline of the Course (Practical):**

| Week No | Practical | Description |
|---|---|---|
| 1 | Basic of Java | Fundamental concepts in Java, Class & Objects , Structure of java program. |
| 2 | Data Types & Variables | Understand and utilize arrays effectively, keywords, Identifiers. |
| 3 | Defining classes, variables and methods | Implement this keyword, static keyword, method overloading, final keyword, creating objects, accessing rules of java |
| 4 | Constructors, Basics of Inheritance & Runtime Polymorphism | Implement Default constructors, Parameterized constructors, passing object as a parameter, constructor overloading. Implement inheritance types and runtime polymorphism |
| 5 | Exeption handling | Able to implements exception handling: types of exceptions, Throwable class, keywords – try, catch, throw, throws and finally, nested try statement, java built in exceptions, user defined exceptions. |
| 6 | Creating packages | Able to used the predefined packages and the use defines package and access protection, String class and its methods, StringBuffer class, StringBuilder class and its methods. |
| 7 | Multithreading | Implementing methods of Thread class. |
| 8 | Creating Multiple Threads | Creating Multiple Threads, Thread class and the Runnable Interface |
| 9 | Thread Priority | Implementing Thread priority |
| 10 | Synchronization & Inter-thread communication | Implementing thread synchronization and inter-thread communication. |
| 11 | Collections Framework | Implementation of Collections Framework, Collection, Accessing a collection using an Iterator, ListIterator and Enumeration. |
| 12 | Interfaces of Collections | List, Set, Sorted Set |
| 13 | Collection classes | Array List, Linked List, Vector, Hash Set, Tree Set, Priority Queue |
| 14 | Map interface | Map, Map Entry, SortedMap, and NavigableMap, HashMap, LinkedHashMap, TreeMap. |

| | | |
|---|---|---|
| **15** | Comparable & Comparator Interface | Implementing Comparable & Comparator Interface |
| | | **Total Hours: 60** |

**Core Books:**

1. Herbert Schildt: The Complete Reference Java J2SE, 5th Edition, TMH Publishing Company Ltd.
2. Sachin Malhotra, SaurabhChodhary: Programming in Java, 2nd Edition, OXFORD press

**Reference Books:**

1. Pravin Jain: The class of JAVA, 3rd Impression, Pearson.
2. Ivor Horton: Beginning Java 2" JDK, 5th Edition, Wiley Computer Publishing 2007. ssss
3. Ken Arnold, James Gosling and David Holmes: The Java Programming Language, 4th Edition, Addison Wesley.

**Web References:**

1. http://www.tutorialspoint.com/java/ [For all JAVA Concepts]
2. http://www.roseindia.net/java/ [For JAVA Tutorials with Example]
3. https://www.javatpoint.com/java-tutorial/[For all JAVA Concepts]

**Course Outcomes:** Upon successful completion of the course, students will be able

| | |
|---|---|
| CO1 : | To understand the object oriented programming concepts & basic building blocks of language |
| CO2 : | To understand basic syntax and semantics of Java, structure of java. |
| CO3 : | To introduce the principles of inheritance and polymorphism; and demonstrate how they relate to the design of abstract classes, Interfaces, |
| CO4 : | To introduce the implementation of Strings, Exception and packages, Strings |
| CO5 : | To introduce the concepts of multithreading |
| CO6: | To introduce collection framework: Interfaces, Implementations, and Algorithms. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | | |
|---|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 | CO6 |
| 1 | Basic Introduction of Java | √ | | | | | |
| 2 | Variable & Data Types | | √ | | | | |
| 3 | Class Objects and Methods | | | √ | | | |
| 4 | Exception Handling & Packages | | | | √ | | |
| 5 | Multithreading | | | | | √ | |
| 6 | Basics of Collections Framework | | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | 3 | 2 | 1 | 3 | 1 | 1 | 1 | 1 | 1 | 2 | 1 |
| CO2 | 3 | 2 | 2 | 3 | 1 | 3 | 1 | 1 | 1 | 2 | 2 | 2 | 2 |
| CO3 | 3 | 3 | 3 | 2 | 2 | 3 | 2 | 2 | 1 | 2 | 2 | 2 | 2 |
| CO4 | 3 | 3 | 2 | 2 | 2 | 3 | 1 | 2 | 2 | 1 | 2 | 2 | 2 |
| CO5 | 3 | 2 | 2 | 2 | 1 | 3 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| CO6 | 3 | 3 | 2 | 1 | 2 | 3 | 1 | 2 | 2 | 2 | 3 | 2 | 2 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

# CTUC302 - Open Source Technology

| Credit | Component | Instruction/Contact Hours( Per Week) | Instruction/Contact Hours( Per Course) | Experiential Learning Hours( Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components (Theory):** Unit Tests, Assignments, Case Study etc.<br>**Experiential Learning Components (Practical):** Lab Assignments, Practical Test, VIVA | | | | | |

**Pre-requisite:** CTUC203 - Working knowledge of HTML and basic knowledge of MYSQL.

**Methodology, Pedagogy & Andragogy:** During theory and practical sessions, students able to install & configure PHP and prerequisite software(s). Also, students will be emphasized to develop dynamic web applications.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Introduction to Web Development and Open Source Technology | 08 |
| 2 | PHP Basics | 10 |
| 3 | Array and Functions | 10 |
| 4 | Handling Form Data | 10 |
| 5 | Database Connectivity with MYSQL | 12 |
| 6 | PHP Utilities | 10 |

**Total Hours (Theory): 60**
**Total hours (Practical): 60**
**Total Contact Hours:120**

**Detail Syllabus:**

**Unit I: Introduction to Web Development and Open Source Technology** **Hours 8**

Introduction of open source software, Development philosophy of open source software, pros and cons of open source software, open source vs. close source software, Community support in open source, Version Control, Introduction to Webpage and Website, Static and Dynamic Website, Client & Server Side Scripting, Web Hosting: Shared / Cloud / VPS / Dedicated.

**Unit II: PHP Basics** **Hours 10**

Introduction to PHP, Installation of Apache, MySQL and PHP (OOP approach), How PHP code is parsed, Embedding PHP and HTML, Executing PHP and viewing in Browser, Comments in PHP, PHP variables: static and global variables, Data types, Operators, Conditional Statements, Looping Structure.

**Unit III: Array and Functions** **Hours 10**

Array, Types of Array, Array Functions, Miscellaneous Functions: define, constant, include, require, header, die, exit, Overview of built in functions of PHP: String functions, Date and Time, Math functions, File handling functions, User Defined Functions.

**Unit IV: Handling Form Data** **Hours 10**

HTML Form element & its attributes, Send Form data using GET Method & POST Method, Receive Form data using $_GET, $_POST & $_REQUEST variables, Super Global Variables, PHP form validations.

**Unit V: Database Connectivity with MYSQL** **Hours 12**

OOPs Concepts, PHP Data Objects (PDO), Introduction to MYSQL DB, CRUD operations, Handling Errors, State Management Techniques: PHP Sessions, starting session, modifying session variables, Un-registering and deleting session variable, PHP Cookies, Types of Cookies.

**Unit VI: PHP Utilities** **Hours 10**

File Uploading: Upload Single and Multiple file using PHP script, Understanding HTTP requests, Exploring and modifying HTTP responses, getting information from web server, Sending mails.

**Outline of the Course:**

| Week No | Practical | Description |
|---|---|---|
| 1 | Student registration page using HTML and CSS | Revised the concepts of HTML and CSS. |
| 2 | Installation of XAMPP & generating basic PHP scripts | Get familiar with echo() and print(). Able to create basic business logic. |
| 3 | Variables, Data Types and Casting | Able to work with variables, data types and able perform type casting. |
| 4 | User Inputs and Conditional Statements | Deal with user inputs. Use of conditional statements. |
| 5 | Looping Structures | Able to work with Looping structures like: for, while, do..while and foreach |
| 6 | Inbuilt Functions: string, math and date & time | Use various built-in functions available in PHP. |
| 7 | PHP Forms | Creating Forms using GET and POST methods. Also use the super global variables $_GET, $_POST and $_REQUEST. |
| 8 | PHP Forms & Validations | Creating Forms with server side validations |
| 9 | PHP Array | How to create an array? Array Functions |
| 10 | Database Connectivity | Understand MYSQL Dashboard. Implement Database connectivity using PDO (PHP Data Object). |
| 11 | Database Connectivity | Implement Database connectivity using PDO (PHP Data Object). |
| 12 | File Upload and Download | Able to upload, download and view various kind of files and store path into database using $_FILES. |
| 13 | State Management: Session | Creating Session, modifying session variables, Un-registering and deleting session variable. |
| 14 | State Management: Cookies. Super Global Variables | Creating Cookies and learning various super global variables like $_SERVER, $GLOBALS and $_ENV |
| 15 | PHP Mail | Sending mail using PHP mail() & PHPMailer |

**Total hours: 60**

**Core Books:**


1. Josh Lockhart, Modern PHP, O'Rielly, 2015.
2. Kevin Tatroe, Peter Macintyre, Programming PHP, 4e: Creating Dynamic Web Pages, 4th Edition, O'Rielly, 2020.


**Reference Books:**


1. Urva Patel, Kaushal Gor, Sanskruti Patel, PHP Hand Book: Crafting Dynamic Web Application, Notion Press Publication, 2024.
2. Matt Zandstra, PHP 8 Objects, Patterns, and Practice: Mastering OO Enhancements, Design Patterns, and Essential Development Tools, Apress, 2021.
3. Luke Welling, Laura Thomson, PHP and MYSQL Web Development, 5th Edition, Pearson Education, 2016.


**Web References:**


1. https://www.php.net/ [Official website of PHP]
2. https://www.geeksforgeeks.org/php-tutorials/ [Lecture notes of PHP]
3. https://www.w3schools.com/php/default.asp [Lecture notes of PHP]
4. https://github.com/PHPMailer/PHPMailer [PHPMailer Code]


**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Gain the skills and knowledge for Open Source and Web Hosting. |
| CO2 : | To utilize knowledge and skills for basics of PHP and configuration of MYSQL Server. |
| CO3 : | Learn the array and built-In functions in PHP. |
| CO4 : | Gain understanding of Form handing and utilities in PHP. |
| CO5 : | Be able to develop dynamic web based application using PHP and MySQL with state management techniques. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Introduction to Web Development and Open Source Technology | √ | | | | |
| 2 | PHP Basics | | √ | | | |
| 3 | Array and Functions | | | √ | | |
| 4 | Handling Form Data | | | | √ | |
| 5 | Database Connectivity with MYSQL | | | | | √ |
| 6 | PHP Utilities | | | | √ | |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 3 | 2 | 3 | - | 1 | - | 1 | 2 | - | - | 2 | - |
| CO2 | 3 | 3 | 3 | 3 | - | 1 | - | 1 | 1 | - | - | 3 | - |
| CO3 | 3 | 3 | 3 | 3 | - | 1 | - | 1 | 1 | - | - | 3 | - |
| CO4 | 3 | 3 | 3 | 3 | - | 1 | - | 1 | 1 | - | - | 3 | - |
| CO5 | 3 | 3 | 3 | 3 | 2 | 1 | 1 | 1 | 2 | 1 | 1 | 3 | 1 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

# CTUE301 - Data Science Essentials

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction/Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 02 | Theory | 02 | 30 | 25 | 55 |
| **Experiential Learning Components:** Unit Tests, Assignments, Case Study etc. | | | | | |

**Pre-requisite:** CTUC101-Introduction to Programming, CTUC102-Relational Database Administration

**Methodology, Pedagogy & Andragogy:** During theory sessions topics related to Data Science will be will be covered with suitable examples. Students will get exposure of model building using machine learning algorithms. Student will get in-death knowledge of Data Science based real applications using case study approach.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Fundamentals of Data Science | 04 |
| 2 | Statistics and Linear Algebra for Data Science | 05 |
| 3 | Data Preprocessing | 06 |
| 4 | Data Visualization | 05 |
| 5 | Machine Learning for Data Science | 05 |
| 6 | Data Science Applications and future trends | 05 |

**Total Hours (Theory):30**
**Total Contact Hours:30**

**Detail Syllabus:**

**Unit I : Fundamentals of Data Science Hours 04**

Introduction to Data Science, Definition and description of Data Science, Need of Data Science, Pre-requisites of Data Scientists, Tools and Skills for Data Science, Data Science Process Life-cycle - Data Discovery, Preparation, Model Planning, Model Building, Communicate Results, Types of Data Analysis- Descriptive, Diagnostic, Predictive, Prescriptive.

**Unit II : Statistics and Linear Algebra for Data Science Hours 05**

Descriptive Statistics- Measure of frequency, Measure of Central Tendency, Measure of Dispersion, Measure of Position. Overview of Inferential Statistics, Linear Algebra for Data Science using Vectors and Matrices- Basic Vector operations, Common Matrix operations.

**Unit III : Data Preprocessing Hours 06**

Reading data using different functions, Introduction to Data Preprocessing, Data types and forms, Possible data error types, Consistent Data, Data Cleaning- Filling Missing Values, smoothing noisy data, Detecting and removing outliers, Data Integration, Data transformation- Rescaling, Normalizing data, Standardizing data Implementation Using Python.

**Unit IV : Data Visualization Hours 05**

Introduction to Data Visualization, Visual encoding, Basic plots using Python, Data Visualization libraries in Python.

**Unit V : Machine Learning for Data Science Hours 05**

Overview of Machine Learning- Supervised, Unsupervised, Semi-Supervised and Reinforcement Learning, Regression Methods – Simple Linear Regression, Multiple Linear Regression, Classification Methods- KNN, Logistic Regression, Naïve Bayes, SVM, Clustering Method- k-means clustering, Fuzzy c-means, Introduction of Reinforcement Learning.

**Unit VI : Data Science Applications and future trends Hours 05**

Applications- Manufacturing, Personalization, Medical Image Analysis, Internet Search, Genomics, Speech Recognition, Advertising, Financial Fraud Detection, Airline Route Planning, Cyber-threats, transportation, gaming, recommendation system.

**Core Books:**

1. Sanjeev J, Wagh, Manisha S. Bhende, Anuradha D. Thakare : Fundamentals of Data Science: First Edition: CRC Press : 2022
2. Dr. Gypsy Nandi, Dr.Rupamkumar Sharma: Data Science Fundamentals and Practical Approaches: BPB Publications:2020
3. Sandhya Arora, Latesh Malik: Data Science and Analytics with Python: Universities Press: 2023, ISBN:978-93-93330-34-5

**Reference Books:**

1. Parikshit Narenda Mahalle, Gitanjali Rahul Shinde, Priya Dudhale Pise, Jyouti Yogesh Deshmukh: Foundation of Data Science for Engineering Problem Solving: Springer:2022
2. K.G.Srinivasa,G.M.Siddesh,ChetanShetty,SowmyaB.J:Statistical Programming in R: First Edition: Oxford Press: 2017.

**Web References:**

1. https://www.cs.cornell.edu/jeh/book.pdf [Machine Learning Concepts]
2. https://www.edureka.co/blog/math-and-statistics-for-data-science/[Statistics for Data Science]
3. https://www.w3schools.com/datascience/ds_analyze_data.asp[Data Preparation]
4. https://www.kdnuggets.com/2022/06/plotting-data-visualization-data-science.html [Data Visualization for Data Science]

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Able to understand fundamentals of Data Science. |
| CO2 : | Able to use statistical data analysis techniques for Data Science Applications |
| CO3 : | Able to apply data cleaning, visualization and exploratory data analysis techniques on data. |
| CO4 : | Able to build model using machine learning techniques. |
| CO5 : | Able to understand applications and future trends of Data Science. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Fundamentals of Data Science | √ | | | | |
| 2 | Statistics and Linear Algebra for Data Science | | √ | | | |
| 3 | Data Preprocessing | | | √ | | |
| 4 | Data Visualization | | | √ | | |
| 5 | Machine Learning for Data Science | | | | √ | |
| 6 | Data Science Applications and future trends | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 1 | 3 | 1 | - | 3 | - | 1 | 2 | 3 | 3 | - | 1 | 1 |
| CO2 | 1 | 3 | 1 | - | 3 | 1 | - | 1 | 3 | 3 | 1 | - | 1 |
| CO3 | 1 | 3 | 1 | 1 | 3 | 3 | - | 2 | 3 | 3 | - | - | 1 |
| CO4 | 1 | 2 | 1 | - | 2 | 3 | 1 | 2 | 2 | 2 | 1 | 1 | 1 |
| CO5 | 1 | 3 | 1 | - | 3 | 3 | - | 1 | 3 | 3 | 1 | - | 1 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUE302 - Basics of Digital Image Processing

| Credit | Component | Instruction/Contact Hours( Per Week) | Instruction/ Contact Hours( Per Course) | Experiential Learning Hours( Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 02 | Theory | 02 | 30 | 25 | 55 |
| **Experiential Learning Components:** Unit Tests, Assignments, Case Study etc. | | | | | |

**Pre-requisite:** None

**Methodology, Pedagogy & Andragogy:** The theory sessions will emphasize fundamentals and applications of digital image processing. The classroom sessions will make students aware of various operations such as colour image processing, image enhancement, segmentation, representation, recognition, and compression, which are performed on digital images. To enhance comprehension, we will provide illustrations of digital image processing operations using a programming language. To bridge the gap between theoretical knowledge and practical application, we have to schedule industrial visits. These visits are designed to offer students a tangible correlation between their studies and real-world practices. Students shall be assigned case studies to facilitate the practical application of their acquired knowledge. These case studies aim to deepen understanding and foster the ability to apply theoretical concepts to real-world scenario of digital image processing and analysis.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Digital Image Fundamentals | 06 |
| 2 | Colour Image Processing | 05 |
| 3 | Image Enhancement | 05 |
| 4 | Image Segmentation | 05 |
| 5 | Image Representations and Recognition | 05 |
| 6 | Image Compression | 04 |

**Total Hours (Theory): 30**
**Total Contact Hours:30**

**Detail Syllabus:**

**Unit I : Digital Image Fundamentals** **Hours 06**

Introduction, History, Components of image processing system, Applications of Digital Image Processing, Image as a 2D data, Image representation, Gray scale and Colour images, image sampling and quantization, Basic Relationships between Pixels.

**Unit II : Colour Image Processing** **Hours 05**

Colour fundamentals, Colour models, Colour transformation, Smoothing and Sharpening, Colour segmentation, Noise Models, Noise Reduction.

**Unit III : Image Enhancement** **Hours 05**

Basic gray level Transformations, Histogram Processing, Spatial Filtering, Extension to functions of two variables, Image Smoothing, Image Sharpening, Frequency domain filtering

**Unit IV : Image Segmentation** **Hours 05**

Edge based segmentation, Region based segmentation, Region split and merge techniques, optimal thresholding, Hough Transform, boundary detection.

**Unit V : Image Representations and Recognition** **Hours 05**

Boundary representation, Polygonal Approximation, Boundary segments, Boundary description, Regional Descriptors, Topological Feature, Texture, Pattern and Patterns classes, Recognition Based on matching

**Unit VI : Image Compression** **Hours 04**

Introduction, Image compression model, Error free compression, Variable length coding, Bit plan coding, Lossless predictive coding, Lossy compression, Compression standards.

**Core Books:**

1. Rafael C. Gonzalez, Richard E. Woods: Digital Image Processing, Fourth edition, Pearson education.
2. Anil K. Jain: Fundamentals of Digital Image Processing, First Edition, Pearson Education.

**Reference Books:**

1. Burger, Wilhelm, Burge, Mark J: Principles of Digital Image Processing, Springer.
2. Milan Sonka, Vaclav Hlavac, Roger Boyle: Image Processing, Analysis, and Machine Vision, CL Engineering
3. Nixon, Mark, and Alberto Aguado. Feature extraction and image processing for computer vision. Academic press, 2019

**Web References:**

1. https://www.tutorialspoint.com/dip/ [ Basics of Image Processing]
2. https://sisu.ut.ee/imageprocessing/book/1 [All concepts of digital image processing]
3. https://www.cs.auckIand.ac.nz/courses/compsci773s1c/lectures/ImageProcessing-html/topic3.htm [Image segmentation]
4. http://www.eIeteI.p.Iodz.pI/mstrzeI/imageproc/enhancement1.PDF [Image enhancement]

**Course Outcomes:** Upon successful completion of the course, students will,

| | |
|---|---|
| CO1 : | Understand the concept of various types of images and analyze various techniques for intensity transformation and spatial filtering with applications. |
| CO2 : | Understand the concepts of color image processing and noise models. |
| CO3 : | Understand the concepts of image enhancement and its techniques. |
| CO4 : | Learn various image segmentation techniques. |
| CO5 : | Be able to represent and recognize images using various pattern recognition methods its compression techniques. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Digital Image Fundamentals | √ | | | | |
| 2 | Colour Image Processing | | √ | | | |
| 3 | Image Enhancement | | | √ | | |
| 4 | Image Segmentation | | | | √ | |
| 5 | Image Representations and Recognition | | | | | √ |
| 6 | Image Compression | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 1 | 1 | 3 | - | 1 | 3 | 1 | 2 | 3 | 1 | 3 | 2 | 1 |
| CO2 | 2 | 2 | 1 | 1 | - | 2 | - | - | 2 | - | 3 | 3 | - |
| CO3 | 2 | 2 | 1 | 1 | - | 2 | - | - | 2 | - | 3 | 3 | - |
| CO4 | 2 | 2 | 1 | 1 | - | 2 | - | - | 2 | - | 2 | 2 | - |
| CO5 | 2 | 2 | 1 | 1 | - | 2 | - | - | 2 | - | 3 | 3 | - |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

# CTUE303 - Object Oriented Analysis and Design

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction/Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| **Experiential Learning Components:** Unit Tests, Assignments, Case Study, Certification courses, etc. | | | | | |

**Pre-requisite:** None.

**Methodology, Pedagogy & Andragogy:** The course focus on providing a clear understanding of dissecting huge and complex systems. Students will be able to understand and analyse such systems. Students will get clarity about the stages of software development by understanding the software development life cycle and will be able to relate it with real life development. Enriching the students with software development using object oriented methods and principles is the goal.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Fundamentals of Information Systems | 10 |
| 2 | Systems Development Life Cycle and Requirements Determinations | 08 |
| 3 | Introduction to Object Oriented Paradigm | 12 |
| 4 | Object Oriented Concepts | 12 |
| 5 | System Design Using UML | 10 |
| 6 | Computer-Aided Systems Tools and Software Quality | 08 |

**Total Hours (Theory): 60**
**Total Contact Hours:60**

**Detail Syllabus:**

**Unit I: Fundamentals of Information Systems Hours 10**

General Architecture of Systems with basic components, Open and Close Systems, TPS, MIS, DSS and ES Types of Systems, Examples of Real-life Systems, Various users of Systems.

**Unit II: Systems Development Life Cycle and Requirements Determination Hours 08**

Phases of the Classical Systems Development Life Cycle (SDLC) Method: Preliminary Investigation, Determination of Requirements, Request Clarification, Design of System, Development of System, System Testing, Implementation, Maintenance and Support.

**Unit III: Introduction to Object Oriented Paradigm Hours 12**

Introduction, Object Oriented System Development Methodology, Difference between traditional approach and OO approach, Objects, Classes, Attributes, Behaviour and methods

**Unit IV: Object Oriented Concepts Hours 12**

Abstraction, Encapsulation, Class Hierarchy, Inheritance, Multiple Inheritance, Polymorphism, Relationships and association, Aggregation

**Unit V: System Design Using UML Hours 10**

Introduction to Class diagrams, Class Diagram and Object Diagram, Activity diagram, Introduction to dynamic system design, Sequence diagrams, Collaboration diagrams

**Unit VI: Computer-Aided Systems Tools and Software Quality Hours 08**

Introduction to CASE Tools, Advantages and Weakness of CASE, Software Quality Parameters, Various Types of Testing and Test Data: Live Data, Artificial Data, Test Libraries, Training Users, Objectives for Training, Documentation

**Core Books:**

1. James A. Senn: Analysis and Design of Information System: Second Edition: McGraw Hill International: 2010.
2. Timothy C. Lethbridge and Robert Laganière: Object-Oriented Software Engineering: Practical Software Development using UML and Java: Second Edition: McGraw-Hill Education: 2005.

**Reference Books:**

1. Grady Booch, James Rambaugh, Ivar Jacobson: The Unified Modeling Language User Guide: Tenth Impression: Pearson Education, Inc : 2011

**Web References:**

1. http://www.cs.uic.edu/~jbell/CourseNotes/OO_SoftwareEngineering

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Understanding of different types and users of system, Details of SDLC phases, Concepts of feasibility study |
| CO2 : | Understanding of object oriented paradigm, fundamentals of Objects, Classes, Attributes, Behavior and methods |
| CO3 : | Gaining knowledge about object oriented concepts such as abstraction,<br>encapsulation, polymorphism, inheritance, etc. |
| CO4: | Understanding analysis and designing with various UML diagrams |
| CO5 : | Understanding modern tools for object oriented systems analysis and design |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Fundamentals of Information Systems | √ | | | | |
| 2 | Systems Development Life Cycle and Requirements Determination | | √ | | | |
| 3 | Introduction to Object Oriented Paradigm | | | √ | | |
| 4 | Object Oriented Concepts | | | | √ | |
| 5 | System Design Using UML | | | | | √ |
| 6 | Computer-Aided Systems Tools and Software Quality | | | | | √ |

## Course Articulation Matrix:

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | - | - | 1 | 2 | 2 | - | - | - | 1 | - | 2 | 1 | 1 |
| CO2 | - | 2 | 2 | - | 3 | 2 | - | - | 1 | 1 | 2 | 2 | 2 |
| CO3 | - | 2 | 3 | 1 | 2 | 2 | - | - | - | 1 | 2 | 3 | 3 |
| CO4 | 1 | 2 | 2 | 2 | 3 | 3 | 1 | - | 1 | - | 3 | 2 | 3 |
| CO5 | 1 | 3 | 3 | 2 | 2 | 3 | - | 1 | - | 2 | 3 | 3 | 3 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUE304 - Advanced Internet of Things

| Credit | Component | Instruction/Contact Hours( Per Week) | Instruction/ Contact Hours( Per Course) | Experiential Learning Hours( Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components:** Lab Assignments, Practical Test, Case Study | | | | | |

**Prerequisites:** NIL

**Methodology, Pedagogy & Andragogy:** This subject focuses Simulation and hands on of various hardware components used in IoT System

**Outline of the Course:**

| Week No | Practical | Description |
|---|---|---|
| 1 | a) Introduction to Tinkercad Tool for various Hardware simulation.<br>b) Designing various Digital Combinational Circuit and Sequential circuit. | Student can implement the digital circuits and verify its truth tables. |
| 2 | a) Introduction to Arduino Board components and various attachments to Board.<br>b) To blink the LED using arduino programming | Student will learn the details of arduino microcontroller and start with basic programming. |
| 3 | a) Design series of LEDs with various blinking pattern.<br>b) To interface potentiometer with arduino and glow LED and sound buzzer if potentiometer value is above given threshold. | Get the details of LEDs and potentiometer and able to program it with arduino |
| 4 | a) To interface seven segment LED with Arduino using Common anode and Common cathode configurations | Learning the theory of common anode and common cathode types of 7 segment display and program it to display desire pattern. |

| | | |
|---|---|---|
| | b) To design 0 to 9 counter and various Characters to be displayed on 7 segment display, | |
| **5** | To interface two seven segment display and make 00 to 99 counter, odd number counter and even number counter. | Student can make series connection of multiple connection of 7 segment displays and program it to make counters. |
| **6** | a) To interface Servomotor with arduino controller and implement following functionality.<br>b) Rotate the servo motor from 0 to 180 degree in forward direction and 180 to 0 degree in reverse direction.<br>c) Use potentiometer (variable resistor) which controls the position of servomotor. | Learning the concepts of actuators and program it to control the output of an embedded system. Students can practice with servomotor and control its motions. |
| **7** | Interface (16 x 2) LCD with arduino. (Without I2C) and perform various text based operation like scrolling the text, move text left , move text right and displaying custom characters. | Study the configurations of LCD and its pins. Making the LCD connections with Arduino and program it for desired patterns. |
| **8** | Interface (16 x 2) LCD with arduino. (With I2C)<br>1. Print "You are the best" message on 1st line.<br>2. Display Time counter on 2nd line. | Learning I2C types of serial communication methods for fast data transmission and program it with specific I2C Library. |
| **9** | Interface LM35 sensor with arduino board. Write a sketch for<br>1. To read the temperature<br>2. To Display the temperature in Celsius<br>3. To Display the temperature in Fahrenheit. | Learning the concepts of sensors and program it to control the input of an embedded system. Students can practice with LM35 temperature sensors and interface with LCD to display the readings. |
| **10** | To interface Ultrasonic sensor with arduino and make three LEDs on /off based on the threshold distance of an object | Studying the concepts and theory of Ultrasonic sensors and program it to find the distance of n object detected by it.. |
| **11** | a) To interface Keypad with Arduino and study its detail concepts, different types and working | Understanding the keypad as an input device and program it to accept user input. |

| | | |
|---|---|---|
| | b) Designing password based door locking system. | |
| **12** | a) To interface two, three and four DC with arduino using L293D motor driver<br>b) Rotate the motor anticlockwise and clockwise based on certain conditions. | Learning the concepts of actuators and program it to control the output of an embedded system. Students can practice with DC motors and control its motions- with directions. |
| **13** | a) To study detail pin diagram of ESP32.<br>b) Design interfacing programs on ESP32 using Wokwi Tool. | Students can learn ESP32and ESP8266 microcontroller which is used to develop an IoT applications |
| **14** | Design interfacing programs on Raspberry PI Pico using Wokwi Tool. | Students can learn Raspberry PI PICO microcontroller which is used to develop an IoT applications and Bluetooth applications |
| **15** | To develop Mobile application for IoT system using MIT App Inventor Tool | Students can design simple mobile applications which can allow users to interface with IoT system. |

**Total hours: 60**

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### Core Books:

1. Arshdeep Bahga, Vijay Madisetti: Internet of Things: A Hands-On Approach, VPT Publication, 2014.
2. Michael Margolis, Brian Jepson, Nicholas Robert Weldin: Arduino Cookbook: Recipes to Begin, Expand, and Enhance Your Projects , O'Reilly Media Publisher, 3rd Edition, Kindle Edition, 2020.
3. M. Richardson, S. Wallace: Getting Started with Raspberry Pi, O'Reilly, 2012
4. Sudip Mishra, Anandarup Mukherjee and Arijit Roy: Introduction to IoT, Cambridge Press University, 2021

### Reference Books:

1. Muhammad Azhar Iqbal, Enabling the Internet of Things, John Wiley & Sons,2020.
2. Massimo Banzi, Michael Shiloh: Getting Started with Arduino: The Open Source Electronics Prototyping Platform ,Make Communication, 3rd Edition, Kindle Edition, 2014.
3. Sudip Misra, Chandana Roy and Anandarup Mukherjee: Introduction to Industrial Internet of Things and Industry 4.0, CRC Press 2021.
4. Dr. Simon Monk: Programming the Raspberry Pi: Getting Started with Python, McGraw Hill Publication

**Web References:**

1. https://www.tutorialspoint.com/arduino/index.htm [Arduino Fundamentals]
2. https://www.arduino.cc/en/Guide/ [Learning Arduino]
3. https://funduino.de/ [Arduino Fundamentals]
4. https://www.tutorialspoint.com/internet_of_things/internet_of_things_tutorial.pdf [IoT Tutorial]
5. https://www.guru99.com/iot-tutorial.html [IoT Tutorial for beginners]
6. https://www.javatpoint.com/iot-internet-of-things [IoT Tutorial]

**Course Outcome:** Upon successful completion of the course, student will:

| | |
|---|---|
| **CO1:** | To study fundamental concepts of any Microcontroller based IoT System. |
| **CO2:** | To understand the programming concepts to develop IoT based system. |
| **CO3:** | To understand and program various sensors and actuators used in IoT system. |
| **CO4:** | Develop Mobile applications for IoT system. |
| **CO5:** | Able to design and develop IoT system |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 1 | 2 | 3 | 3 | - | 1 | 1 | 1 | 1 | 1 | 2 | 2 | 2 |
| CO2 | 1 | 3 | 3 | 3 | - | 1 | 1 | 1 | 1 | 1 | 2 | 2 | 2 |
| CO3 | 1 | 2 | 3 | 3 | - | 1 | 1 | 1 | 1 | 1 | 2 | 3 | 2 |
| CO4 | 2 | 2 | 3 | 3 | - | 1 | 1 | 1 | 2 | 1 | 2 | 3 | 2 |
| CO5 | 2 | 2 | 3 | 3 | - | 1 | 1 | 1 | 2 | 1 | 2 | 3 | 2 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

# CTUE305 - Introduction to Game Development

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components (Practical):** Lab Assignments, Practical Test, VIVA, certification courses, development of projects, etc. | | | | | |

**Pre-requisite:** CTUC201-Programming the Internet.

**Methodology, Pedagogy & Andragogy:** The course employs a hands-on, project-based methodology where learners incrementally progress from foundational concepts like JavaScript and HTML5 Canvas to advanced Phaser 3 game development. The pedagogy emphasizes engagement through live coding, Q&A sessions, and gamified learning, while using visual aids and scaffolding to build concepts gradually. It ensures immediate feedback and encourages collaboration via pair programming and group discussions. Andragogy focuses on real-world relevance, self-directed learning, and creativity through open-ended tasks and case studies. Problem-solving, and debugging challenges are incorporated to enhance professional skills. The sessions blend concepts with extensive practice, leveraging tools like GIMP and VS IDE. Outcomes are linked to creating mini-game projects, supported by quizzes and peer-reviewed assignments. Overall, the approach ensures interactive, outcome-oriented learning with ample resources for self-paced study.

**Outline of the course:**

| Week No | Practical | Description |
|---|---|---|
| 1 | Javascript – One and Two Dimensional Arrays, Objects and Classes and Inheritance and DOM | Revision of Javascript to access DOM Elements with Object Oriented Programming Concepts of Objects, Classes and Inheritance to facilitate Phaser Game Coding in further units – 4, 5 and 6 of the Syllabus. |
| 2 | HTML5 Canvas Tag, Co-ordinates, Basic Shapes | Implementing <canvas> Tag, Fetching it in JavaScript, Drawing Rectangles with *strokeRect()*, *fillRect()* and *clearRect()* and Basic shapes using |

| | | |
|---|---|---|
| | | Path methods like *beginPath()*, *closePath()*, *moveTo()*, *lineTo()*, *quadraticCurveTo()*, *bezierCurveTo()*, *arc()*, *ellipse()*, *fill()* and *stroke()*. |
| **3** | HTML5 Canvas – Complex Shapes and Solid Colours | Drawing Complex Shapes like Rectangular Spiral, Kite, Smiley, House, etc.<br>Changing and filling Colour using *strokeStyle* and *fillStyle* properties. |
| **4** | HTML5 Canvas – Gradients, Transparency and Text | Implementing shading of two or more colours for using Gradients, Implementing Transparency using *globalAplha* property, Adding Text using *strokeText()* and *fillText()* methods. |
| **5** | Canvas Transformations, Animations | Implementation of Canvas Transformations using *translate(), rotate(), scale(), transform(), resetTransform()* and *setTransform()* methods.<br>Implementation of Animation effects using *setInterval()*, *setTimeOut()* and *requestAnimationFrames()* methods. |
| **6** | Remaining Canvas Topics | Implementation of remaining Canvas topics like Clipping Region, Adding Images, Scaling and Slicing Images, Setting Background of Canvas, downloading and saving Canvas drawings. |
| **7** | Phaser Game Scene, Config, Game Loop, Loading the Asset Cache, Rendering Sprites and Images to Screen | Create a Game Project and explain its basic directory structure, Traversing through the Game Scene life-cycle, Loading the Assets in *preload()* method, Adding Sprites and Images to Screen in *create()* method, Game Loop Implementation using *update()* method. |
| **8** | GIMP Editor for Sprite creation, Sprite-sheet creation and Sprite-Atlas creation | Using GIMP Editor for Designing 2D Sprites, Sprite-Sheets and Sprite-Atlas. |
| **9** | Performing Animations using SpriteSheets, SpriteAtlas and Tweens | Animating the above Sprite-Sheets, Sprite-Atlas in Phaser Game Project with Javascript code. Implementing Tweening Animations separately in Phaser using Javascript code. Implementation of **CoinCollector** Game. |
| **10** | Working with Tile Maps, GameObjects and transformations like | Implementing GameObjects, it's types, properties for performing transformations on GameObjects. Implementation of Tile Maps. |

| | | |
|---|---|---|
| | scaling, flipping, rotating and moving, Adding Group of Sprites | Implementing Group of Sprites and the concept of Children. |
| **11** | User Interactions and Event handling using Keyboard and Pointer events, Adding Sounds | Keyboard and Mouse Pointer Event Handling with *callback* method implementation, Adding sounds of supported File-formats both for background music as well as user-input click or key-press. Implementation of **CatCatcher** Game. |
| **12** | Working with Arcade Physics System in Phaser-3 | Implementing concepts of Speed, Velocity, Drag, Gravity, Acceleration, SlideFactor, Friction, rotation, collision, bounce, mass, immovable, etc. |
| **13** | Advanced Physics Concepts in Phaser-3 | Implementing Collision and Overlap of:<br>(i) Sprite versus Sprite<br>(ii) Sprite versus Group<br>(iii) Group versus Group |
| **14** | Particle System in Phaser-3 | Implementing Emitter and Signals. |
| **15** | Transitioning through Scenes, Concept of World, Camera and Camera-effects | Calling a Scene Function, Listening for Scene Events, Exchange Data via Game Registry, Scene Updating and rendering, Multi-Scene Demo.<br>Concept of World and Camera, Implementing Camera effects. |

**Total hours: 60**

**Core Books:**

1. Travis Faas - An Introduction to HTML5 Game Development with Phaser.js, CRC Press.
2. David Geary – Core HTML5 Canvas, Prentice Hall Publications.
3. Emanuele Feronato – HTML5 Cross Platform Game Development Using Phaser 3, GumRoad.

**Reference Books:**

1. Ernest Adams - Fundamentals of Game Design, New Riders Publication.
2. Jeannie Novak – Game Development Essentials, Delmar Cengage Learning.
3. Steve Fulton and Jeff Fulton – HTML5 Canvas, O'Reilly.
4. Aditya Ravi Shankar – Pro HTML5 Games, Apress Publications.
5. Earle Castledine – HTML5 Games Novice to Ninja, SitePoint Pty. Ltd.

**Web References:**

1. https://docs.phaser.io/phaser/getting-started/what-is-phaser
2. https://developer.mozilla.org/en-US/docs/Web/API/Canvas_API/Tutorial

3. https://www.tutorialspoint.com/html/html_canvas.htm
4. YouTube – How to make Sprite-sheets using GIMP
5. https://www.spriters-resource.com/
6. https://www.freepik.com/free-photos-vectors/sprite-sheets
7. https://opengameart.org/art-search?keys=sprite-sheet
8. https://www.vecteezy.com/free-vector/sprite-sheet

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Able to understand Web Game Development Technologies |
| CO2 : | Capable of Designing Game Graphics/Sprites using HTML5 Canvas and GIMP Editor |
| CO3 : | Learn Animation of Game Graphics/Sprites using HTML5 Canvas and Phaser-3 |
| CO4 : | Capable of Development of Game Logic in Phaser-3 |
| CO5 : | Learn to Work with Physics Concepts of Phaser-3 Game Engine |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 2 | 1 | 3 | - | 2 | 2 | - | - | 1 | - | 1 | - |
| CO2 | 3 | 3 | 3 | 2 | - | 1 | 2 | - | - | 1 | - | 1 | - |
| CO3 | 2 | 1 | 2 | 3 | - | 3 | 1 | - | - | 2 | - | 1 | - |
| CO4 | 3 | 3 | 3 | 2 | 1 | 2 | 2 | 2 | - | 3 | 1 | 3 | 2 |
| CO5 | 3 | 3 | 2 | 2 | - | 2 | 1 | - | - | 1 | - | 3 | - |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# TEACHING SCHEME & DETAILED SYLLABUS

# FOR

# BSc-IT PROGRAMME (6th SEMESTER) AS PER NEP 2020

# EFFECTIVE FROM ACADEMIC YEAR 2025-26

## B.Sc. (IT) Semester-VI

| Course Code | Course Name | Theory / Practical / Others | Marks (Theory) | | Marks (Practical) | | Credit | NEP - 2020 Classification |
|---|---|---|---|---|---|---|---|---|
| | | | CCE | SEE | CCE | SEE | | |
| **CTUC303** | Project Work | VIVA | 150 | 150 | - | - | 12 | Major Core |
| **CTUE306**<br>**CTUE307** | (Select any one)<br>3. Multi-paradigm Programming Language<br>4. Frameworks and Applications | Theory and Practical | 25 | 25 | 25 | 25 | 04 (02-Theory) (02-Practical | Minor |
| **CTUS301** | Mobile Application Development | Theory and Practical | 25 | 25 | 25 | 25 | 04 (02-Theory) (02-Practical | SEC |
| **HSUA302** | Professional Communication, Soft Skills and Personality Development | Practical | - | - | 25 | 25 | 02 | AEC |

*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)
*SEE- Semester –End- Evaluation

# CTUC303 - Project Work

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction/ Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 12 | VIVA | 12 | 180 | 50 | 110 |
| **Experiential Learning Components (Practical):** VIVA | | | | | |

**Pre-requisite:** None

**Introduction:**

The Project work constitutes a major component in most Professional Programmes. It needs to be carried out with due care, and should be executed with seriousness by the students. The project work is not only a partial fulfilment of the course requirements, but also provides a mechanism to demonstrate your skills, abilities and specialization. The Project work is one that involves practical work for understanding and solving problems in the field of computing. The objective of the project is to help the student develop the ability to apply theoretical and practical tools/techniques to solve real life problems related to industry, academic institutions and research laboratories.

**Methodology, Pedagogy & Andragogy:** The core objective of the course is to provide project and practical based on deep learning, concept understanding, its analysis, deriving inferences and documenting it towards productive outcome of the subject. The main objective of this mini project is to let the students apply the programming knowledge to a real-world situation/problem and exposed the students with specific programming skills and help in developing a working model in terms of application.

**Objective**:

Student should be able to:

- demonstrate a systematic understanding of project contents;
- understand methodologies and professional way of documentation;
- know the meaning of different project contents, and
- understand established techniques of project report development.

**Importance of the project**

The Project is not only a part of the coursework, but also a mechanism to demonstrate student's abilities and specialization. It provides the opportunity for them to demonstrate originality, teamwork, inspiration, planning and organization in a software project, and to put into practice some of the techniques they have been taught throughout the previous courses. The Project is important for a number of reasons. It provides students with:

- Opportunity to specialize in specific areas of computer science;
- Future employers will most likely ask about project at interview;
- Opportunity to demonstrate a wide range of skills and knowledge learned,
- Encourages integration of knowledge gained in the previous course units.

**Course Outcomes:** Upon successful completion of the course, students will,

| | |
|---|---|
| CO1 : | Understand the implementation of concepts of SDLC and Software Engineering. |
| CO2 : | The programming concepts they learn during their academics, it will be converted in to the actual implementations. |
| CO3 : | Be exposed to understand the requirement of proposed software and implement these requirements in terms of programming logic and methods. |
| CO4 : | Understand the difference between a program and professional application/product/software. |
| CO5 : | Learn different categories of applications like Desktop application, Web applications, etc. |

**Guidelines:**

Mini Project is in house project development. Every student is required to carry out Mini Project work under the supervision of a guide provided by the placement Coordinator. The guide shall monitor progress of the student continuously. A candidate is required to present the progress of the Mini Project work during the semester as per the schedule provided by the placement Coordinator.

Mini Project proposal should be prepared in consultation with project guide. It should clearly state the objectives and environment of proposed Mini Project to be undertaken. Project documentation must be with the respect to the project only. Project report should strictly follow the points suggested in format of project report. Placement coordinator will provide the format of project report. Student has to submit one copy of

Mini Project to the institute. Each Student is required to make a copy of Mini Project and submit along with Mini Project report.

**Evaluation:**

The project must be evaluated in two aspects:

a. Internal (100 Marks):

i. Reporting to internal project guide

ii. Incorporation of suggestions by project guide

iii. Internal Project viva examination

b. External (200 Marks):

i. Project Report Preparation & Evaluation

ii. External Project Viva Examination

**Web References:**

1. http://techwhirl.com/writing-software-requirements-specifications/[For effective SRS]
2. http://www.ibm.com/developerworks/websphere/library/techarticles/0306_perks/perks2.ht ml [ For best practices of Software Project Development]
3. http://www.uacg.bg/filebank/acadstaff/userfiles/publ_bg_397_SDP_activities_and_steps.pd f [ Requirement analysis guidelines]
4. http://www.cs.wustl.edu/~schmidt/PDF/design-principles4.pdf [Software Design Principles and Guidelines]
5. http://www.cse.hcmut.edu.vn/~hiep/KiemthuPhanmem/Tailieuthamkhao/Effective%20Soft ware%2 0Testing%20-2050%20specific%20ways%20to%20improve%20your%20testing.pdf [ForEffective Software Testing]
6. http://www.cs.uics.edu/~jbell/CourseNotes/OO_SoftwareEngineering/SE_Project_Report_Template .pdf [ For guidelines to prepare software project report]

## Course Articulation Matrix:

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 3 | 3 | 2 | 3 | 3 | 3 | 3 | 3 | 2 | 2 | 2 | 3 | 2 |
| CO2 | 3 | 3 | 3 | 2 | 3 | 3 | 3 | 3 | 3 | 2 | 2 | 2 | 3 | 2 |
| CO3 | 3 | 3 | 3 | 2 | 3 | 3 | 3 | 3 | 3 | 2 | 2 | 2 | 3 | 2 |
| CO4 | 3 | 3 | 3 | 2 | 3 | 3 | 3 | 3 | 3 | 2 | 2 | 2 | 3 | 2 |
| CO5 | 3 | 3 | 3 | 2 | 3 | 3 | 3 | 3 | 3 | 2 | 2 | 2 | 3 | 2 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

# CTUE306 - Multi-Paradigm Programming Language

| Credit | Component | Instruction/Contact Hours( Per Week) | Instruction/Contact Hours( Per Course) | Experiential Learning Hours( Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 02 | Theory | 02 | 30 | 30 | 60 |
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components (Theory):** Unit Tests, Assignments, Case Study etc. **Experiential Learning Components (Practical):** Lab Assignments, Practical Test, VIVA | | | | | |

**Pre-requisite:** CTUC103-Introduction to Object Oriented Programming.

**Methodology, Pedagogy & Andragogy:** Theory sessions are required to address computational power of Python through its ability to deploy programs using functional, object oriented and web-based aspect. Practical sessions demonstrate the implementation of the concepts which are taught during the theory sessions. Case study will help the students to come out with one working module in any of the paradigm through Python.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Introduction to Python | 02 |
| 2 | Python Programming - I | 05 |
| 3 | Python Programming - II | 05 |
| 4 | Object Oriented Programming with Python | 06 |
| 5 | Exception Handling | 06 |
| 6 | Web based programming with Python | 06 |

**Total Hours (Theory): 30**
**Total hours (Practical): 60**
**Total Contact Hours: 90**

**Detail Syllabus:**

**Unit I: Introduction to Python** **Hours 02**

Overview of programming Paradigms: -Imperative, Functional, logic and object oriented, Python As programming Language, Features and comparison of python with other programming language, Downloading and Installing, Running Python, Python Documentation, Python Programming structure, Python various Statement & its Syntax, Python Identifiers, Memory Management scheme adopted in Python, Conditional & Branching Statement in python, Different looping constructs available in Python: While, for, continue, break and pass statement.

**Unit II: Python Programming - I** **Hours 05**

Python objects and its variations: List, Tuple, Dictionary & Set. File open, close and other file manipulation functions and Command Line Argument.

**Unit III: Python Programming - II** **Hours 05**

What are functions, calling functions, creating functions, passing functions, formal arguments, variable length arguments, default arguments, returning values from the functions, returning multiple values from the functions, functional programming, variable scope and recursion. Built In functions for List, Tuple, Set and Dictionary.

**Unit IV: Object Oriented Programming with Python** **Hours 06**

Basics of class, object and instance. Class level attribute and instance level attribute. Constructor and other magic methods. Bound and unbound methods. Built in functions for python class and objects

**Unit V: Exception Handling with Python** **Hours 06**

Exceptions in Python, Detecting & handling exception, Raising exception, Standard exception & creating exception.

**Unit VI: Web based Programming with Python** **Hours 06**

Introduction of Django and concept of MVT programming. Developing applications using Django: Database connectivity and CRUD operations and Designing Templates for User Interfaces.

**Outline of the Course:**

| Week No | Practical | Description |
|---|---|---|
| 1 | Downloading and Installation: Installation of Python and Python documentation. | Student will be familiarizing with downloading and installing Python. |
| 2 | Primary and secondary prompts, basic statements and expressions, Datatypes and Operators | Students will understand executing programs using Python. They also learn various data types supported by Python. |
| 3 | Conditional statements and Iterative statements | Students will to learn conditional and looping statements in Python. |
| 4 | Introduction to sequence types supported by Python: String, List, Tuple | Students will learn how to implement String, List, and Tuple types supported by Python. |
| 5 | Mapping and other types: Dictionary, set | Students will learn how to implement Dictionary and Set types supported by Python. |
| 6 | Exposure of python built-in Functions | Student will be familiarize with built-in functions in Python. |
| 7 | Python User-Define Functions | Students will to learn to create own function for reusability of the code. |
| 8 | Python Comprehension and filter() and map() | Students will learn to optimize the conditional and looping statements. |
| 9 | Introduction to Object Oriented Programming | Student will be familiarize with the concept of Object Oriented Programming using Python. |
| 10 | Introduction to Inheritance and concept of method overriding. | Student will be familiarize with the concept of Inheritance, method overriding using Python |
| 11 | Introduction to Polymorphism, Encapsulation, Abstraction and magic methods | Student will be familiarize with the concept of Polymorphism, Encapsulation, Abstraction and magic methods. |
| 12 | Exception Handling in python | Student will be able to understand exception handling using Python. |
| 13 | Python Module and Libraries | Student can able to learn python module and libraries to use. |

| | | |
|---|---|---|
| 14 | Introduction to Web Framework: Django. | Students understand MVT based web application development using Django. |
| 15 | Creating web application based on MVT and database. | Student can apply concept of MVT and develop applications for CRUD operations. |

**Total hours (Practical): 60**

**Core Books:**


1. Wesley J. Chun: Core Python Programming, 2nd edition, Pentice Hall,2006. 2. Megnus Lie Hetland: Beginning Python from novice to professional, 2nd edition, Apress,2009.
2. Megnus Lie Hetland: Beginning Python from novice to professional, 2nd edition, Apress,2009.
3. Julia Elman, Mark Lavin: Lightweight Django: Using REST, WebSockets, and Backbone, 1st edition, O'Reilly Media, Inc, 2014


**Reference Books:**


1. Richard L. Halterman: Learning to Program with Python, Southern Adventist University 2011.
2. Mark Lutz: Programming Python, 4th Edition,O'reilly, 2011. 3.Steve Holden: Python Web Programming, 1st edition,2002.


**Web References:**


1. https://www.python.org/about/gettingstarted/ [For beginners of Python]
2. htttp://ocw.mit.edu/courses/electrical-engineering-and-computerscience/6-189-agentleintroduction-to-programming-using-pythonjanuary-iap-2008/ [Python Materials]
3. http://people.cs.aau.dk/~normark/prog303/html/notes/paradigms_themesparadigmoverview-section.html [Overview of Programming Paradigms]
4. https://docs.python.org [Python Documentation]
5. https://www.tutorialspoint.com/django/index.htm [Overview of Django Framework]

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Able to understand scripting and the contributions of scripting languages. |
| CO2 : | Able to aware the built-in objects of Python and object-oriented concepts |
| CO3 : | Able to understand and implement functional and Object Oriented Programming |
| CO4 : | Able to learn to exception handling |
| CO5 | Able to grasp Python for web based programming |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Introduction to Python | √ | | | | |
| 2 | Python Programming - I | | √ | | | |
| 3 | Python Programming - II | | | √ | | |
| 4 | Object Oriented Programming with Python | | | √ | | |
| 5 | Exception Handling | | | | √ | |
| 6 | Web based programming with Python | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | - | - | - | 2 | - | - | - | - | - | 3 | 1 |
| CO2 | - | 3 | 3 | - | - | - | 3 | 1 | - | - | - | 3 | 1 |
| CO3 | 2 | - | 2 | - | 3 | - | 1 | - | - | - | - | 3 | 1 |
| CO4 | - | - | 3 | 3 | 1 | - | 2 | - | - | - | - | 3 | 1 |
| CO5 | 1 | 2 | 3 | 3 | 1 | 2 | 2 | 1 | 1 | 1 | 1 | 3 | 1 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUE307 - Frameworks and Applications

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction/ Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 02 | Theory | 02 | 30 | 30 | 60 |
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Component (Theory):** Unit Tests, Assignments, Case Study etc.<br>**Experiential Learning Components (Practical):** Lab Assignments, Practical Test, VIVA | | | | | |

**Pre-requisite:** CTUC205 - Basic knowledge of C#.Net.

**Methodology, Pedagogy & Andragogy**: During theory lectures, illustrations will be provided to emphasize the importance of foundational and advanced features of the .NET framework, including its application in developing web-based and console-based solutions using C#. Practical sessions will focus on hands-on learning, where students will be tasked with creating solutions such as collections-based applications, file handling utilities, threading demonstrations, and minimal web APIs.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Introduction to C#.NET | 03 |
| 2 | Collections and Generics | 05 |
| 3 | File Handling and Serialization | 05 |
| 4 | Threading and Asynchronous Programming | 05 |
| 5 | Web Applications and API Development | 07 |
| 6 | Networking and API Integration | 05 |

**Total Hours (Theory): 30**
**Total hours (Practical): 60**
**Total Contact Hours:90**

**Detail Syllabus:**

**Unit I: Introduction to C#.NET** **Hours 03**

introduces the .NET ecosystem, including .NET Framework, .NET Core, and .NET 5+, focusing on their differences and cross-platform capabilities, development environment with Visual Studio, the structure of a C# program, and exploring data types, variables, and constants.

**Unit II: Collections and Generics** **Hours 05**

starting with arrays and dynamic collections like List<T> and Dictionary<TKey, TValue>. Introduction of LINQ for querying collections, emphasizing its query and method-based syntax. The advantages of generics for type safety and code reusability, with examples of creating generic methods and classes

**Unit III: File Handling and Serialization** **Hours 05**

File handling techniques using System.IO to create, read, write, and delete files and directories. Working with text and binary files, along with serialization and deserialization using System.Text.Json and Newtonsoft.Json.

**Unit IV: Threading and Asynchronous Programming** **Hours 05**

Explores multithreading and parallel execution, concepts like System.Threading.Thread, Task Parallel Library (TPL), and async/await for asynchronous programming. Introduces background workers and thread pools for running tasks efficiently in the background, demonstrating practical implementations for performance optimization.

**Unit V: Web Applications and API Development** **Hours 07**

Introduces the development of web applications and APIs using ASP.NET. build lightweight web APIs using minimal APIs, set up web API projects, define routes, and manage controllers for handling HTTP requests and responses. Key topics include middleware integration, dependency injection, and securing APIs with authentication and authorization techniques. Reducing boilerplate and streamlining the development process for efficient and functional web applications.

**Unit VI: Networking and API Integration** **Hours 05**

Understanding of basic networking concepts in C#, including sockets and HTTP protocols. Work with HttpClient for sending HTTP requests, consuming REST APIs, and handling responses. The unit also covers JSON parsing with libraries like Newtonsoft.Json and System.Text.Json for working with structured data.

**Outline of the Course (Practical) :**

| Week No | Practical | Description |
|---|---|---|
| 1 | Console Environment without IDE | To set the path and environment variable to execute a console-based program without any IDE |
| 2 | Introduction to Visual Studio IDE and .NET | Familiarize with the Visual Studio IDE and the basic programming structure of C#.NET . |
| 3<br>4 | Working with Collections | Using built-in collections like List<T>, Dictionary<TKey, TValue>, and arrays. Implementing basic operations like adding, removing, and accessing items from these collections. |
| 5 | Introduction to Generics and LINQ | Creating generic methods and classes for better type safety. Using LINQ queries to filter, sort, and manipulate collections, learning both query and method syntax. |
| 6 | File Handling and Serialization | Using operators, control statements, and working with arrays in C#. |
| 7<br>8<br>9 | Threading and Asynchronous Programming | Implementing simple threading using System.Threading.Thread. Creating background tasks and understanding the impact of multithreading on application performance. Using async and await for performing asynchronous operations. Implementing methods that run tasks in the background and return results asynchronously. |
| 10<br>11 | Networking with HttpClient | Using HttpClient to make HTTP requests, consume REST APIs, and display responses in the console. Basic error handling and JSON parsing from API responses. |
| 12<br>13 | Working with Forms in Web Applications | Developing simple HTML forms for user input. Using ASP.NET to process form data in the backend, validate user input, and display results dynamically in web pages. |
| 14<br>15 | Building Minimal Web APIs | Creating simple APIs using ASP.NET Core minimal API. Defining routes, handling HTTP GET requests, and returning data in JSON format. Introduction to Web API architecture. |

**Total hours (Practical): 60**

**Core Books:**

1. Christian Nagel : Professionals C# and .NET 2021 Edition , Wiley – India, WROX
2. Albahani Josheph: C# 4.0 Pocket Reference, Shroff publication,2010.
3. Michele Leroux Bustamante: Microsoft .NET: Architecting Applications for the Enterprise, Microsoft Press
4. Mark Michaelis: C# in Depth, Manning Publications, 2019

**Reference Books:**

1. Jon Skeet, C# in Depth, Managing Publication,2010
2. Drayton David: C# Language Pocket Reference, Shroff publication,2005
3. Andrew Troelsen: Pro C# 8 with .NET Core, Apress, 2020
4. K. Scott Allen: Professional C# 7 and .NET Core, Wiley – India, WROX

**Web References:**

1. .http://csharp.net-informations.com/ [to learn C# and .NET fundamentals]
2. https://www.tutorialspoint.com/csharp/csharp_socket_programming.htm
   [to learn socket programming]
3. https://dotnet.microsoft.com/en-us/learn/csharp [tutorial and videos to learnC#]
4. https://dotnettutorials.net/course/csharp-dot-net-tutorials/
5. https://www.w3schools.com/cs/ [ C# tutorial for Linq]

**Course Outcomes:** Upon successful completion of the course, the students will:

| | |
|---|---|
| CO1 : | Mastery of C# and .NET fundamentals, including syntax, control structures, and data types to develop console-based applications. |
| CO2 : | Proficiency in working with collections and generics to efficiently manage and manipulate data in C#. |
| CO3 : | Capability to handle file operations, serialization, and deserialization, ensuring data persistence and format conversion. |
| CO4 : | Understanding of threading and asynchronous programming, enabling the development of efficient, responsive, and concurrent applications. |
| CO5: | Ability to design and implement web applications and APIs using ASP.NET, integrating networking capabilities for data communication. |

**Course Outcomes Mapping :**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Introduction to C#.NET | √ | | | | |
| 2 | Collections and Generics | | √ | | | |
| 3 | File Handling and Serialization | | | √ | | |
| 4 | Threading and Asynchronous Programming | | | | √ | |
| 5 | Web Applications and API Development | | | | | √ |
| 6 | Networking and API Integration | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | P10 | P11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 1 | 1 | 2 | 3 | - | 3 | 1 | - | - | 1 | - | 1 | - |
| CO2 | 1 | 1 | 2 | 2 | - | 3 | 2 | - | - | 2 | - | 1 | - |
| CO3 | 1 | 1 | 3 | 2 | - | 2 | 2 | - | - | 2 | 1 | 1 | - |
| CO4 | 2 | 3 | 3 | 3 | 1 | 3 | 3 | - | - | - | 1 | 3 | 1 |
| CO5 | 2 | 3 | 3 | 3 | 1 | 3 | 3 | - | - | - | 1 | 3 | 1 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUS301 - Mobile Application Development

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 02 | Theory | 02 | 30 | 30 | 60 |
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components (Theory):** Unit Tests, Quizzes, Assignments, Case Studies etc.<br>**Experiential Learning Components (Practical):** Lab Assignments, Practical Test, VIVA | | | | | |

**Pre-requisite:** CTUC103 - Object Oriented Programming Concepts.

**Methodology, Pedagogy & Andragogy:** The teaching methodology integrates theoretical and practical approaches to provide a comprehensive understanding for mobile application development subject for android application development using Kotlin language. Pedagogy involves interactive lectures, real-world examples, and guided coding sessions to build foundational knowledge and problem-solving skills. Andragogic strategies focus on self-directed learning through projects, case studies, and hands-on lab sessions, enabling students to apply theoretical concepts in developing functional mobile applications. Regular assessments, collaborative group tasks, and instructor feedback ensure an engaging and practice-oriented learning environment. **Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1. | Introduction Android with Kotlin | 05 |
| 2. | Android Layouts and Views | 07 |
| 3. | Android Widgets and Adapters | 05 |
| 4. | Advanced Designing: Material Design | 04 |
| 5. | Data Storage: SQLite, Firebase | 06 |
| 6. | Deploying Web Services | 03 |

**Total Hours (Theory): 30**
**Total hours (Practical): 60**
**Total Contact Hours: 90**

**Detail Syllabus:**

**Unit I: Introduction Android with Kotlin** **Hours 05**

Introduction to Mobile Applications and Android, Basics of Kotlin, setting up development environment, Dalvik Virtual Machine, Android Application Structure, DDMS, Android Manifest File, Gradle, Android permissions, APK, Basic Building blocks – Activities, Services, Broadcast Receivers & Content providers.

**Unit II: Android Layouts and Views** **Hours 07**

Working with Containers: LinerLayout, RelativeLayout, TableLayout, ScrollView, and FrameLayout, Fragment, Views: Toast, TextView, EditText, Button, ImageButton, CheckBox, ToggleButton, RadioButton, RadioGroup, ProgressBar, AutoComplete TextView, Components for communication Intents & Intent Filters Intents: Explicit Intents, Implicit Intents, switching between activities and passing data between activities using Intents, Logging.

**Unit III: Android Widgets and Adapters** **Hours 05**

Calendar: Manage Date and Time, Adapters: ArrayAdapter, BaseAdapter ListView and ListActivity, GridView, Gallery Menus: Option menu, Context menu, Sub menu Tabs and TabActivity Alert Dialogs, Android Toolbar.

**Unit IV: Advanced Designing: Material Design** **Hours 04**

Android Design Support Library, Vector Drawables, Basics of material design components and principles, Android styles and themes, Animations and Transitions.

**Unit V: Data Storage: SQLite, Firebase** **Hours 06**

Basics of Storage in Android, Shared Preference: Save key-value data, Store Data in SQLite Database, store data in a firebase, save files on Device, Sharing of data.

**Unit VI: Deploying Web Services** **Hours 03**

Working with Web Service in Android, Deploying Web Service.

**Outline of the Course (Practical):**

| Week No | Practical | Description |
|---|---|---|
| 1 | Kotlin, Toast | Kotlin Code Programs, Toast Messages "Hello World" |
| 2 | TextView, EditText, Activity, Intent | Use of Textview, EditText and Toast with Operations. Use of Activity and Intent for values. |
| 3 | Views, EditText, ViewBinding, FindViewById, layouts | Create the calculator app and use proper controls, Handle the views without using FindViewById |
| 4 | Implicit, Explicit Intent, Activity Life Cycle | Manage Student Information using Intent and also create the App to learn the activity life cycle. |
| 5 | Different Activities, UI, toggle button, progress bar | Splash Screen, Login and user creation as well as create the application for addition operation of two variables. |
| 6 | EditText, Button | Create an app to perform the arithmetic operations using 4 buttons. |
| 7 | Implicate Intent, Explicit Intent, UI with functions | Open CHARUSAT website using Implicit Intent. Create an application to share data with native mobile apps for browser, calling app with optimized code. |
| 8 | Shared Preference and UI with complete application, Material design | Use shared Preferences for the session management and also create application for one college event using radio button, check boxes and spinners. |
| 9 | Images, Alert Dialog box | Display Toast message by clicking on Image Button and open CHARUSAT website.<br>Use alert dialog for the messages in the application |
| 10 | AutoCompeteTextView | Create the application using AutocompleteTextView for Countries list. |
| 11 | Adapters, Array Adapter, Base Adapter, ListView | Use the Spinner using Array Adapter. Create the app with student photos using base adapter |
| 12 | Alert Dialog, Back Press | Manage the back press event using alert dialog. |
| 13 | Menu | Create an application with different types of menues.<br>Context Menu<br>Popup Menu<br>Option Menu |
| 14 | Recyclerview, SQLite | Create app with Photos and names using recyclerview. Also create the app for CRUD operations using SQLite. |

| | | |
|---|---|---|
| **15** | Firebase | Create the app for authentication using firebase. Also create the app for data management in Firestore. |

**Total hours: 60**

**Core Books:**

1. Wei-Meng Lee: Beginning Android 4 Application Development, Wiley India Pvt Ltd.
2. Reto Meier: Professional Android 2 Application Development, Wrox publication
3. Mark L. Murphy: The Busy Coder's Guide to Android Development.

**Reference Books:**

1. Dawn Griffiths, David Griffiths: Head First Android Development, O'REILLY publication
2. Mark L. Murphy: Beginning Android 2, APRESS publication
3. Mark L. Murphy: The Busy Coder's Guide to Advanced Android Development, Commons Ware, LLC.

**Web References:**

1. https://kotlinlang.org/ [Kotlin Tutorials]
2. https://developer.android.com/guide/ [Android Tutorials]
3. www.vogella.com/tutorials/android.html [Android Practical Tutorials]
4. https://www.androidhive.info/ [Android Practical Tutorials]

**Course Outcomes:** Upon successful completion of the course, students will,

| | |
|---|---|
| CO1 : | Be acquainted with Android OS |
| CO2 : | Be aware with Android Development Tools |
| CO3 : | Be aware with Android Development Tools |
| CO4 : | Be capable to create interactive mobile application that handle data |
| CO5 : | Be able to create mobile application that works web service |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Introduction Android with Kotlin | √ | | | | |
| 2 | Android Layouts and Views | | √ | | | |
| 3 | Android Widgets and Adapters | | | √ | | |
| 4 | Advanced Designing: Material Design | | | | √ | |
| 5 | Data Storage: SQLite, Firebase | | | | √ | |
| 6 | Deploying Web Services | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | 2 | 1 | 3 | - | 2 | 1 | - | - | 2 | - | - | - |
| CO2 | 2 | - | 3 | - | 2 | 1 | - | - | 2 | 2 | 3 | 2 | 1 | 1 |
| CO3 | - | 3 | - | 2 | - | 1 | 3 | 2 | - | - | - | 3 | 3 | 2 |
| CO4 | 2 | 1 | 2 | 2 | 2 | 1 | 2 | - | 2 | - | 2 | 2 | 3 | 2 |
| CO5 | - | 2 | 2 | 3 | 1 | - | 2 | 2 | 2 | 2 | - | - | 2 | 3 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

# TEACHING SCHEME & DETAILED SYLLABUS

# FOR

# BSc-IT PROGRAMME
# ($7^{th}$ SEMESTER)
# AS PER NEP 2020

# EFFECTIVE FROM ACADEMIC YEAR 2025-26

## B.Sc. (IT) Semester-VII

| Course Code | Course Name | Theory / Practical/ Others | Marks (Theory) | | Marks (Practical) | | Credit | NEP -2020 Classification |
|---|---|---|---|---|---|---|---|---|
| | | | CCE | SEE | CCE | SEE | | |
| **CTUC401** | Enterprise Computing using Java EE | Theory and Practical | 50 | 50 | 25 | 25 | 06<br>(04-Theory)<br>(02-Practical | Major Core |
| **CTUC402** | Database Technologies | Theory and Practical | 50 | 50 | 25 | 25 | 06<br>(04-Theory)<br>(02-Practical | Major Core |
| <br>**CTUE401**<br>**CTUE402**<br><br>**CTUE403** | (Select any one)<br>1. Cloud Computing<br>2. Cryptography and Network Security<br>3. Blockchain Essentials | Theory | 50 | 50 | - | - | 04 | Minor |
| <br>**CTUP401**<br>**CTUR401** | (Select anyone)<br>1. On-Job Training-I<br>2. Research Project-I | VIVA | - | - | 75 | 75 | 06 | Research Project / Dissertation |

*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)
*SEE- Semester –End- Evaluation

# CTUC401 - Enterprise Computing using Java EE

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components (Theory):** Unit Tests, Assignments, Case Study etc. | | | | | |
| **Experiential Learning Components (Practical):** Lab Assignments, Practical Test, VIVA | | | | | |

**Pre-requisite:** Object Oriented Programming Using Java (CTUC301 )

**Methodology, Pedagogy & Andragogy:** Advanced Java fundamentals like JDBC, Servlet, JSP and Servlet Security provides capabilities and features for developing web application which is essential for developer in order to understand entire web application development life cycle. During the theory sessions of the syllabus, fundamentals of the mentioned technologies will be explained in detail with real time examples. Moreover, practical session will provide hands-on exposure about JDBC, Servlet, JSP and Servlet Security with variety of practical assignment. Also, MVC architecture using these technologies will be covered during theory as well as practical sessions.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1. | JDBC Programming | 10 |
| 2. | Web Application Components Fundamentals | 08 |
| 3. | Servlet API | 12 |
| 4. | Advanced Servlet Features and Security | 10 |
| 5. | Java Server Pages | 12 |
| 6. | Packaging and Deploying Web Applications | 8 |

**Total Hours (Theory) : 60**
**Total hours (Practical) : 60**
**Total Contact Hours: 120**

**Detail Syllabus:**

**Unit I: JDBC Programming Hours 10**

The JDBC Connectivity Model, Database Programming: Connecting to the Database, creating a SQL Query, Getting the Results, Updating Database Data, Error Checking and the SQLException Class, the SQLWarning class, the Statement Interface, PreparedStatement, CallableStatement, ResultSet Interface, Updatable Result sets, JDBC Types, Executing SQL Queries, ResultSetMetaData, Executing SQL Updates, Transaction Management.

**Unit II: Web Application Components Fundamentals Hours 08**

Basics of web application development and web components, Introduction of MVC design pattern, Fundamentals of: Containers, Packaging Web Applications, Web Application Structure, JAR Files, WAR Files, HTTP, HTTP request methods, HTML Form Processing, HTTP request response cycle.

**Unit III: Servlet API Hours 12**

Servlet Model: Introducing Servlet, Introducing Servlet & the MVC Pattern, Introducing javax.servlet Packages, Introducing HTTP & Servlets, Understanding the Request/ Response Cycle, Input & Output Streams, Introducing Servlet/ Container Communication, Introducing ServletContext and ServletConfig, RequestDispatcher Interface. The Filter API: Filter, Filter Chain and Filter Config.

**Unit IV: Advanced Servlet Features and Security Hours 10**

Cookies and Session Management: Understanding state and session, Understanding Session Timeout and Session Tracking, URL Rewriting. Understanding the Stateless nature of HTTP, Why Track Client Identity & State Maintain Sessions, Session Management Using the Servlet API.

**Unit IV: Java Server Pages Hours 12**

JSP Overview: The Problem with Servlets, Life Cycle of JSP Page, JSP Processing, JSP Application Design with MVC, Setting Up the JSP Environment JSP Directives, JSP Action, JSP Implicit Objects JSP Form Processing, JSP Session and Cookies Handling, JSP Session Tracking JSP Database Access, JSP Standard Tag Libraries, JSP Custom Tag, JSP Expression Language, JSP Exception Handling, JSP XML Processing.

**Unit – VI: Packaging and Deploying Web Applications Hours 08**

Basics of server container, build a WAR artifact for the web application, configure the Docker connection node and deploy the application to server container.

**Outline of the Course (Practical):**

| Week No | Practical | Description |
|---|---|---|
| 1 | Basics JDBC | Configuration of Java SDK, NetBeans and Oracle 11 G for creating database. |
| 2 | JDBC Operations | Understanding and executing jdbc queries for creating and managing database. |
| 3 | Statement types | Implementing Statement, PreparedStatement, Callable Statement for handing dynamic queries and calling stored procedures and functions in database. |
| 4 | ResultSet, ResultSetMetaData and Types of ResultSet | Implementing ResultSetMetaData to know the more specific information about the records and to update the records even if the records are fetched in ReslultSet. |
| 5 | Web Application Components | Able to implement and deploy the applications on web server while configuring the Tomcat server. Understanding JAR Files, WAR Files, HTTP, HTTP request response cycle. |
| 6 | Web Application Components | Understanding the concept of Model View Controller(MVC) with proper code. |
| 7 | Servlet API | Able to configure the Tomcat Server, Understanding servlet, MVC architecture. Implementing ServletContext and ServletConfig, RequestDispatcher Interface and Filters in servlets. |
| 8 | JDBC Connectivity with Servlet | Creating servlet application for implementing CRUD operations using JDBC with servlets. |
| 9 | Session management using Servlet | Implementing the concepts of session techniques such as : Hidden Form Fields, URL Rewriting, Session and Cookies. |
| 10 | Java Server Pages (JSP) | Understanding JSP for web application, JSP Life cycle, MVC design with JSP, JSP directives. |
| 11 | JSP Actions | Understanding JSP implicit object of JSP, Actions in JSP and JDBC using JSP. |
| 12 | JSP Session management | Managing the session of the user using JSP session and cookies. |

| | | |
|---|---|---|
| **13** | JSP Standard Tag Libraries | JSP Custom Tag, JSP Expression Language, JSP Exception Handling. |
| **14** | Packaging and Deploying Web Applications | Building a WAR artifact for the web application, configure the Docker connection node and deploy the application to server container |
| **15** | Packaging and Deploying Web Applications | Deploying and configuring docker for web applications. |
| | | **Total Hours: 60** |

**Core Books:**

1. Cay S Horstmann, Gary Cornell: Core Java, Volume II – Advanced Features, 8th Edition, Pearson Education.

2. Marty Hall, Larry Brown: Core Servlets and JavaServer Pages, Volume 2, Advanced Technologies, 2nd Edition, Pearson Education, 2008.

**Reference Books:**

1. Bryan Basham, Kathy Sierra and Bert Bates: Head First Servlet and JSP, O'Reilly Publication, 1st Edition

2. James Keogh: Complete Reference J2EE, Mcgraw publication

**Web References:**

1. http://courses.coreservlets.com/Course-Materials/csajsp2.html [Servlet Basics]

2. http://www.ceit.es/asignaturas/InteInfo/Recursos/Servlets/JavaServlets.pdf

3. http://www.msuniv.ac.in/AdvancedJavaProgrammingwithDatabaseApplication.pdf

4. www.doc.ic.ac.uk/~rcheung/teaching/2720/ppt/lecture12.ppt [JSP Tutorial Slides]

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Able to connect Java application with various RDBMS using JDBC |
| CO2 : | Have understanding of web application architecture and terminologies. |
| CO3 : | Able to develop real time web application using Servlet and JSP technologies. |
| CO4 : | Be familiar with the concepts of JSPs, Servlets and related security issues. |
| CO5 : | Apply Model-View-Controller architecture to build complex client-server applications, packaging and deployment of web applications. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | JDBC Programming | √ | | | | |
| 2 | Web Application Components Fundamentals | √ | √ | | | |
| 3 | Servlet API | | √ | √ | | |
| 4 | Advanced Servlet Features and Security | | | √ | √ | |
| 5 | Java Server Pages | | | √ | √ | √ |
| 6 | Packaging and Deploying Web Applications | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | 3 | 2 | 1 | 3 | 1 | 1 | 1 | 1 | 1 | 2 | 1 |
| CO2 | 3 | 2 | 2 | 3 | 1 | 3 | 1 | 1 | 1 | 2 | 2 | 2 | 2 |
| CO3 | 3 | 3 | 3 | 2 | 2 | 3 | 2 | 2 | 1 | 2 | 2 | 2 | 2 |
| CO4 | 3 | 3 | 2 | 2 | 2 | 3 | 1 | 2 | 2 | 1 | 2 | 2 | 2 |
| CO5 | 3 | 2 | 2 | 2 | 1 | 3 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

# CTUC402 - Database Technologies

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components (Theory):** Unit Tests, Assignments, Case Study etc.<br>**Experiential Learning Components (Practical):** Lab Assignments, Practical Test, VIVA | | | | | |

**Pre-requisite: Relational Database Management and administration**

**Methodology, Pedagogy & Andragogy:** During theory lectures, illustrations of certain complex real world applications, which emphasis the use of advanced concepts of databases, will be discussed. The fundamental concepts regarding database development activities, various database management systems and other advanced issues in database management will also be discussed. In addition, there may be announced or unannounced quizzes/assignments. During Practical sessions, students will be required to carry out case studies using the concepts and techniques they have learnt during theory sessions.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Introduction to DBA | 10 |
| 2 | User Access and Security | 12 |
| 3 | Data Storage Management and Migration | 08 |
| 4 | Managing Database Backup and Recovery | 10 |
| 5 | Database Security and Auditing | 08 |
| 6 | Query Processing and Optimization | 12 |

**Total Hours (Theory): 60**

**Total Hours (Practical): 60**

**Total Contact Hours:120**

**Detail Syllabus:**

**Unit I: Introduction to DBA** **Hours 10**

Types of Oracle Database Users, DBA Roles and Responsibilities; Database Architecture; ORACLE logical and physical database structure; Memory and Process Structure, SQLPLUS Overview, creating a database.

**Unit II: User Access and Security** **Hours 12**

Administrative User Accounts, Database Administrator Authentication, Administrative Privileges and its operations, managing Password authentication, Granting and Revoking Administrative Privileges.

**Unit III: Data Storage Management and Migration** **Hours 08**

Disk Storage, Basic File structure and Indexing, Indexing structure for Files, Data Migration, Data Utilities (SQL loader and Import Export)

**Unit IV: Managing Database Backup and Recovery** **Hours 10**

Backup and Recovery Overview, Database backup, restoration and recovery, defining a backup and recovery strategy, Backup and Recovery options; Data Dump; User-Managed Backup and Recovery; Configuring RMAN; RMAN Backups, Restore and Recovery.

**Unit V: Database Security and Auditing** **Hours 08**

Database Security and Auditing; Database Authentication and Authorization Methods; Data Encryption Techniques, Virtual Private Database

**Unit VI: Query Processing and Optimization** **Hours 12**

Query Processing, Steps of Query Processing, Query Optimization, Distributed Database System, Centralized Database Management System, Difference of Distributed Database System and Centralized Database System, Advantage and Disadvantage of Distributed and Centralized Database Management System.

**Outline of the practical course:**

| Week No | Practical | Description |
|---|---|---|
| 1 | Basics of DBMS | Understanding the basic Data Definition Language (DDL), Data Manipulation Language (DML) statements |
| 2 | Data Querying | Understanding the Data Query Language (DQL) statements |
| 3 | Advanced Querying – I | Understanding the concepts of Join and its types |
| 4 | Advanced Querying – II | Understanding the concepts of nested queries |
| 5 | Basics of PL-SQL | Understanding the basic structure of PL-SQL, datatypes used in PL-SQL |
| 6 | PL-SQL Cursors | Understanding the use of cursor and its types (implicit and explicit) |
| 7 | Stored procedures in PL-SQL | Understanding the significance of stored procedures and its uses |
| 8 | User defined functions in PL-SQL | Understanding the significance and uses of user defined functions |
| 9 | Triggers in PL-SQL | Understanding the use and types of triggers to maintain the integrity of the data, enforce business rules, and automate tasks |
| 10 | Packages in PL-SQL | Understanding the use and implementation of packages in PL-SQL to organize and encapsulate related procedures, functions, variables, triggers and other PL-SQL items into a single item |
| 11 | Data Control Language (DCL) statements | Understanding the data control language statements to control the access to the data within the database |
| 12 | User creations and alteration | Understanding the process of user creation and modifying users |
| 13 | Managing user privileges and roles | Simplifying the database administration by allowing you to group related privileges together, making it easier to manage access rights and maintain consistent security policies. |
| 14 | SQL Views, Administrative Views | Understanding the use of views to simplify the process of accessing and analyzing data stored in the database |
| 15 | Tablespaces and Data-Dictionary | Creation, organizing and managing the tablespaces |

**Total Contact Hours:60**

**Core Books:**

1. Oracle® Database Database Administrator's Guide by Mark Doran, Padmaja Potineni, Rajesh Bhatiya, Oracle Press.

**Reference Books:**

1. Abraham Silberschatz, Henry F.Koeth, S.Sudarshan: Database System Concepts, 6th edition, McGraw Hill Publication.

2. Ramez Elmasri,Shamkant B.Navathe:Fundamentals of Database Systems, 5th Edition, PearsonPublication.

3. C.J. Date, An Introduction to Database Systems (eighth edition), Addison Wesley, 2000

**Web References:**

1. A Practical Guide to Oracle Database Administration for DBA & Developers (oracletutorial.com)

**Course Outcomes:** Upon successful completion of the course, students will,

| | |
|---|---|
| CO1 | Create and manage databases |
| CO2 | Be familiar with database storage management |
| CO3 | Manage backup and recovery |
| CO4 | Control user security |
| CO5 | Manage database query optimization |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Introduction to DBA | ✓ | | | | |
| 2 | User Access and Security | ✓ | ✓ | | | |
| 3 | Data Storage Management and Migration | ✓ | ✓ | ✓ | | |
| 4 | Managing Database Backup and Recovery | | ✓ | | ✓ | ✓ |
| 5 | Database Security and Auditing | | | ✓ | | ✓ |
| 6 | Query Processing and Optimization | | | | ✓ | ✓ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 3 | 1 | 3 | 1 | 3 | 1 | 2 | - | - | - | 3 | - |
| CO2 | 2 | 2 | 3 | 2 | 1 | 3 | 3 | 2 | - | - | - | 3 | - |
| CO3 | 3 | 3 | 2 | 1 | 2 | 3 | 2 | 1 | - | - | - | 3 | 3 |
| CO4 | 3 | 3 | 3 | 1 | 2 | 3 | 3 | 1 | - | - | - | 3 | 3 |
| CO5 | 3 | 3 | 2 | 3 | 2 | 2 | 3 | 1 | - | - | - | 3 | 3 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

# CTUE401 - Cloud Computing Essentials

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| **Experiential Learning Components (Theory):** Unit Tests, Assignments, Case Study etc. | | | | | |

**Pre-requisite:** Operating System Concepts and Network Technology

**Methodology, Pedagogy & Andragogy:** During theory lectures the emphasis will be given on the fundamentals of cloud computing. Students will be introduced basic types, architecture, service providers, mechanism, security issues and some hidden aspects of cloud computing. Students will give practical exposure in form of case study and by showing cloud infrastructure of university.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Evolution of Cloud Computing | 09 |
| 2 | Understanding Cloud Computing and basic types | 10 |
| 3 | Fundamentals of Cloud Architecture & Service Providers | 11 |
| 4 | Cloud Computing Mechanisms | 11 |
| 5 | Cloud Computing Security and Business Use | 10 |
| 6 | Intensive analysis of Cloud Computing with case studies | 09 |

**Total Hours (Theory): 60**
**Total Contact Hours: 60**

**Detail Syllabus:**

**Unit I: Evolution of Cloud Computing Hours 09**

Introduction of Cloud Computing, Growth of technology, Paradigm Shift in Computing, distributed nature of service Provisioning, Support entrepreneurship using Cloud Computing.

**Unit II: Understanding Cloud Computing and basic types Hours 10**

Advantages and drawbacks of Cloud Computing, Essential component for Cloud contract, Major outage of Cloud Computing and Enhancers for Cloud Computing. Introduction to SaaS, PaaS, IaaS. Introduction to Public Cloud, Private Cloud, Hybrid Cloud and Community Cloud, Storage Services for Cloud Computing.

**Unit III: Fundamentals of Cloud Architecture and Service Providers Hours 11**

Workload Distribution Architecture, Resource Pooling Architecture, Dynamic Scalability Architecture, Elastic Resource Capacity Architecture, Service Load Balancing Architecture, Cloud Bursting Architecture, Elastic Disk Provisioning Architecture, Redundant Storage Architecture. Introduction to major Cloud Service Provider: Amazon Web Services, Google Cloud. Microsoft Windows Azure and Office 365, Hp Cloud, RackSpace, CSC Corp, Verizon Terrimark, DropBox.

**Unit IV: Cloud Computing Mechanisms Hours 11**

Introduction to Cloud Infrastructure Mechanisms: Logical Network Perimeter, Virtual Server, Cloud Storage Device, Cloud Storage Levels, Network Storage Interfaces, Object Storage Interfaces, Database Storage Interfaces, Relational Data Storage, on-Relational Data Storage, Cloud Usage Monitor, Monitoring Agent, Resource Agent, Polling Agent, Resource Replication. Introduction to Cloud Management Mechanisms: Remote Administration System, Resource Management System, SLA Management System, Billing Management System.

**Unit V: Cloud Computing Security and Business Use Hours 10**

Introduction to Encryption, Symmetric Encryption, Asymmetric Encryption, Hashing, Digital Signature, Public Key Infrastructure (PKI), Identity and Access Management (IAM), Single SignOn (SSO), Cloud-Based Security Groups. Overview of Compliance and Certification, Access Control, Organizational Control. Benefits of Business using Cloud Computing, Risk of Cloud Computing, Cost factor in Cloud Computing.

**Unit VI: Intensive analysis of Cloud Computing with case studies Hours 09**

Overview of Cloud services, Designing Solutions for the Cloud, Implement & Integrate Solutions, Emerging Markets and the Cloud.

**Core Books:**

1. Kevin L. Jackson, Scott Goessling: Architecting Cloud Computing Solutions: Packt Publication: 2018.
2. Thomas Erl, Zaigham Mahmood and Ricardo Puttini: Cloud Computing Concepts, Technology & Architecture, PHI, 2013.
3. S. Srinivasan: Cloud Computing Basics, Springer, 2014.

**Reference Books:**

1. Derrick Rountree, Ileana Castrillo: The Basics of Cloud Computing, Syngress, 2013.
2. Rajkumar Buyya, James Broberg, Andrzej M. Goscinsk: Cloud Computing-Principles and Paradigms, John Wiley &Sons, 2011.

**Web References:**

1. http://whatisCloud.com/basic_concepts_and_terminology/Cloud [For basic terminology of Cloud Computing].
2. http://www.tutorialspoint.com/Cloud_Computing/ [For cloud computing lecture notes].
3. http://www.intel.in/content/dam/www/public/us/en/documents/guides/cloudcomputingvirtualization-building-private-iaas-guide.pdf [For cloud computing virtualization].
4. https://www.cs.purdue.edu/.../Anya-Kim-Bhargava-MCCWorkshop.ppt [Security issues PPTs].

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Students will learn basics of cloud computing. |
| CO2 : | Students will be familiar with various cloud architectures and services. |
| CO3 : | They will get the knowledge of various network mechanics. |
| CO4 : | Students will get various business aspects of cloud computing with security aspects. |
| CO5 : | They will be able to design and deploy Cloud Infrastructure. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Evolution of Cloud Computing | √ | | | | |
| 2 | Understanding Cloud Computing and basic types | | √ | | | |
| 3 | Fundamentals of Cloud Architecture & Service Providers | | √ | | | |
| 4 | Cloud Computing Mechanisms | | | √ | | |
| 5 | Cloud Computing Security and Business Use | | | | √ | |
| 6 | Intensive analysis of Cloud Computing with case studies | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 1 | - | - | 1 | - | 3 | 1 | 2 | 2 | - | - | 1 |
| CO2 | 3 | 1 | 1 | 1 | 1 | 1 | 3 | 1 | 2 | 2 | 1 | 3 | 2 |
| CO3 | 3 | 2 | 1 | 2 | 1 | 3 | 3 | 2 | 2 | 2 | - | 3 | 2 |
| CO4 | 3 | 3 | 2 | 3 | 1 | 3 | 3 | 2 | 2 | 2 | 3 | - | 1 |
| CO5 | 3 | 3 | 3 | 3 | 1 | 3 | 3 | 3 | 2 | 2 | 2 | - | 3 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUE402 - Cryptography and Network Security

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| **Experiential Learning Components (Theory):** Unit Tests, Assignments, Case Study etc. | | | | | |

**Pre-requisite:** Basic concepts of computer network and digital communication

**Methodology, Pedagogy & Andragogy:** During the lectures the students will learn about basic cryptographic concepts. The teacher will also present applications of cryptography with focus on network security. Further the teacher will also introduce system level security issues and their solutions.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Introduction to Network Security | 06 |
| 2 | Symmetric and Public Key Cryptography Systems and Related Concepts | 11 |
| 3 | Key Distribution | 10 |
| 4 | Transport Level Security | 11 |
| 5 | Wireless and Internet Security | 11 |
| 6 | Advanced Security Concepts | 11 |

**Total Hours (Theory): 60**
**Total Contact Hours: 60**

**Detail Syllabus:**

**Unit I: Introduction to Network Security** **Hours 06**

Network security concepts, Need for security, OSI security architecture, Security attacks, Security services, Security mechanism, Model for network security, Network security standards

**Unit II: Symmetric and Public Key Cryptography Systems and Related Concepts Hours 11**

Symmetric and Asymmetric Key Cryptography, Encryption system and algorithm, Concepts of random and pseudo numbers, Operations on stream ciphers and block ciphers. Message authentication concepts, Secure hash function, Message authentication codes, Public key cryptography, Water marking

**Unit III: Key Distribution Hours 10**

Basics of key distribution, Symmetric and asymmetric key distribution, Kerberos algorithm, Public key infrastructure, Federated identity management, Personal identity verification

**Unit IV: Transport Level Security Hours 11**

World Wide Web security, Web Security Threats, SSL Architecture, SSL Protocols, Transport layer security, Alert Codes, HTTPS, Secure Shell.

**Unit V: Wireless and Internet Security Hours 11**

Wireless security: IEEE 802.11 standard, Wireless LAN security, Wireless application protocol (WAP), Transport layer security in wireless network, WAP end-to-end security, Mobile Device Security

Internet security: pretty-good-privacy, Multipurpose Internet Mail Extensions (MIME), Domain key based mail identification. IP security policy, Integrated security features in IP packet, Internet key exchange (IKE), Cryptographic suits.

**Unit VI: Advanced Security Concepts Hours 11**

Intruders, Intrusion detection, Password management, Malicious software: Viruses, Worms, Backdoor, Logic Bomb, Trojan Horses, Zombie, Virus Countermeasures, Firewalls, Design principles of firewall, Characteristics of firewall, Types of firewall, Firewall location and configuration, Trusted systems.

**Core Books:**

1. William Stallings: Network Security Essentials: 4th Edition, Pearson Publication: 2018.
2. William Stallings: Cryptography and network security: 6th Edition, Pearson Publication: 2019

**Reference Books:**

1. Atul Kahate: Cryptography And Network Security: 2nd Edition, Tata McGraw Hill Education Private Limited: 2011.
2. Alphred J. Menezes: Handbook of applied cryptography: 3rd Edition:CRC Press : 2010.

3. David Pointcheval: Applied cryptography and network security: 1st Edition: Springer: 2012
4. W. Chen, H.-W.LI et al.; Applied Cryptography and Network Security, 1st Edition, Magnum Publication, 2016

**Web References:**

1. https://www.edureka.co/blog/what-is-network-security/ [Introduction to Network Security]
2. https://sectigostore.com/blog/5-differences-between-symmetric-vs-asymmetric-encryption/ [Symmetric and Asymmetric Key Difference]
3. https://www.cloudflare.com/en-in/learning/ssl/transport-layer-security-tls/ [Transport Level Security]
4. https://www.checkpoint.com/cyber-hub/network-security/what-is-firewall/ [Firewall]

**Course Outcomes:** Upon successful completion of the course, students will,

| | |
|---|---|
| CO1 : | Be able to understand network security |
| CO2 : | Be familiar with symmetric and public key cryptography and key distribution |
| CO3 : | Understand transport level security |
| CO4 : | Be familiar with wireless and Internet security |
| CO5 : | Be able to understand advanced security concepts |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | **CO1** | **CO2** | **CO3** | **CO4** | **CO5** |
| 1 | Introduction to Network Security | √ | | | | |
| 2 | Symmetric and Public Key Cryptography Systems and Related Concepts | | √ | | | |
| 3 | Key Distribution | | √ | | | |
| 4 | Transport Level Security | | | √ | | |
| 5 | Wireless and Internet Security | | | | √ | |
| 6 | Advanced Security Concepts | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 3 | 3 | 2 | 3 | 1 | 2 | 2 | 3 | 2 | 3 | 3 | 3 |
| CO2 | 3 | 2 | 3 | 3 | 3 | 2 | 3 | 3 | 2 | 3 | 2 | 2 | 3 |
| CO3 | 2 | 2 | 3 | 3 | 3 | 2 | 2 | 2 | 3 | 2 | 3 | 3 | 3 |
| CO4 | 2 | 3 | 3 | 2 | 3 | 1 | 2 | 3 | 2 | 3 | 2 | 3 | 2 |
| CO5 | 3 | 3 | 3 | 3 | 3 | 2 | 3 | 3 | 2 | 2 | 3 | 3 | 3 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUE403 - Blockchain Essentials

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| **Experiential Learning Components (Theory):** Unit Tests, Assignments, Group Discussion, Case Study etc. | | | | | |

**Pre-requisite:** None

**Methodology, Pedagogy & Andragogy:** During the lecture sessions, a blended instructional approach will be adopted to ensure both conceptual clarity and practical relevance. Students will be introduced to various subsystems that work in an integrated manner to make blockchain technology function effectively, with emphasis on major platforms such as Bitcoin, Ethereum, and Hyperledger. The teaching methodology will involve interactive lectures, real-world case studies, and visual demonstrations to explain key blockchain components like consensus algorithms, cryptographic techniques, and distributed ledgers. From an andragogical perspective, the teaching will acknowledge the learners' prior knowledge and focus on real-world applicability, career relevance, and goal-oriented learning. This integrated approach ensures that learners not only understand the technology but also develop the skills to apply blockchain concepts in diverse application domains.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Introduction to Blockchain | 08 |
| 2 | Blockchain Functionality and Access | 11 |
| 3 | Bitcoin and Crypto Essentials | 11 |
| 4 | Ethereum and related concepts | 11 |
| 5 | Hyperledger and related concepts | 10 |
| 6 | Applications of blockchain | 09 |

**Total Hours (Theory): 60**
**Total Contact Hours: 60**

**Detail Syllabus:**

**Unit I: Introduction to blockchain** **Hours 09**

History of blockchain, Centralized, decentralized and distributed systems, Layers of blockchain, Importance of Blockchain.

**Unit II: Blockchain Functionality and Access** **Hours 10**

Cryptographic concepts- Symmetric key cryptography, Cryptographic hash function, MAC and HMAC, Asymmetric key cryptography, Diffie-Hellman key exchange, Byzantine general's problem, Merkel tree, Structure of block. Access diagram of the blockchain, ledger creation.

**Unit III: Bitcoin and Crypto Essentials** **Hours 11**

What is bitcoin, Working with bitcoin, Bitcoin network, Bitcoin script, Bitcoin mining, Bitcoin wallets, Types of blockchain networks, Consensus algorithms.

**Unit IV: Ethereum and related concepts** **Hours 11**

Design of ethereum, Ethereum blockchain, Smart contracts in ethereum, Ethereum virtual machine, Ethereum eco system.

**Unit V: Hyperledger and related concepts** **Hours 10**

Introduction to hyperledger and hyperledger fabric, hyperledger fabric transaction flow, membership and identity management, fabric network setup, hyperledger fabric demo, hyperledger composer demo.

**Unit VI: Applications of blockchain** **Hours 09**

Applications of blockchain in domains – Finance, Healthcare, Insurance, Supply chain management, Internet of things.

**Core Books:**

1. Bikramaditya Singhal, Gautam Dhameja, Priyansu Sekhar Panda: Beginning Blockchain - A Beginner's Guide to Building Blockchain Solutions: 1st Edition: APress Publication: 2018.
2. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, Steven Goldfeder: Bitcoin and Cryptocurrency Technologies - A Comprehensive Introduction: 1st Edition: Princeton University Press: 2016.
3. Tiana Laurence: Introduction to blockchain technology – the many faces of blockchain technology in the 21st century: 1st Edition: Van Heren Publishing: 2019

**Reference Books:**

1. S. Ashraf, S. Adarsh: Decentralized Computing Using Blockchain Technologies and Smart Contracts - merging Research and Opportunities: 1st Edition: IGI Global, 2017
2. Fabian Schar, Aleksander Berentsen: Bitcoin, Blockchain, and Cryptoassets: 1st Edition: MIT Press: 2020

**Web References:**

1. https://andersbrownworth.com/cms/460/blockchain/demo [Blockchain Demo]
2. https://ethdocs.org/en/latest/introduction/what-is-ethereum.html [Ethereum Introduction]
3. https://www.hyperledger.org/wp-content/uploads/2018/07/HL_Whitepaper_IntroductiontoHyperledger.pdf [Hypderledger Introduction]

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | The students will learn about underlying technologies of blockchain and various applications of blockchain technology. |
| CO2 : | The students will understand working of blockchain in detail. |
| CO3 : | The students will learn about how the blockchain is used to power bitcoin. |
| CO4 : | The students will see how ethereum utilizes blockchain. |
| CO5 : | The students will recognize the role of blockchain in hyperledger platform. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Introduction to Blockchain | √ | | | | |
| 2 | Functioning of Blockchain | | √ | | | |
| 3 | Bitcoin and related concepts | | | √ | | |
| 4 | Ethereum and related concepts | | | | √ | |
| 5 | Hyperledger and related concepts | | | | | √ |
| 6 | Applications of blockchain | √ | | | | |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 1 | 2 | - | 1 | - | - | 3 | - | - | - | - | 1 | - |
| CO2 | - | 1 | 2 | 1 | 1 | - | 3 | - | - | - | - | 1 | - |
| CO3 | 2 | 2 | 3 | - | 3 | - | - | - | - | - | - | 2 | 1 |
| CO4 | 3 | 2 | 3 | - | 3 | - | - | - | - | - | - | 2 | 1 |
| CO5 | 3 | 1 | 3 | 1 | 3 | - | - | - | - | - | - | 3 | 2 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUP401 - On-Job Training-I

| Course Code | Course Name | Theory / Practical / Others | Marks (Theory) | | Marks (Practical) | | Credit | NEP-2020 Classification |
|---|---|---|---|---|---|---|---|---|
| | | | CCE | SEE | CCE | SEE | | |
| CTUP401 | On-Job Training-I | Viva | - | - | 75 | 75 | 06 | Research Project/Dissertation |

| Credit | Component | Instruction /Contact Hours (Per Week) | Instruction / Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| **06** | **VIVA** | **12** | **180** | **270** | **450** |

**Pre-requisite:** None

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Apply foundational IT skills (e.g., basic programming, database querying, or system configuration) to perform structured tasks in an industry setting. |
| CO2 : | Demonstrate effective communication and teamwork under supervision in professional IT environments. |
| CO3 : | Utilize industry-standard IT tools (e.g., IDEs, DBMS software) to complete assigned tasks with guidance. |
| CO4 : | Identify workplace practices, workflows, and professional ethics relevant to IT roles. |
| CO5 : | Document and reflect on foundational IT tasks through logbooks and reports to enhance learning outcomes. |

- **Refer Annexure-I for more information.**

# CTUR401 - Research Project-I

| Course Code | Course Name | Theory / Practical / Others | Marks (Theory) | | Marks (Practical) | | Credit | NEP-2020 Classification |
|---|---|---|---|---|---|---|---|---|
| | | | CCE | SEE | CCE | SEE | | |
| CTUR401 | Research Project-I | Viva | - | - | 75 | 75 | 06 | Research Project/Dissertation |

| Credit | Component | Instruction /Contact Hours (Per Week) | Instruction / Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| **06** | **VIVA** | **12** | **180** | **270** | **450** |

**Pre-requisite:** None

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Formulate a clear research question or problem statement in the IT domain using appropriate methodologies. |
| CO2 : | Conduct a systematic literature review to identify relevant research gaps and existing solutions. |
| CO3 : | Apply research tools and techniques (e.g., data analysis software, programming libraries) to design exploratory studies or experiments. |
| CO4 : | Demonstrate ethical research practices in data handling, citation, and adherence to intellectual property guidelines. |
| CO5 : | Prepare a research proposal or methodology report, articulating the research plan with clarity and precision. |

- **Refer Annexure-I for more information.**

# TEACHING SCHEME & DETAILED SYLLABUS

# FOR

# BSc-IT PROGRAMME
# (8th SEMESTER)
# AS PER NEP 2020

# EFFECTIVE FROM ACADEMIC YEAR 2025-26

## B.Sc. (IT) Semester-VIII

| Course Code | Course Name | Theory / Practical / Others | Marks (Theory) | | Marks (Practical) | | Credit | NEP - 2020 Classification |
|---|---|---|---|---|---|---|---|---|
| | | | CCE | SEE | CCE | SEE | | |
| **CTUC403** | Full Stack Web Development | Theory and Practical | 50 | 50 | 25 | 25 | 06 (04-Theory) (02-Practical | Major Core |
| **CTUC404** | Advanced Mobile Programming | Theory and Practical | 50 | 50 | 25 | 25 | 06 (04-Theory) (02-Practical | Major Core |
| **CTUE404** | Web Development using Open Source Technologies | Practical | - | - | 25 | 25 | 2 | Minor |
| **CTUE405**<br>**CTUE406**<br>**CTUE407** | (Select any one)<br>1. Python Web framework<br>2. HTTP Web Service for Enterprise Applications<br>3. Game Development using Unity | Practical | - | - | 25 | 25 | 2 | Minor |
| **CTUP402**<br>**CTUR402** | (Select any one)<br>On Job Training-II<br>Research Project-II | VIVA | - | - | 75 | 75 | 6 | Research Project / Dissertation |

*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)
*SEE- Semester –End- Evaluation

# CTUC403 - Full Stack Web Development

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components (Theory):** Unit Tests, Assignments, Case Study etc.<br>**Experiential Learning Components (Practical):** Lab Assignments, Practical Test, VIVA, Certificate Course etc. | | | | | |

**Pre-requisite:** Working knowledge of HTML and JavaScript

**Methodology, Pedagogy & Andragogy:** During the theory sessions students will understands the various JavaScript frameworks and its architecture, also able to comprehend MEAN stack application concepts. During the practical sessions students will learn how to reduce the amount of code you write to build rich user interface applications, Modularizing code, retrieving data from back-end- server and manipulate it.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Introduction to JavaScript and JavaScript Frameworks. | 07 |
| 2 | Directives in AngularJS | 11 |
| 3 | MVW-The AngularJS way | 12 |
| 4 | Introduction to NodeJS | 10 |
| 5 | Introduction to ExpressJS Framework | 11 |
| 6 | Working with Database | 09 |

**Total Hours (Theory): 60**
**Total Hours (Practical): 60**
**Total Contact Hours:120**

**Detail Syllabus:**

**Unit I: Introduction to JavaScript Frameworks. Hours 07**

Introduction JavaScript and its DOM concepts, Arrow function, JSON Objects, JavaScript Frameworks & Libraries, MEAN.JS introduction, Architecture of MEAN.JS.

**Unit II: Introduction to AngularJS. Hours 11**

Introduction to AngularJS, Features of AngularJS, Data Binding in angularJS, working with Expressions in angularJS, Introduction to Directives, Directive Lifecycle, Using Angular JS built-in directives, creating a custom directive, Working with $scope, $rootScope, Lifecycle of $scope.

**Unit III: MVW-The Angular JS way. Hours 12**

MVW Architecture: Model-View-Controller and Model-View-View-Model Architecture, Introduction to AngularJS Modules – Application module & Controller modules, Attaching Properties and functions to scope, Controller in external files, AngularJS Filters, working with Angular Forms, Form events, validating Angular forms.

**Unit IV : Introduction to NodeJS Hours 10**

Introduction to NodeJS, Features of NodeJS, Traditional web Server Model, Node.js process Model, Installation of NodeJS, working with Node Package Manager, Command Line options, NodeJS modules – Core Modules, Creating Local modules & Exporting Local modules, Third-party modules, Creating NodeJS web server.

**Unit V: Introduction to ExpressJS Framework. Hours 11**

Introduction to ExpressJS, Installing ExpressJS using NPM, Advantages of ExpressJS, Installing Express.js, building web server, configure routes, ExpressJS middleware's, Serve Static Resources using Express.js, working with HTTP methods of ExpressJS, ExpressJS routing, Creating RESTful API.

**Unit VI: Working with Database. Hours 09**

Introduction to Mongo DB, Installation of MongoDB – MongoDB Server & MongoDB Shell, MongoDB Compass, MongoDB Database, Collections, and documents, Basic operations using MongoDB – Create, Insert, Update and Delete, Access MongoDB in Node.js, setting up mongoose, Connecting MongoDB, Insert, update and delete document.

**Outline of the course (Practical):**

| Week No. | Practical | Description |
|---|---|---|
| 1 | JavaScript function, event and basic DOM manipulation methods. | Introduction JavaScript DOM |
| 2 | AngularJS application – Auto initialization and manual initialization, Basic directives, AngularJS Expression | Basic concepts of AngularJS application |
| 3 | Working with AngularJS conditional directives | Familiar with frontend condition using AngularJS. |
| 4 | Working with Complex datatype - array and object, Working with AngularJS loop directives. | Get familiar with frontend loop with complex datatype. |
| 5 | Working with MVC and MVVM architecture – Creating module and controller | Able to identify the difference between One-way and two-way data binding methods with architecture. |
| 6 | Creating and using custom directives of AngularJS | Understand the reusability component in AngularJS. |
| 7 | Working with NodeJS – Basic NodeJS program, NPM Commands, Working with Core modules in NodeJS – Creating HTTP server, Handling files, URL & Path | Understand the NodeJS environment and its Core modules, Get familiar with Node Package Manager repository. |
| 8 | Creating custom local module in NodeJS | Able to create user defined module in NodeJS |
| 9 | Working with NodeJS third-party module – ExpressJS, Basic concept & methods. | Able to create server using ExpressJS module. |
| 10 | Working with files, middleware's of ExpressJS. | Able to access .html & its dependency file on a server. |
| 11 | Creating restful APIs in ExpressJS – GET, POST, PUT, DELETE. | Able to create restful APIs to read, Insert, Update and delete data from JSON file. |
| 12 | Working with MongoDB Command & MongoDB Atlas | Get able to create database and apply CRUD operation through MongoDB shell & MongoDB Atlas. |
| 13 | Working with Mongoose – NodeJS third-party module for Database connection. | Able to access data of MongoDB database in ExpressJS Application. |

| | | |
|---|---|---|
| 14-15 | Working with MEAN Stack application | Able to create full stack application with MongoDB, ExpressJS, AngularJS and NodeJS. |

**Core Books:**

1. Jeffry Houser : "Learn With: Angular 5, Bootstrap, and NodeJS", Kindle Edition,
2. Shyam Seshadri Brad Green:"AngularJS – Up and Running, Brad Green", Second Edition, O'REILLY
3. Basarat Ali Syed: "Beginning Node.js", Apress Publication.
4. Greg Lim : "Beginning MEAN Stack (MongoDB, Express, Angular, Node.js)", kindle Edition.

**Reference Books:**

1. Agus Kurniawan: "AngularJS Programming by Example 2017 Edition", Kindle Edition.
2. Adam Freeman: " Pro AngularJS 2017 Edition", Apress.
3. Krasimir Tsonev: "Node.js By Example", Packt Publishing
4. Ethan Brown: "Web Development with Node and Express", O'REILLY

**Web References:**
1. http://www.w3schools.com/angular/default.asp [Tutorial link for AngularJS]
2. http://www.tutorialspoint.com/angularjs/ [Tutorial link for AngularJS]
3. https://www.tutorialspoint.com/angularjs/angularjs_tutorial.pdf [E-book for AngularJS]
4. http://www.tutorialsteacher.com/nodejs/nodejs-modules [Tutorial link for NodeJS]
5. https://www.javatpoint.com/mean-stack-tutorial [MEAN stack development]

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Familiar with JavaScript and JavaScript frameworks. |
| CO2 : | Able to understand basic fundamental of AngularJS and implement Model-View-View-Model Architecture. |
| CO3 : | Able to create NodeJS and ExpressJS applications. |
| CO4 : | Able to create Restful API to access data from MongoDB. |
| CO5 : | Able to create Single page application and MEAN stack application. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Introduction to JavaScript and JavaScript framework | √ | | | | √ |
| 2 | Introduction to AngularJS | | √ | | | √ |
| 3 | MVW-The Angular JS way | | √ | | | √ |
| 4 | Introduction to NodeJS | | | √ | √ | √ |
| 5 | Introduction to ExpressJS | | | √ | √ | √ |
| 6 | Working with Database | | | | √ | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | - | - | - | - | 2 | - | 1 | - | - | - | - | 1 | - |
| CO2 | 1 | 2 | 2 | 1 | 2 | - | 1 | 1 | - | 1 | 1 | 2 | 1 |
| CO3 | 1 | - | 3 | 1 | 3 | - | 2 | 1 | - | - | 1 | 2 | 1 |
| CO4 | 2 | 2 | 3 | 2 | 3 | 2 | 2 | 3 | - | 2 | 2 | 3 | 3 |
| CO5 | 2 | 3 | 3 | 2 | 3 | 2 | 2 | 3 | 2 | 2 | 3 | 3 | 3 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUC404 - Advanced Mobile Programming

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components (Theory):** Unit Tests, Assignments, Case Study etc.<br>**Experiential Learning Components (Practical):** Lab Assignments, Practical Test, VIVA etc. | | | | | |

**Pre-requisite:** Object-oriented Programming Concepts

**Methodology, Pedagogy & Andragogy:** During the theory session illustrations emphasizing the need for Cross-platform Mobile application development will be given. Further, fundamental to intermediate concepts of programming in Flutter will also be discussed. During Practical sessions, students will be required to develop mobile applications using Dart programming language. Student shall also develop mobile applications with elegant user interface and applications that interact with popular cloud-based database.

**Outline of the course (Theory):**

| Unit Number | Title of the Unit | Minimum Number of Theory Hours |
|---|---|---|
| 1 | Introduction to Cross-platform application development with Flutter & Dart | 09 |
| 2 | UI Designing with Flutter | 11 |
| 3 | State management with App creation | 09 |
| 4 | Architecting Flutter applications and Its packages | 11 |
| 5 | Introduction to Backend to Flutter-I | 10 |
| 6 | Introduction to Backend to Flutter-II | 10 |

**Total Hours (Theory): 60**

**Total Hours (Practical): 60**
**Total Contact Hours: 120**

**Detailed Syllabus:**

**Unit I Introduction to Cross-platform application development with Flutter & Dart Hours 09**

Set up a new Flutter project using Android Studio. Widget tree, Interface design: pre-made Flutter Widgets. Image and Text Widgets. App Icons for iOS and Android. Add and load image assets to Flutter projects. Run Flutter apps on iOS Simulator, Android Emulator and physical devices.

**Unit II: UI Designing with Flutter Hours 11**

Hot Reload and Hot Restart, Use of Pubspec.yaml file, custom assets and fonts. An introduction to the Widget build(), layout widgets: Columns, Rows, Containers and Cards. Material icons, Icons class. Theme widgets. Refactoring widgets. Dart annotations and modifiers. Immutability of Stateless and Stateful Widgets. Update screen with the build(). Custom Flutter Widgets.Difference between final and const in Dart. Maps, enums and the ternary operator. Functions and arguments in Dart. Multi-screen Flutter apps, routes and the Navigator widget. Flutter favours composition vs. inheritance (customizing widgets).

**Unit III: State management with App creation Hours 09**

About Stateful and Stateless Widgets, callbacks. Declarative style of UI programming, Flutter widgets react to state changes. Import Dart libraries. Variables, data types and functions in Dart. Build flexible layouts. Understand the relationship between setState(), State objects and Stateful Widgets.

**Unit IV: Architecting Flutter applications and its packages Hours 11**

Dart package manager, Flutter compatible packages. The structure of the pubspec.yaml file. Incorporate the audio-players package to play sound. Functions in Dart and arrow syntax. Refactor widgets, Flutter's philosophy of UI as code. The lists and conditionals in Dart. Classes and objects. Understand Object Oriented Dart. Dart Constructors for Flutter widgets. Design patterns to structure Flutter apps. Structuring and organizing Flutter apps.

**Unit V: Introduction to Backend to Flutter-I Hours 10**

Asynchronous programming in Dart and use of async/await. Stateful Widget lifecycle methods, handling exceptions in dart. Null aware operators. Location data from both iOS and Android. Http package and live data from open APIs. Parse JSON data using the dart:convert library.

**Unit VI: Firebase and State Management with Flutter-II Hours 10**

State objects via the Stateful Widget. Use the TextField Widget to take user input. Pass data backwards using the Navigator widget. Hero animations in Flutter apps. Animation controller, custom animations. The use of Dart mixins. Firebase Cloud Firestore into your Flutter apps. Authentication with Firebase Auth package.

**Outline of the course (Practical):**

| Week No. | Practical | Description |
|---|---|---|
| 1 | Setting up the development environment and understanding fundamentals of Flutter applications | Installing Flutter and configuring Android Studio for Flutter application development, understanding Flutter app directory structure, setting target device, creating and executing a simple Flutter app. |
| 2 | Widgets and widget tree | Using simple widgets to display text and images, Stateless vs. Stateful widgets, understanding the concept of widget tree. |
| 3 | More about Flutter application execution | Hot Reload, Hot Restart and Full Restart, pubspec.yaml, importance of the build() method |
| 4 | Designing UI in Flutter applications | Top-level widgets-MaterialApp, Scaffold, AppBar, Container, Drawer, Row, Column, Center Applying theme |
| 5 | Common widgets-I | Text, TextField, ElevatedButton, ListTile, RadioListTile,CheckboxListTile, DropdownList , CircleAvatar, Icon, Image, Using callbacks. |
| 6 | Commons widgets-II | ListView, GridView, Slider, DatePicker, TimePicke, Badge |
| 7 | Navigation and routing | Navigation in a multi-screen app with the help of push(), pop(), pushNamed() and named routs, passing data in navigation, using Tabs |
| 8 | Dart programming language fundamentals | Variables, constants, functions, object-oriented programming in Dart. |
| 9 | Using dependencies | Adding dependencies in pubspec.yaml, importing packages and using API. |
| 10 | Asynchronous programming | Use of await, async, Stateful Widget lifecycle methods, handling exceptions in dart. |

| | | |
|---|---|---|
| **11** | Accessing location and HTTP service | Getting location of a device, fetching data from HTTP service |
| **12** | Using animation | Hero animation, Animation controller, custom animations, the use of Dart mixins. |
| **13** | Introduction to cloud-based database | Creating a Firebase project and configuring it for Flutter app, creating a Firestore database, understating collection, document, fields. |
| **14** | Database operations from Flutter app | Performing CRUD operations from a Flutter app. |
| **15** | Firebase Authentication | Configuring Firebase for authentication and using it for authentication in Flutter app. |

**Core Books:**

1. Marco L. Napoli: Beginning Flutter: A Hands On Guide to App Development: Wrox publication: 2019.

**Reference Books:**

1. Eric Windmill: Flutter in Action: Edition: Manning Publication: January 2020.
2. Alessandro Biessek: Flutter for Beginners: An introductory guide to building cross-platform mobile applications with Flutter and Dart 2: Packt publication: September 2019

**Web References:**

1. https://docs.flutter.dev/reference/tutorials
2. https://www.tutorialspoint.com/flutter/index.htm
3. https://www.javatpoint.com/flutter
4. https://fluttertutorial.in/

**Course Outcomes:** Upon successful completion of the course, students will be

| | |
|---|---|
| CO1 : | able to clear all object oriented programming and cross platform concepts. |
| CO2 : | able to learn Flutter and Dart step by step. |
| CO3 : | able to learn the reduce the code through native app performance, animated UI with material design and least testing. |
| CO4 : | able to use Firebase to authenticate the users and use the remote database. |
| CO5 : | able to build engaging native mobile apps for both Android and iOS. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Introduction to Cross-platform application development with Flutter & Dart | √ | | | | |
| 2 | UI Designing with Flutter | | √ | | | √ |
| 3 | State management with App creation | | | | | √ |
| 4 | Architecting Flutter applications and Its packages | | | √ | | √ |
| 5 | Introduction to Backend to Flutter-I | | √ | | | |
| 6 | Introduction to Backend to Flutter-II | | √ | √ | √ | |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | - | - | 3 | 2 | 1 | 3 | - | 1 | - | 1 | - | 1 |
| CO2 | 3 | 1 | 3 | 2 | 3 | 1 | 2 | 1 | 1 | - | 1 | 1 | 2 |
| CO3 | 3 | - | 3 | 1 | 3 | 1 | 1 | 2 | 1 | - | 1 | 1 | 2 |
| CO4 | 3 | 3 | 3 | 3 | 3 | 1 | 1 | 3 | 1 | - | 2 | 3 | 1 |
| CO5 | 3 | 3 | 3 | 1 | 3 | 1 | 1 | 3 | 1 | 1 | 2 | 3 | 2 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

# CTUE404 - Web Development using Open Source

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 04 | Theory | 04 | 60 | 50 | 110 |
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components (Theory):** Unit Tests, Assignments, Case Study etc.<br>**Experiential Learning Components (Practical):** Lab Assignments, Practical Test, VIVA | | | | | |

**Pre-requisite:** Basic understanding of HTML and MySQL.

**Methodology, Pedagogy & Andragogy:** During theory and practical sessions, students able to install & configure PHP and prerequisite software(s). Also, student will be emphasized to develop dynamic web applications.

**Outline of the course:**

| Unit Number | Title of the Unit | Minimum Number of Hours |
|---|---|---|
| | | Theory |
| 1 | Basics of HTML and JavaScript. | 08 |
| 2 | Introduction to Open Source Software and PHP, Basics of PHP | 10 |
| 3 | Control Structure, Function, Array and Overview of OOP in PHP | 10 |
| 4 | Form Handling Using PHP | 10 |
| 5 | PHP Utilities | 10 |
| 6 | Relational Database Using PHP | 12 |

**Total Hours (Theory): 60**
**Total Hours (Practical): 60**
**Total Contact Hours: 120**

**Detail Syllabus:**

**Unit I: Basics of HTML and Java Script Hours 08**

Basics of HTML, Introduction of JavaScript, Variables, Overview of operators, Control statements
and looping statement, Overview of DOM, function declaration and calling with event.

**Unit II: Introduction to Open Source Software and PHP, Basics of PHP Hours 10**

Overview of Open Source Software, Installation & Configuration of PHP, Introduction to PHP, Working of HTML with PHP, PHP language Basics: Lexical Structure, Data types, Variables, Expressions and Operators.

**Unit III: Control Structure, Function, Array and Overview of OOP in PHP Hours 10**

PHP language Basics: Control and Looping statements. Functions: Function Definition, Function Parameters, Returning Values. Strings: Usages and String Functions, Arrays: Types of Arrays and its Usages, Array functions. Overview: Objects, Declaring Class, Properties, Methods, Exception Handling, Examples.

**Unit IV: Form Handling Using PHP Hours 10**

Capturing data with PHP Using HTML Form Elements, Send Form data using GET Method & POST Method, Receive Form data using $_GET, $_POST & $_REQUEST variables, Super Global Variables, State Management Techniques: Concept of Session, starting session, modifying session variables, Un registering and deleting session variable, Concept of Cookies.

**Unit V: PHP Utilities Hours 10**

File Uploading: Upload Single and Multiple file using PHP script, Understanding HTTP requests, Exploring and modifying HTTP responses, getting information from web server.

**Unit VI: Relational Databases Using PHP Hours 12**

Relational Databases and SQL, Using PHP to access Databases, CRUD operations, Handling Errors.

**Outline of the course (Practical):**

| Week No. | Practical | Description |
|---|---|---|
| 1 | HTML Basics | Create web pages using headings, paragraphs, lists, links, images, forms, and tables. |
| 2 | JavaScript Fundamentals | Use JavaScript for variables, operators, control statements, and loops. |
| 3 | JavaScript DOM and Events | Manipulate web page elements using the DOM and trigger actions via events. |
| 4 | Intro to PHP & OSS Setup | Install XAMPP/WAMP, configure PHP environment, and explore open-source concepts. |
| 5 | PHP Syntax & Data Types | Learn PHP lexical structure, variables, data types, and basic expressions. |
| 6 | PHP Control Structures | Use conditional statements (if, else, switch) and loops (for, while, foreach). |
| 7 | PHP Functions & Strings | Define and call functions, pass parameters, return values, and use string functions. |
| 8 | Arrays & OOP Overview | Use indexed, associative arrays, array functions, and basic OOP (class, object, method). |
| 9 | Form Handling with PHP | Capture HTML form data using GET/POST, and process it using PHP superglobals. |
| 10 | Sessions and Cookies | Implement session start, update, destroy; work with setting and retrieving cookies. |
| 11 | File Uploading | Upload single and multiple files with validations using PHP scripts. |
| 12 | HTTP Request/Response | Handle HTTP requests/responses, set headers, and retrieve server info using PHP. |
| 13 | Intro to PHP & MySQL | Connect PHP with MySQL, create database/tables, and execute queries. |
| 14 | CRUD Operations & Error Handling | Perform Create, Read, Update, Delete operations using MySQL and handle errors gracefully. |
| 15 | Project / Revision | Integrate form, file upload, session, and database in a mini-project; revise full syllabus. |

**Core Books:**

1. Vikram Vaswani, PHP: A Beginner’s Guide: Indian Edition, First Edition, McGraw Hill, 2009
2. Matt Doyle , Beginning PHP 5.3 , Wrox , 2010

3. Ballard and Moncur, Teach Yourself Javascript in 24 Hours, Sams Publishing, 2015

**Reference Books:**

1. Timothy Boronczyk, Elizabeth Naramore, Jason Gerner, Yann Le Scouarnec, Jeremy Stolz, Michael K. Glass: Beginning PHP6, Apache, MySQL Web Development: Wrox , 2009.
2. Lynn Beighley and Michael Morrison, Head First PHP & MySQL, First Edition, O'Reill Publication, 2009.

**Web References:**

1. https://github.com/PHPMailer/PHPMailer [PHP Mailer Code]
2. https://www.php.net/ [Official website of PHP]
3. https://www.geeksforgeeks.org/php-tutorials/ [Lecture notes of PHP]
4. https://www.w3schools.com/php/default.asp [Lecture notes of PHP]

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Gain understanding of HTML and Java Script. |
| CO2 : | To utilize knowledge and skills for basics of PHP and functions of PHP |
| CO3 : | Acquire the knowledge of array and OOP in PHP |
| CO4 : | Learn Form handling and utilities in PHP. |
| CO5 : | Be able to develop dynamic web-based application using PHP and MySQL with state management techniques. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name | Course Outcomes | | | | |
|---|---|---|---|---|---|---|
| | | CO1 | CO2 | CO3 | CO4 | CO5 |
| 1 | Basics of HTML and Java Script | √ | | | | |
| 2 | Introduction to Open Source Software and PHP, Basics of PHP | | √ | | | |
| 3 | Control Structure, Function, Array and Overview of OOP in PHP | | | √ | | |
| 4 | Form Handling Using PHP | | | | √ | |
| 5 | PHP Utilities | | | | √ | |
| 6 | Relational Database Using PHP | | | | | √ |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 3 | 3 | 3 | - | 1 | - | 1 | 1 | - | 3 | - | 3 |
| CO2 | 3 | 3 | 3 | 3 | - | 1 | - | 1 | 1 | - | 3 | - | 3 |
| CO3 | 3 | 3 | 2 | 3 | - | 1 | - | 1 | 2 | - | 2 | - | 3 |
| CO4 | 3 | 3 | 3 | 3 | - | 1 | - | 1 | 1 | - | 3 | - | 3 |
| CO5 | 3 | 3 | 3 | 3 | 2 | 1 | 1 | 1 | 2 | 2 | 3 | 3 | 3 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUE405 - Python Web Framework

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components (Practical):** Lab Assignments, Practical Test, VIVA | | | | | |

**Prerequisites:** Basic concepts of Python Programming.

**Methodology, Pedagogy & Andragogy:** This course focuses on providing hands-on experience to students for design and develop web application using Django framework and MVT pattern integration.

## A. Outline of the Course:

| Week No | Practical | Description |
|---|---|---|
| 1 | Downloading and Installation: Installation of Python and Python documentation. Creating a virtual environment and installing Django for creating first web project. | Learn basic concepts virtual environment then creating and execute first web project of Django and understanding directory structure. |
| 2 | Creating first web application in web project. | Get familiar with different files (settings.py, urls.py, manage.py, models.py etc.) with their usages and then executing first application. |
| 3 | Creating multiple web applications within same web project. | Learning mappings of Django web project. |
| 4 | Django views: Writing different types of views. | Able to display the output using different types of views. |

| | | |
|---|---|---|
| **5** | Creating first template and developing MVT based application. | Learn cerate and add a template in web project then understand the mapping of MVT based web application. |
| **6** | Integrating templates and views for developing MVT based applications. | Able to write Python code in a template language and developing MVT based applications for dynamic content generation. |
| **7** | Model and Database Setup for Django App | Setting super user credentials and using Djang's admin panel for database applications |
| **8** | Creating web application with database connectivity. | Using code first approach, developing web application with database connectivity and managing it with Django Admin Panel. |
| **9** | Django Forms | Learn Django forms for taking user input from the web applications. |
| **10** | Django Forms | Developing full fledge web applications for CRUD operation implementation. |
| **11** | Django & REST APIs | Understanding Django REST Framework for building web APIs. |
| **12** | Unit Testing with Django | First, learning how to write tests with Django then executing them using Python standard library module: unittest. |
| **13** | Unit Testing with Django | Writing different test cases for various applications, ensuring the reliability and correctness. |
| **14** | Admin, Security, Performance & Optimization | Applying different ways like django-debug-toolbar, Third-party services etc. for Performance Optimization. |
| **15** | Admin, Security, Performance & Optimization | Learning different types of security measures like XSS Protection, CSRF Protection, SQL injection protection etc. |

**Total hours: 60**

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**Core Books:**

1. Adrian Holovaty, Jacob K. Moss: The Definitive Guide to Django:Web Development Done Right

2. Jeff Forcier, Paul Bissex, Wesley Chun: Python Web Development with Django, Pearson Education,2008.

3. Dusty Philips: Python 3 Object oriented Programming, PACKT publishing, 2010

**Reference Books:**

1. Adrian Holovaty, Jacob Kaplan-Moss, et al.: The Django Book, 2013
2. Nathan Yergler: Effective Django, 2015
3. Steve Holden: Python Web Programming, 1st edition,2002.

**Web References:**

1. https://www.djangoproject.com/start/ [For Django Projects]
2. https://djangoforbeginners.com/initial-setup/ [For Django Setups]
3. https://docs.djangoproject.com/en/2.1/intro/tutorial01/ [For Django app]

**Course Outcome:** Upon successful completion of the course, student will:

| | |
|---|---|
| **CO1 :** | Understand Django fundamentals and use its concepts to build and deploy robust web applications and apps. |
| **CO2 :** | Learn about Django URL patterns and views and deploy Django applications |
| **CO3 :** | Learn how to configure Django for powerful databases and create the Django admin interface |
| **CO4 :** | Learn about Django's security implications and how to create safe web applications with it |
| **CO5 :** | To get some idea about unit testing concepts and its implementation through Django |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 2 | 3 | 2 | 2 | 2 | 3 | 2 | 1 | 1 | 1 | 3 | 2 |
| CO2 | 2 | 3 | 3 | 3 | 3 | 2 | 3 | 2 | 1 | 1 | 1 | 3 | 2 |
| CO3 | 2 | 3 | 3 | 3 | 3 | 2 | 3 | 3 | 2 | 1 | 2 | 3 | 2 |
| CO4 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 2 | 2 | 3 | 3 | 2 |
| CO5 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 1 | 2 | 2 | 3 | 3 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUE406 - HTTP Web service for Enterprise Applications

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components (Practical):** Lab Assignments, Practical Test, VIVA | | | | | |

### Prerequisites:

Expected to have foundational knowledge from **CTUC104 – Fundamentals of Web Designing, CTUC205 – Fundamentals of Visual Programming, CTUC302 – Open Source Technology and Web Application Development using .NET Framework.** This skillset will ensure a smoother transition into advanced Web API concepts and practical implementations.

### Methodology & Pedagogy:

In the sessions, students will actively develop Web APIs using C#, Entity Framework, and SQL Server through structured, real-world assignments. They will explore API testing and debugging using industry tools like Postman, Fiddler, and browser-based tools. JQuery and AJAX integration will enhance client-side interaction and API consumption skills. Each session fosters hands-on learning, problem solving, and application of best practices in a project-based environment.

### Outline of the Course:

| Week No | Practical | Description |
|---|---|---|
| 1 | **Basic Web API with GET Method:** A Web API using the GET method retrieves data from the server. The GET method does not change the server state and is used to fetch data. | Create an API endpoint that returns a simple welcome message. Test it via browser, Postman, and Fiddler to observe the HTTP request and response. |

| | | |
|---|---|---|
| | **Web API with POST Method:** POST is used to submit data to be processed to a specified resource. | Develop an API endpoint that accepts name and age from the user and returns a response confirming the received data. |
| | **GET/POST Methods with Postman:** Postman is a tool for testing API requests including GET, POST, PUT, and DELETE. | Use Postman to manually send GET and POST requests to your API, and observe headers, request body, and response. |
| 2 | **GET/POST Methods with Fiddler:** Fiddler is a web debugging proxy tool to inspect traffic between computer and internet. | Capture and analyze Web API HTTP requests and responses using Fiddler. |
| | **Web API Project Explaining REST Principles:** REST (Representational State Transfer) is an architectural style using stateless communication and resource-based URLs. | Develop a RESTful Web API that demonstrates statelessness, a uniform interface, and resource-based routing. |
| | **Return Custom HTTP Status Codes from Web API Controller:** Configure controller methods to return specific HTTP status codes based on operation outcomes. | Enhances understanding of how to deliver precise client feedback using return types like HttpResponseMessage or IHttpActionResult with custom status codes like 201 (Created), 204 (No Content), or 409 (Conflict). |
| | **Web Services, WCF, and Web API:** Compare different technologies used for building web-based communication services. | Create a comparative table highlighting architecture, protocol support, performance, and hosting model differences. |
| 3 | **Web API with GET, POST, PUT, DELETE Methods:** These are the primary HTTP verbs used for CRUD operations. | Implement a controller with all CRUD methods without database persistence. |
| | **Integrate Web API with Entity Framework (Code First):** EF Code First approach allows defining models in code, which EF uses to generate database schema. | Develop a Web API project with CRUD operations using EF Code First and connect to SQL Server. |
| | **Integrate Web API with EF (Database First):** EF Database | Build a Web API that connects to an existing database schema and provides CRUD functionalities. |

| | | |
|---|---|---|
| | First allows generating model classes from an existing database. | |
| 4 | **Use ActionResult for Different Return Types:** ActionResult allows an API method to return different content types like string, JSON, or XML. | Implement multiple methods that return various types of responses depending on Accept headers. |
| | **Web API and MVC Controller:** Web API and MVC are both controllers, but Web API is RESTful, while MVC serves web pages. | Create both controller types and explain differences in routing, responses, and return types. |
| | **Use [Route] Attribute in API Controller:** The [Route] attribute allows defining custom URL paths for action methods. | Apply [Route] and [HttpGet]/[HttpPost] to define and test clean and descriptive routes. |
| 5 | **Action Method Naming Conventions in Web API:** Method names like Get, Post, Put, Delete map to corresponding HTTP verbs. | Demonstrate method naming patterns and how they impact API behavior. |
| | **MVC App Consuming Web API (Read-Only View):** MVC application can act as a front-end client to Web API. | Use MVC controller to call Web API and display GET data in a view. |
| | **Implement Convention-Based Routing:** Routing based on URL patterns configured in WebApiConfig. | Set default routes like "api/{controller}/{id}" and test endpoint behavior. |
| 6 | **Implement Attribute-Based Routing:** Routing defined directly above action methods using [Route] attribute. | Use attribute routing to explicitly map methods to specific URLs. |
| | **Custom Parameterized Routing in Web API:** URL segments can be used as parameters for dynamic API behavior. | Create APIs like /api/product/{id} to fetch or process data based on route values. |

| | | |
|---|---|---|
| | **Multiple Route Templates in One Controller:** Use multiple [Route] attributes on a single method for flexible API access. | Define alternate route paths that map to the same logic, improving usability. |
| **7** | **Use JSON and XML Formatters:** Formatters serialize data into requested formats based on Accept header. | Return different formats from API methods by testing with varied Accept headers. |
| | **Register a Custom Media Type Formatter:** Custom formatter defines how data is serialized and returned to clients. | Implement and register a formatter for a custom content type like CSV. |
| | **Handle Validation Errors with Custom Error Response:** Design a Web API method that returns custom structured validation error messages when data model validation fails. | This activity intercept ModelState errors and format them into a consistent JSON error object using HttpError or a custom response class, improving client-side error handling. |
| **8** | **Implement HttpResponseException in Web API:** This exception is thrown for HTTP error scenarios. | Use HttpResponseException to return meaningful status codes and error messages. |
| | **Exception Filter Attribute:** Filters catch unhandled exceptions at a global or controller level. | Write a custom ExceptionFilter and apply it for centralized error handling. |
| | **Implement Logging in Custom Exception Filter:** Create an exception filter that logs errors into a text file or database before returning the response. | Web API by integrating logging in exception filters using tools like ILogger, enabling real-time monitoring of unhandled exceptions with user-friendly messages. |
| **9** | **Use HttpError to Return Custom Error Message:** HttpError sends detailed error information in the response body. | Use HttpError inside catch blocks for structured error responses. |

| | | |
|---|---|---|
| | **Register Exception Filter Globally in WebApiConfig:** Global filters apply across all controllers. | Register custom ExceptionFilter globally to handle exceptions uniformly. |
| 10 | **Model Validation using Data Annotations:** Validation rules defined using attributes like [Required], [Range]. | Apply data annotations to model properties and return validation errors. |
| | **Custom Validation Attributes for Complex Rules:** User-defined attribute for validating complex input logic. | Implement attributes like [EmailDomain] and validate with business logic. |
| | **Implement Global Validation Error Response Structure:** Create a centralized response structure for handling all model validation errors in a unified JSON format. | Configure Web API to return consistent validation messages using filters or middleware, improving API usability and making it easier for clients to process error responses programmatically. |
| 11 | **Use Custom Exception Class:** Custom exception extends base Exception class for domain-specific errors. | Throw and catch these exceptions for controlled error reporting. |
| | **Handle ModelState Errors in Web API:** Model State contains validation error information. | Check ModelState.IsValid and return all validation errors if false. |
| 12 | **Backend Validation using Try-Catch and HttpError:** Exception handling using try-catch ensures robust error response. | Wrap logic in try-catch and return error details using HttpError. |
| | **Global Error Logging with Custom Exception Filter:** Logs unhandled exceptions centrally using logging frameworks. | Extend ExceptionFilter to log exception details using tools like ILogger. |
| 13 | **Consume Web API GET using jQuery AJAX:** AJAX allows asynchronous web requests using JavaScript. | Use jQuery's $.ajax or $.get to fetch data and show it in an HTML table. |

| | | |
|---|---|---|
| | **Consume Web API POST using jQuery AJAX:** Use AJAX to send form data to Web API without page reload. | Collect form inputs and use $.post or $.ajax with type 'POST'. |
| **14** | **Perform PUT and DELETE Operations via AJAX:** PUT updates data, DELETE removes it. AJAX can perform both. | Create buttons triggering AJAX requests to update or delete records. |
| | **Display API JSON Response in HTML Format:** JSON is the default data exchange format. | Fetch JSON data from API and parse/display it using jQuery DOM manipulation. |
| **15** | **AJAX Form Submission with Validation Feedback:** Combines client-side validation with AJAX submission. | Validate form, show errors or success, and post data to API via jQuery. |
| | **Use jQuery to Dynamically Populate DropDown from Web API:** Populate UI controls with data from API dynamically. | On page load, call API to get options and populate <select> using jQuery. |
| | | **Total hours: 60** |

**Core Books:**


1. Jamie Kurtz, Brian Wortman: ASP.NET Web API 2: Building REST Services with .NET Framework, Apress
2. Glenn Block, Pablo Cibraro: Professional ASP.NET Web API, Wrox
3. Adam Freeman: Pro ASP.NET MVC 5, Apress
4. Dino Esposito: Programming Microsoft ASP.NET MVC, Microsoft Press
5. Jon P. Smith: Entity Framework Core in Action, Manning Publications


**Reference Books:**


1. Leonard Richardson, Mike Amundsen, Sam Ruby: RESTful Web APIs, O'Reilly Media
2. Dino Esposito: Microsoft ASP.NET and AJAX: Architecting Web Applications, Microsoft Press
3. Badrinarayanan Lakshmiraghavan: Pro ASP.NET Web API Security, Apress

**Web References:**

1. https://learn.microsoft.com/en-us/aspnet/web-api/ **[ASP.NET Web API (Legacy) Documentation]**
2. https://learn.microsoft.com/en-us/aspnet/core/web-api/ **[ASP.NET Core Web API (Latest Version)]**
3. https://learn.microsoft.com/en-us/ef/ **[Entity Framework Documentation (EF Core & EF 6)]**
4. https://learn.microsoft.com/en-us/training/modules/build-web-api-aspnet-core/ **[Microsoft Learn – Web API Beginner's Learning Path]**
5. https://www.tutorialsteacher.com/webapi **[TutorialsTeacher – Web API Tutorials]**
6. https://www.dotnettricks.com/learn/webapi **[Dot Net Tricks – ASP.NET Web API]**
7. https://www.c-sharpcorner.com/technologies/web-api **[C# Corner – ASP.NET Web API Articles]**

**Course Outcome:** Upon successful completion of the course, student will:

| | |
|---|---|
| **CO1 :** | Understand the basics of Web API, REST principles, and testing using Postman and Fiddler |
| **CO2 :** | Develop CRUD operations in Web API with and without Entity Framework integration |
| **CO3 :** | Configure routing in Web API using convention-based, attribute-based, and parameterized routes. |
| **CO4 :** | Implement formatters, filters, and exception handling for structured Web API responses |
| **CO5 :** | Integrate jQuery and AJAX to consume Web API services with client-side validation |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 1 | 3 | 3 | 1 | 3 | - | - | 2 | 2 | 2 | 2 | 2 | 2 |
| CO2 | 2 | 1 | 2 | 2 | 3 | 2 | 2 | 3 | - | 3 | 2 | 3 | 3 |
| CO3 | 2 | 2 | 2 | 2 | 3 | 2 | 2 | 2 | - | 1 | 2 | 3 | 2 |
| CO4 | 3 | 3 | 3 | 3 | 2 | 3 | 2 | 2 | - | 2 | 3 | 3 | 2 |
| CO5 | 2 | 3 | 3 | 1 | 3 | 2 | 2 | 3 | 2 | 2 | 3 | 3 | 2 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUE407 - Game Development Using Unity

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| 02 | Practical | 04 | 60 | 40 | 100 |
| **Experiential Learning Components (Practical):** Lab Assignments, Practical Test, VIVA | | | | | |

**Prerequisites:** Basic concepts of object oriented programming language - c#.

**Methodology, Pedagogy & Andragogy:** This course focuses on experiential learning using the Unity game engine. Students will engage in hands-on activities to design and develop 2D games using Unity and basic C# scripting. Emphasis is on visual development, animation, game logic, and interactive content creation.

## B. Outline of the Course:

| Week No | Practical | Description |
|---|---|---|
| 1 | Unity setup | Install Unity and Visual Studio; Explore Unity Interface; Create a 2D Project; Add GameObjects. |
| 2 | C# Basics | Introduction to C# syntax: variables, data types, functions, and basic movement script. |
| 3 | Object Manipulation | Manipulate GameObjects using Transform; Work with Vector2/Vector3; Group and prefab objects |
| 4 | Physics Setup | Rigidbody2D and Collider2D setup; Physics Materials and gravity simulation |
| 5 | Player Control | Implement player movement and jump with keyboard input; Clamp player position |
| 6 | Sprite Handling | Import sprite sheets; Setup sorting layers; Create parallax backgrounds. |

| | | |
|---|---|---|
| **7** | Animation | Create basic animations with Animator; Transitions between states; Trigger with script |
| **8** | Collision & Scoring | Detect collisions and triggers; Create collectible objects; Implement scoring system. |
| **9** | Audio & UI | Add background music and sound effects; Design UI elements like score and health. |
| **10** | Game Loop | Develop basic game states (start/play/pause/over); Manage game loop and UI controls. |
| **11** | GameObject Lifecycle | Unity GameObject lifecycle; Script order (Awake, Start, Update); Optimize asset usage |
| **12** | Game Build | Build and export game (Windows/WebGL); Create splash screen and credits screen |
| **13** | Project Planning | Mini project planning and design approval |
| **14** | Project Development | Mini project development and internal testing |
| **15** | Final Submission | Final game polishing, presentation, and viva |

**Total hours: 60**

**Core Books:**


1. John Horton: Beginning C# Game Programming, Packt Publishing, 2016.
2. Harrison Ferrone: Learning C# by developing Games with Unity 2021, Packt Publishing, 5th Edition.
3. Jesse Freeman: Introduction to Game Development using C# and Unity, Apress, 2014.

**Reference Books:**

1. Eric Lengyel: Foundations of Game Enigne Development – Vol 1: Mathematics, 2017
2. Alan Thorn: Unity Game Development Cookbook, Packt Publishing.
3. Jonathan Weinberger: Unity 2021 Mobile Game Development, Unity certification associate guide.

**Web References:**

1. https://learn.unity.com/ [For Unity tutorials and learning paths]
2. https://docs.unity3d.com/Manual/index.html [For Unity Documentation]
3. https://www.raywenderlich.com/unity [For Unity game development guide and articles]
4. https://geeksforgeeks.org/unity/ [For beginner tutorials for Unity]

**Course Outcome:** Upon successful completion of the course, student will:

| | |
|---|---|
| **CO1 :** | Understanding the Unity editor interface and core game development workflow |
| **CO2 :** | Develop basic 2D games using GameObjects, physics and C# scripting |
| **CO3 :** | Animate and control objects with logic and input |
| **CO4 :** | Design Game UI, scoring systems and simple audio integration |
| **CO5 :** | Export a playable game prototype and present it with confidence |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 3 | 2 | 3 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 3 | 3 |
| CO2 | 3 | 3 | 3 | 3 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 3 | 3 |
| CO3 | 3 | 3 | 3 | 3 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 3 | 3 |
| CO4 | 2 | 2 | 2 | 3 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 3 |
| CO5 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

# CTUP402 - On-Job Training-II

| Course Code | Course Name | Theory / Practical / Others | Marks (Theory) | | Marks (Practical) | | Credit | NEP-2020 Classification |
|---|---|---|---|---|---|---|---|---|
| | | | CCE | SEE | CCE | SEE | | |
| CTUP402 | On-Job Training-II | Viva | - | - | 75 | 75 | 06 | Research Project/Dissertation |

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| **06** | **VIVA** | **12** | **180** | **270** | **450** |

**Pre-requisite:** None

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Apply advanced IT skills (e.g., system integration, full module development, or research methodologies) to solve industry-relevant problems independently or in teams. |
| CO2 : | Lead or contribute significantly to team-based IT projects, demonstrating advanced collaboration and problem-solving skills. |
| CO3 : | Optimize and implement IT solutions using advanced tools or techniques (e.g., cloud platforms, APIs, or emerging frameworks). |
| CO4 : | Adhere to professional standards, including project management, confidentiality, and ethical considerations in IT environments. |
| CO5 : | Deliver comprehensive project documentation and presentations, showcasing employability and technical expertise. |

- **Refer Annexure-I for more information.**

# CTUR402 - Research Project-II

| Course Code | Course Name | Theory / Practical / Others | Marks (Theory) | | Marks (Practical) | | Credit | NEP-2020 Classification |
|---|---|---|---|---|---|---|---|---|
| | | | CCE | SEE | CCE | SEE | | |
| CTUR402 | Research Project-II | Viva | - | - | 75 | 75 | 06 | Research Project/Dissertation |

| Credit | Component | Instruction/Contact Hours (Per Week) | Instruction /Contact Hours (Per Course) | Experiential Learning Hours (Per Course) | Total Notional Hours |
|---|---|---|---|---|---|
| **06** | **VIVA** | **12** | **180** | **270** | **450** |

**Pre-requisite:** None

**Course Outcomes:** Upon successful completion of the course, students will be,

| | |
|---|---|
| CO1 : | Execute a research project or dissertation, applying methodologies developed in Research Internship-I to address the research question. |
| CO2 : | Analyze research data or results using appropriate IT tools (e.g., Python, R, statistical software) to derive meaningful conclusions. |
| CO3 : | Communicate research findings effectively through a dissertation/thesis and oral presentation, adhering to academic standards. |
| CO4 : | Ensure originality and ethical conduct in research outputs, including compliance with plagiarism policies and intellectual property rights. |
| CO5 : | Evaluate the significance and potential impact of research findings on IT practices or interdisciplinary applications. |

- **Refer Annexure-I for more information.**

# Annexure - I

## Guidelines for Research Internship/On-Job Training (OJT) for BSc IT Programmes

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### 1.0 Introduction

In alignment with the National Education Policy (NEP) 2020 and the University Grants Commission (UGC) Guidelines for Internship/Research Internship/On-the-Job Training (OJT), the Bachelor of Science in Information Technology (B.Sc. IT) programmes at CMPICA, CHARUSAT, integrates practical training and research opportunities in Semesters VII and VIII. This document outlines the structure, objectives, and operational framework for:

- **On-the-Job Training (OJT)**: For B.Sc. (IT) Honours, students to enhance employability through industry exposure.
- **Research Internship**: For B.Sc. (IT) Honours with Research, students to develop research aptitude and innovation in computer applications.

These guidelines ensure that the students are equipped for professional careers or advanced studies, contributing to NEP 2020's vision of a research-oriented, skill-driven education system.

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### 2.0 Objectives

### 2.1 OJT Objectives

1. Enhance employability through practical IT skills (e.g., programming, web development).
2. Bridge classroom learning with workplace practices.
3. Develop professional competencies like teamwork and decision-making.
4. Expose students to emerging technologies and industry trends.
5. Encourage entrepreneurial thinking and job readiness.

### 2.2 Research Internship Objectives

1. Develop research aptitude using tools and methodologies in computer applications.
2. Enable students to solve complex IT problems through inquiry and analysis.
3. Enhance analytical skills (e.g., data analysis, model development).
4. Promote ethical research practices and awareness of intellectual property rights.
5. Prepare students for postgraduate research or doctoral studies.

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## 3.0 Categories

As per UGC guidelines, the research internship/OJT is classified into two types for B.Sc. (IT) students:

1. **OJT for Enhancing Employability**: Practical training in IT industries (e.g., software firms, start-ups) or in-house at CHARUSAT to build job-ready skills (B.Sc. (IT) Honours).
2. **Research Internship for Developing Research Aptitude**: Research projects/dissertations under faculty or industry mentors, including multidisciplinary topics (B.Sc. (IT) Honours with Research).

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## 4.0 Structure

The internship structure for B.Sc. (IT) students applies to Semesters VII and VIII with the following general requirements:

- **Credit Allocation**: 12 credits (6 credits/semester, 15-week semester); 1 credit = 30 hours.
- **Eligibility**: Completion of Semesters I-VI (132 credits) and prerequisite courses.
- **Flexibility:** Summer/winter breaks can extend hours beyond the 6-week period without affecting credits
- **Portal:** CMPICA digital portal with API integration (to be developed)
- **Mode:** Physical, digital, or hybrid

| Aspect | OJT (B.Sc. (IT) Honours) | Research Internship (B.Sc. (IT) Honours with Research) |
|---|---|---|
| **Total Duration** | Semesters VII – 6 Credit Course<br>Semesters VIII – 6 Credit Course<br><br>360 hours<br>(180 hours/semester, including a 6-week intensive period at 30 hours/week) | Semesters VII – 6 Credit Course<br>Semesters VIII – 6 Credit Course<br><br>360 hours<br>(Semester VII: methodology, 180 hours over 6 weeks; Semester VIII: dissertation/thesis, 180 hours over 6 weeks) |
| **Specifics** | Hands-on training in software development, database management, networking; includes in-house options via CHARUSAT IT Cell | Research methodology (Semester VII) and dissertation/thesis (Semester VIII); may include multidisciplinary topics |

| | | |
|---|---|---|
| **Collaboration** | MOUs with IT firms (local/MNCs) and in-house CHARUSAT IT Cell | Partnerships with research labs (e.g., DA-IICT) and CHARUSAT Institutions (e.g., Depstar, PDPIAS) |
| **Learning Outcomes** | - Technical proficiency (e.g., coding, networking).<br>-Problem-solving.<br>- Teamwork.<br>- Professional ethics etc. | - Research techniques (e.g., data analysis, tool usage).<br>- Complex problem-solving.<br>- Communication of findings.<br>- Ethical conduct etc. |

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## 5.0 Roles and Responsibilities

5.1 Role of Internship Providing Organization (IPO)

- **OJT**: IT firms, startups, service providers, or in-house CHARUSAT IT Cell offering practical training and mentorship.
- **Research Internship**: HEIs, research labs, IT companies, or in-house CHARUSAT units providing research opportunities and resources.
- Issue completion certificates for both tracks.

5.2 Role of Nodal Officer

- Coordinate with IPOs (external and in-house), sign MOUs, and maintain the CMPICA internship portal.
- Register students, supervisors, and mentors for both OJT and Research Internship.
- Oversee both tracks, ensuring smooth execution and addressing stakeholder concerns.

5.3 Role of Supervisor

- **OJT**:
  - Monitor attendance and task completion (external or CHARUSAT-based supervisor).
  - Evaluate performance and certify completion.
- **Research Internship**:
  - Oversee research progress (e.g., milestones, methodology adherence) by faculty or research mentor.
  - Assess reports and conduct viva-voce.

5.4 Role of Mentor

- **OJT**:

  - Guide students in technical tasks and validate skill acquisition (industry or CHARUSAT mentor).
- **Research Internship**:
  - Provide research guidance, validate outcomes, and certify reports (faculty or interdisciplinary mentor).

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### 6.0 Mechanism

1. **Application**: Students apply via the *CMPICA portal or directly to IPOs (including CHARUSAT IT Cell)* with mentor approval.
2. **Selection**: IPOs select based on their criteria; digital or in-house alternatives provided if needed.
3. **Execution**:
   - **OJT**: Practical training (Semesters VII and VIII, 180 hours each), including in-house training through CHARUSAT IT Cell or similar units.
   - **Research Internship**: Methodology coursework (Semester VII) and dissertation (Semester VIII, 360 hours total), with potential multidisciplinary topics explored via consultation with *in-house CHARUSAT organizations (e.g., engineering, pharmacy, or sciences).*
   - Fortnightly progress reports for both tracks.
4. **Group Projects**: Allowed (2-5 students) (e.g., OJT: app development; Research: AI-healthcare integration).
5. **Digital Backup**: Virtual training or simulation-based research if physical options are unavailable.

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### 7.0 Evaluation

Internships will be evaluated as follows:

| Component | CCE (Internal) - 75 Marks | SEE (Final Exam) - 75 Marks |
|---|---|---|
| Weekly Logs & Attendance | 14 Marks (20% of CCE) | - |
| Mid-Term Report & Evaluation | 23 Marks (30% of CCE) | - |
| Final Report | 38 Marks (50% of CCE) | 50 Marks (Report: OJT Training Report / Research Dissertation) |
| Presentation & Viva-Voce | - | 25 Marks (Assessed by a committee with **one internal and one external examiner**) |

- **Assessment Criteria**:
    - **OJT**: Skill acquisition, report quality, attendance, and project outcomes.
    - **Research Internship**: Innovativeness, originality, research skills, and significance of findings.
- **Research Specifics**: Report includes an originality certificate (no plagiarism), signed by student and mentor.
- **Passing Criteria**: Minimum 36% in each component (*CHARUSAT norms as per NEP 2020*).

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### 8.0 Technical Specifications for Report

- **Content (*detailed template for the final report will be provided to the students*)**:
    - **OJT**: Introduction, tasks performed, skills learned, and outcomes.
    - **Research Internship**: Research question, methodology, results, and conclusions.
    - **Format**: A4 size, 12 pt. Times New Roman, 1.5-line spacing, 1-inch margins, spiral-bound, printed on both sides.
- **Submission**: Two hard copies (one for FCA - CMPICA, one for student) + digital copy, 10 days before viva-voce.

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